DATA STRUCTURES AND ALGORITHMS

23CSE203

LAB MANUAL

SUBMITTED BY	
NAME	Varshith P
ROLL NO	AV.SC.U4CSE25232
YEAR/SEM/SECTION	2ND YEAR/SEM-3/CSE-C
CREDITS	05

SUBMITTED TO	Dr Gayathri Parasa
DESIGNATION	Asst. Professor (Sr. Gr)
DEPARTMENT	SCHOOL OF COMPUTING

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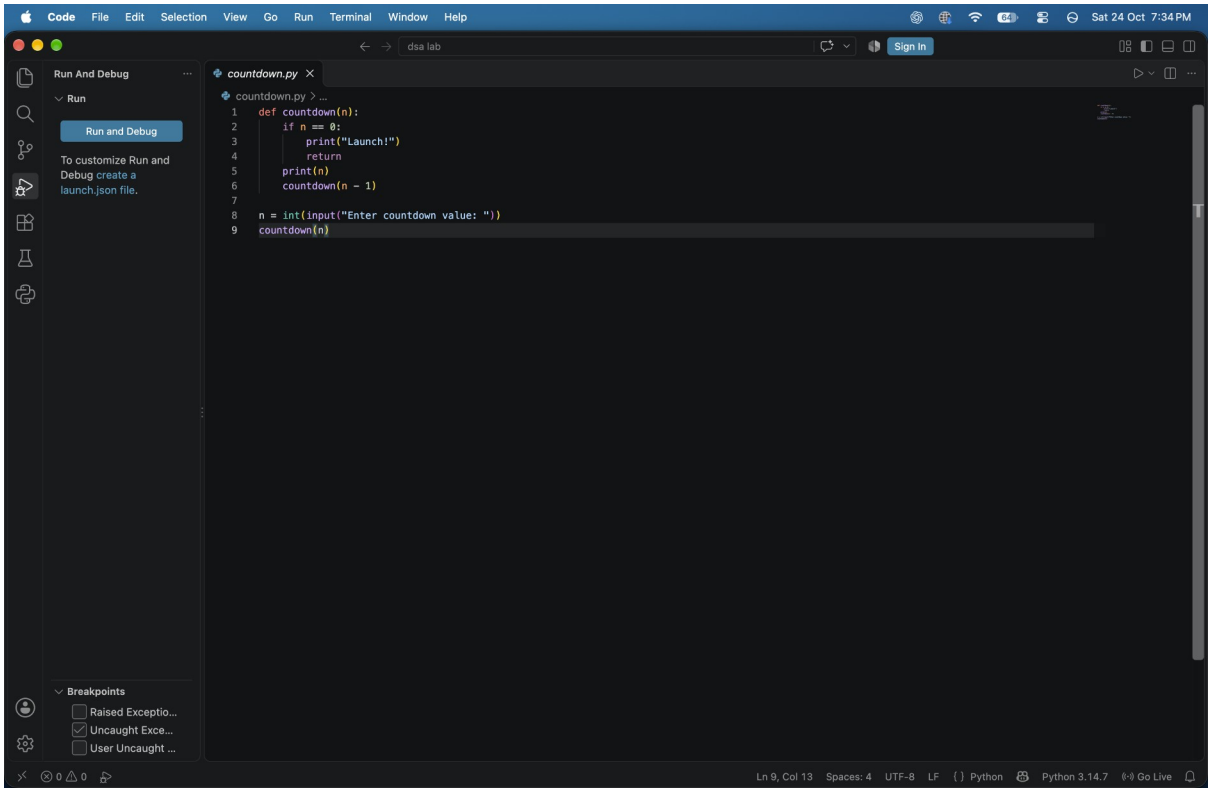
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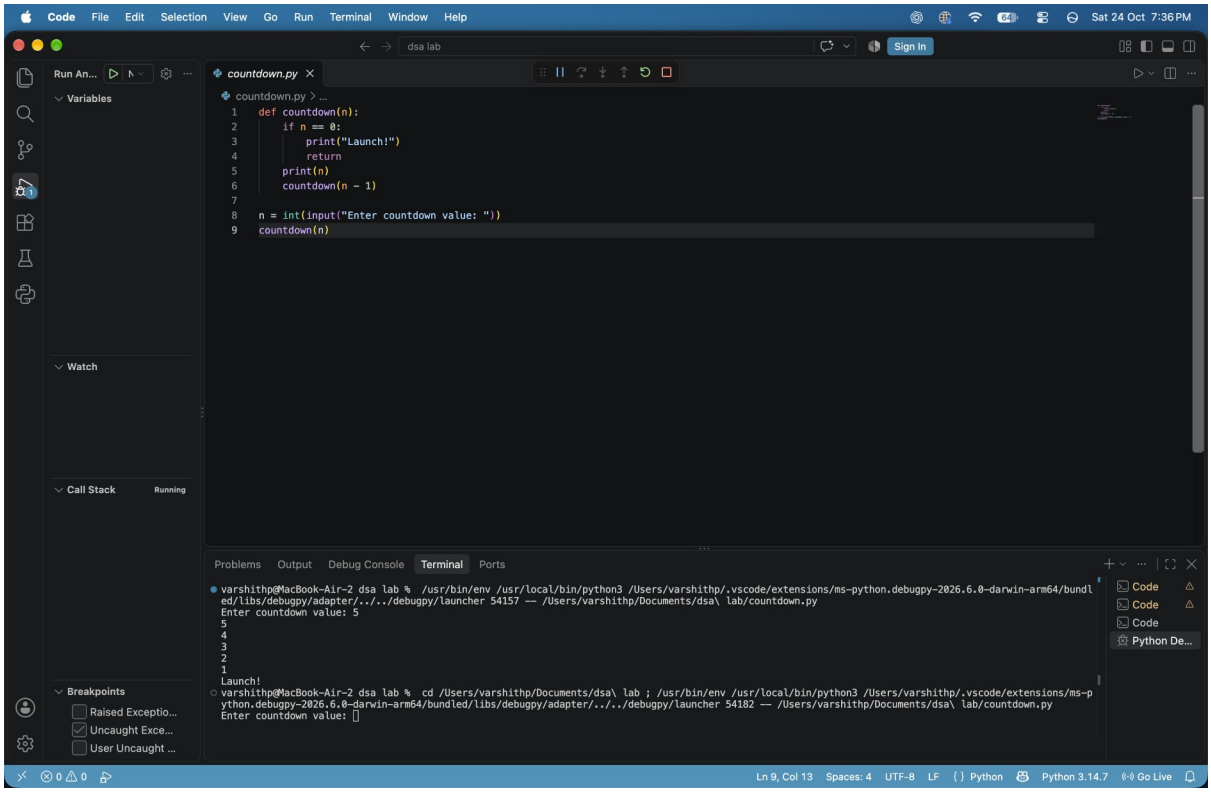
WEEK – 1

1(A). A rocket launch system displays a countdown before launch. Instead of using a loop, implement the countdown using recursion. Write a recursive python program to display a countdown from n to 1, followed by “Launch!”.

Code:-

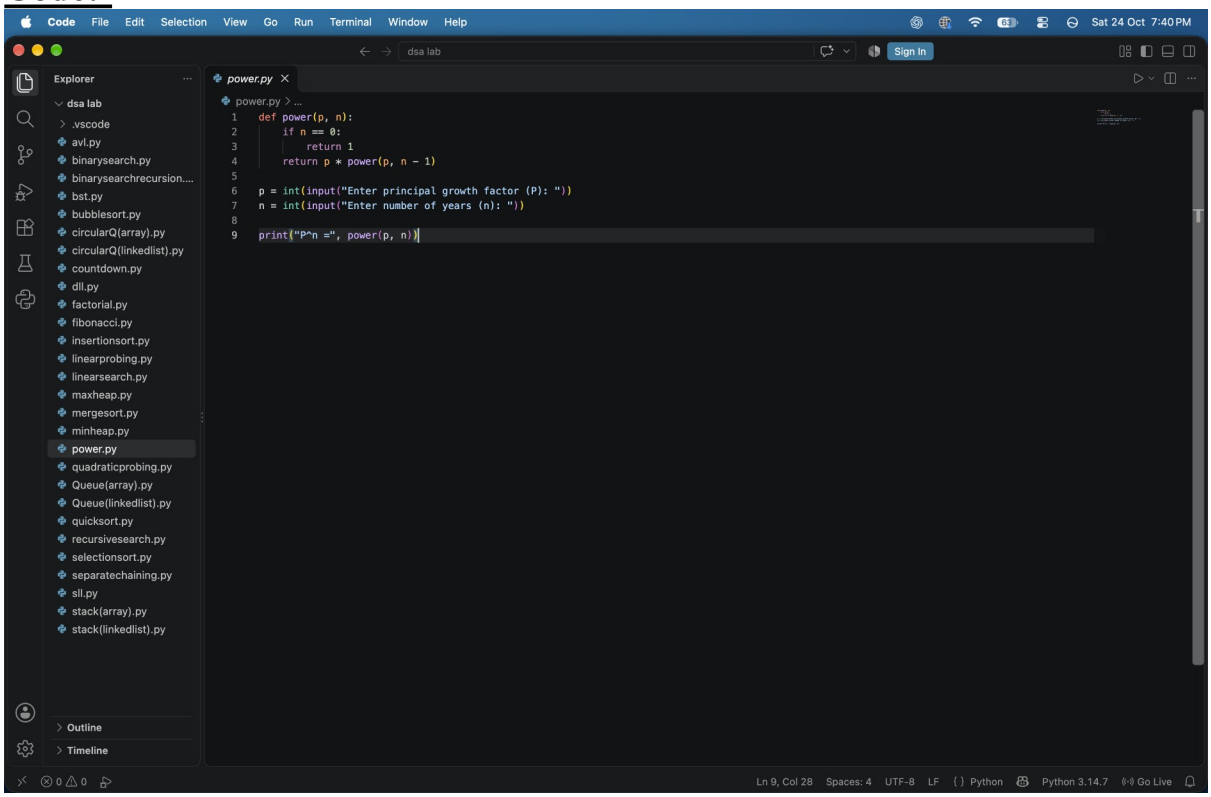


Output:-

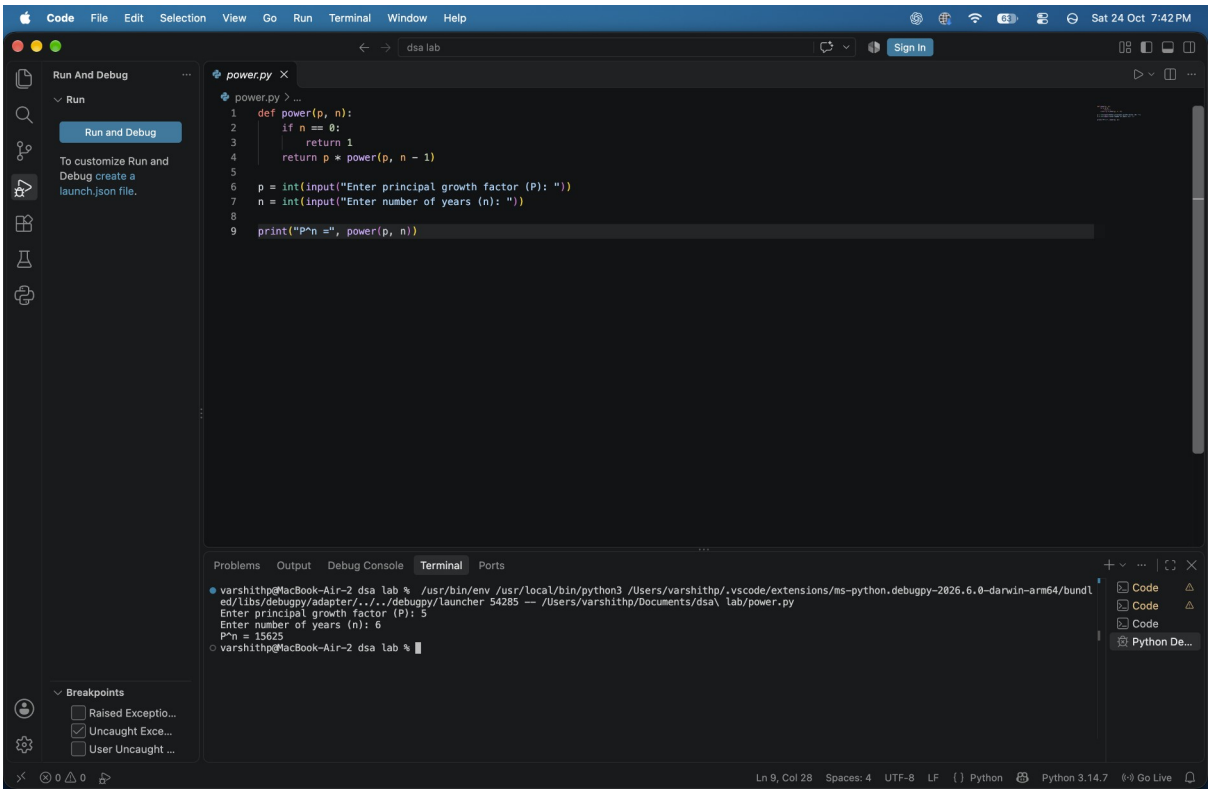

countdown.py
def countdown(n):
if n == 0:
print("Launch!")
return
print(n)
countdown(n - 1)
n = int(input("Enter countdown value: "))
countdown(n)
Problems Output Debug Console Terminal Ports
Enter countdown value: 5
5
4
3
2
1
Launch!
Enter countdown value:

1(B). Banks calculate the growth of an investment using compound interest. The formula requires calculating powers, which can be implemented recursively. Write a recursive program to calculate P^n(power) where: P= principal growth factor, n= number of years.

Code:-

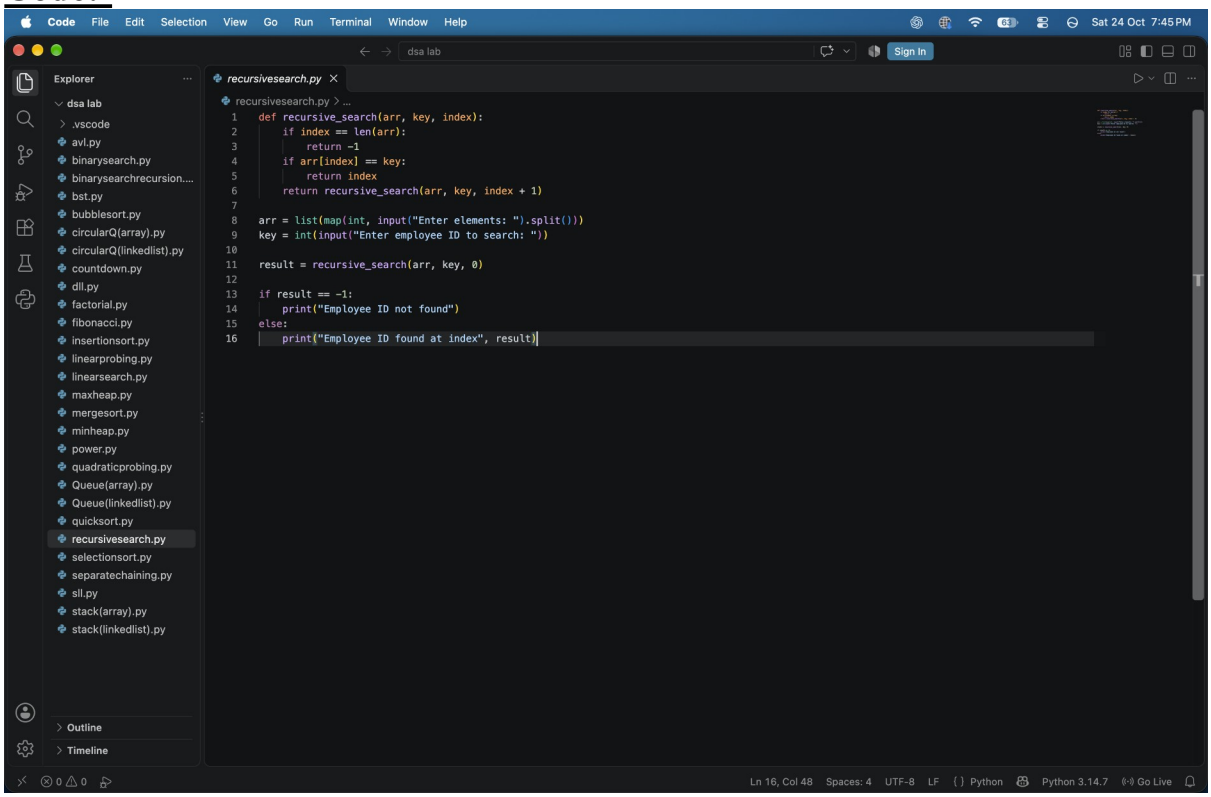


Output:-

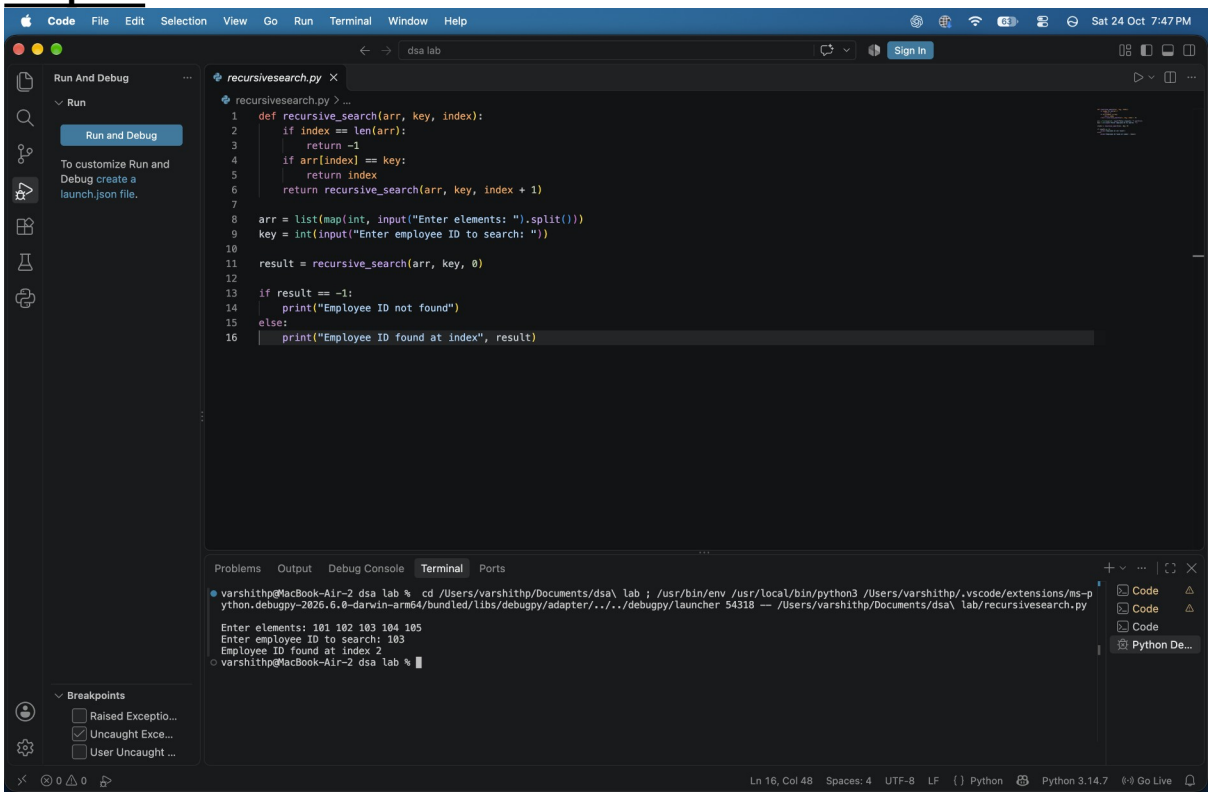

Code File Edit Selection View Go Run Terminal Window Help
Sat 24 Oct 7:42 PM
dsa lab
Sign In
Run And Debug
Run
Run and Debug
To customize Run and Debug create a launch.json file.
power.py
power.py > ...
def power(p, n):
if n == 0:
return 1
return p * power(p, n - 1)
p = int(input("Enter principal growth factor (P): "))
n = int(input("Enter number of years (n): "))
print("P^n =", power(p, n))
Problems Output Debug Console Terminal Ports
varshithp@MacBook-Air-2 dsa lab % /usr/bin/env /usr/local/bin/python3 /Users/varshithp/.vscode/extensions/ms-python.debugpy-2026.6.0-darwin-arm64/bundled/libs/debugpy/adapter/../../debugpy/launcher 54285 -- /Users/varshithp/Documents/dsa\ lab/power.py
Enter principal growth factor (P): 5
Enter number of years (n): 6
P^n = 15625
varshithp@MacBook-Air-2 dsa lab %
Code
Code
Code
Python De...
Breakpoints
Raised Exceptio...
Uncaught Exce...
User Uncaught ...
Ln 9, Col 28 Spaces: 4 UTF-8 LF Python Python 3.14.7 Go Live

1(C). A HR department stores employee ID's in a list. Develop a recursive program to search for a given employee ID. Write a recursive function to search for an employee ID in a list.

Code:-

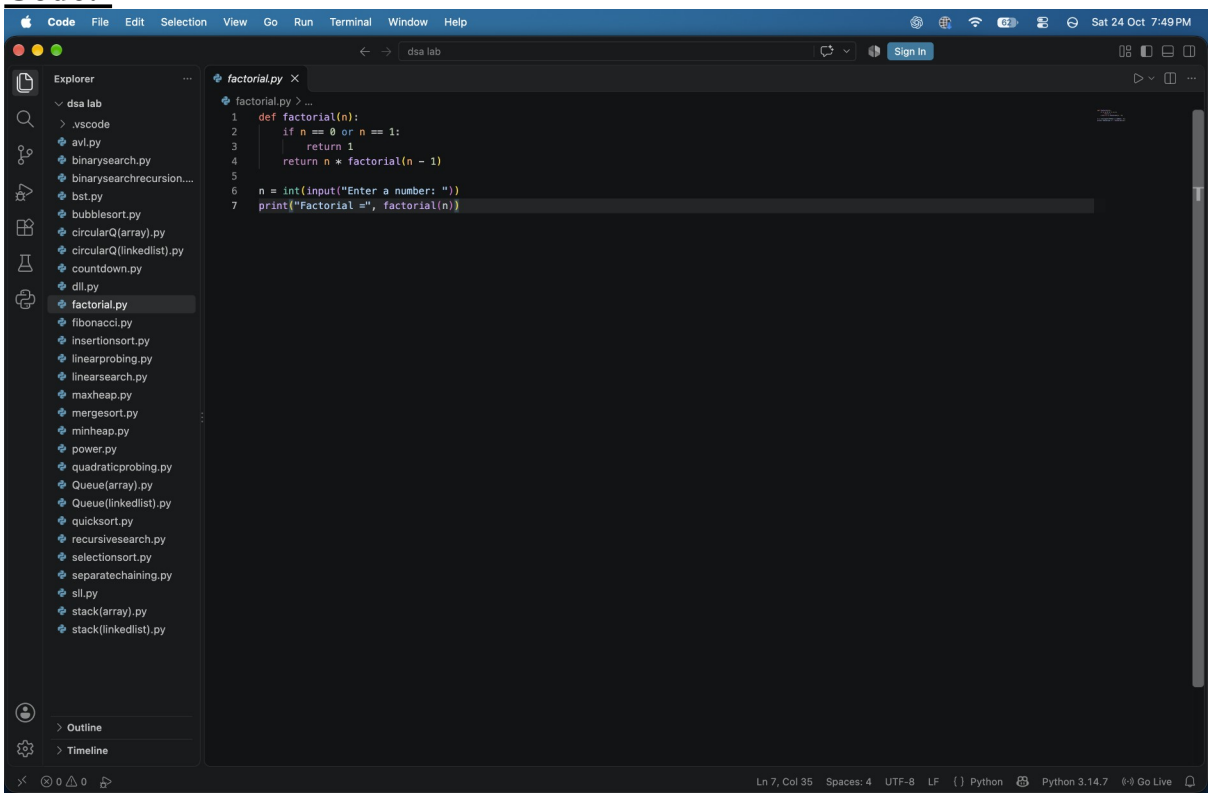


Output:-

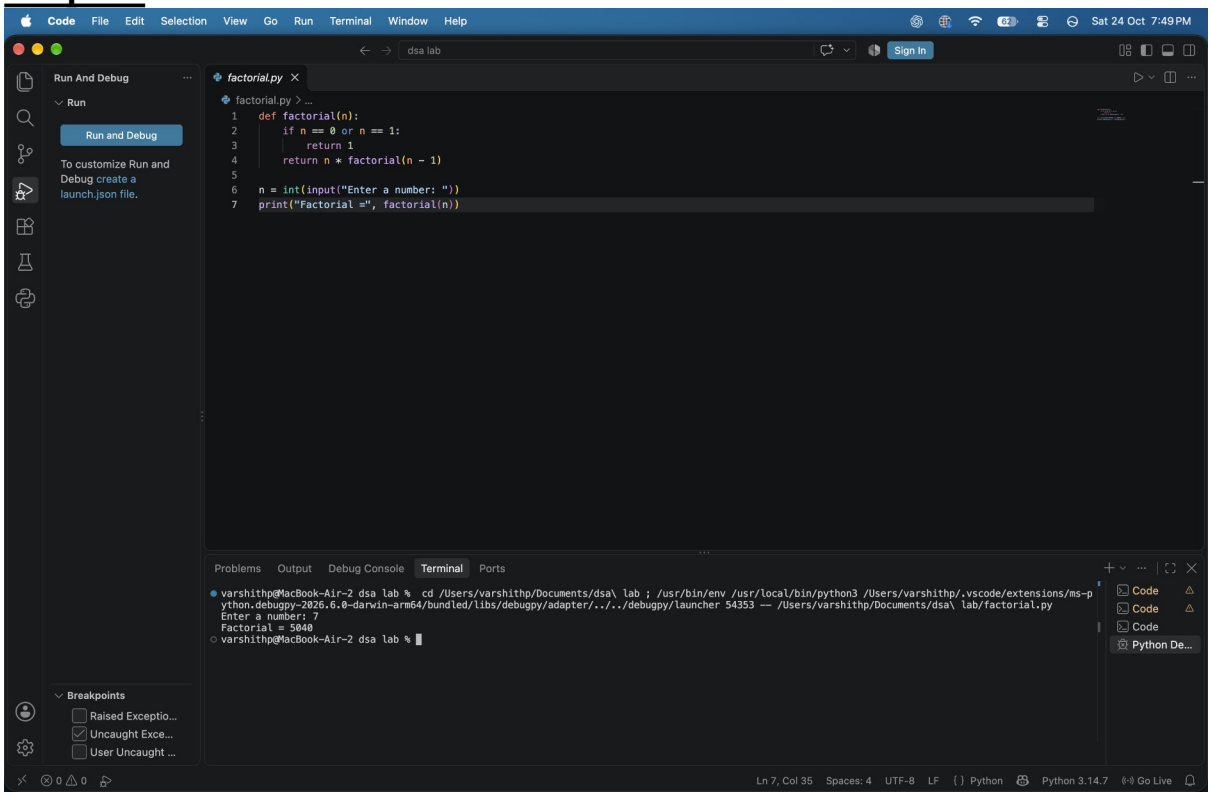

Code File Edit Selection View Go Run Terminal Window Help
Sat 24 Oct 7:47 PM
dsa lab
Sign In
Run And Debug
Run
Run and Debug
To customize Run and Debug create a launch.json file.
recursivesearch.py
def recursive_search(arr, key, index):
if index == len(arr):
return -1
if arr[index] == key:
return index
return recursive_search(arr, key, index + 1)
arr = list(map(int, input("Enter elements: ").split()))
key = int(input("Enter employee ID to search: "))
result = recursive_search(arr, key, 0)
if result == -1:
print("Employee ID not found")
else:
print("Employee ID found at index", result)
Problems Output Debug Console Terminal Ports
varshithp@MacBook-Air-2 dsa lab % cd /Users/varshithp/Documents/dsa\ lab ; /usr/bin/env /usr/local/bin/python3 /Users/varshithp/.vscode/extensions/ms-python.debugpy-2026.6.0-darwin-arm64/bundled/libs/debugpy/adapter/../../debugpy/launcher 54318 -- /Users/varshithp/Documents/dsa\ lab/recursivesearch.py
Enter elements: 101 102 103 104 105
Enter employee ID to search: 103
Employee ID found at index 2
varshithp@MacBook-Air-2 dsa lab %
Breakpoints
Raised Exceptio...
Uncaught Exce...
User Uncaught ...
Ln 16, Col 48 Spaces: 4 UTF-8 LF Python Python 3.14.7 Go Live

1(D). In a logistics company, the number of possible ways to arrange n different parcels are calculated using the factorial formula (n!). Develop a recursive program to calculate the factorial of a given number.

Code:-

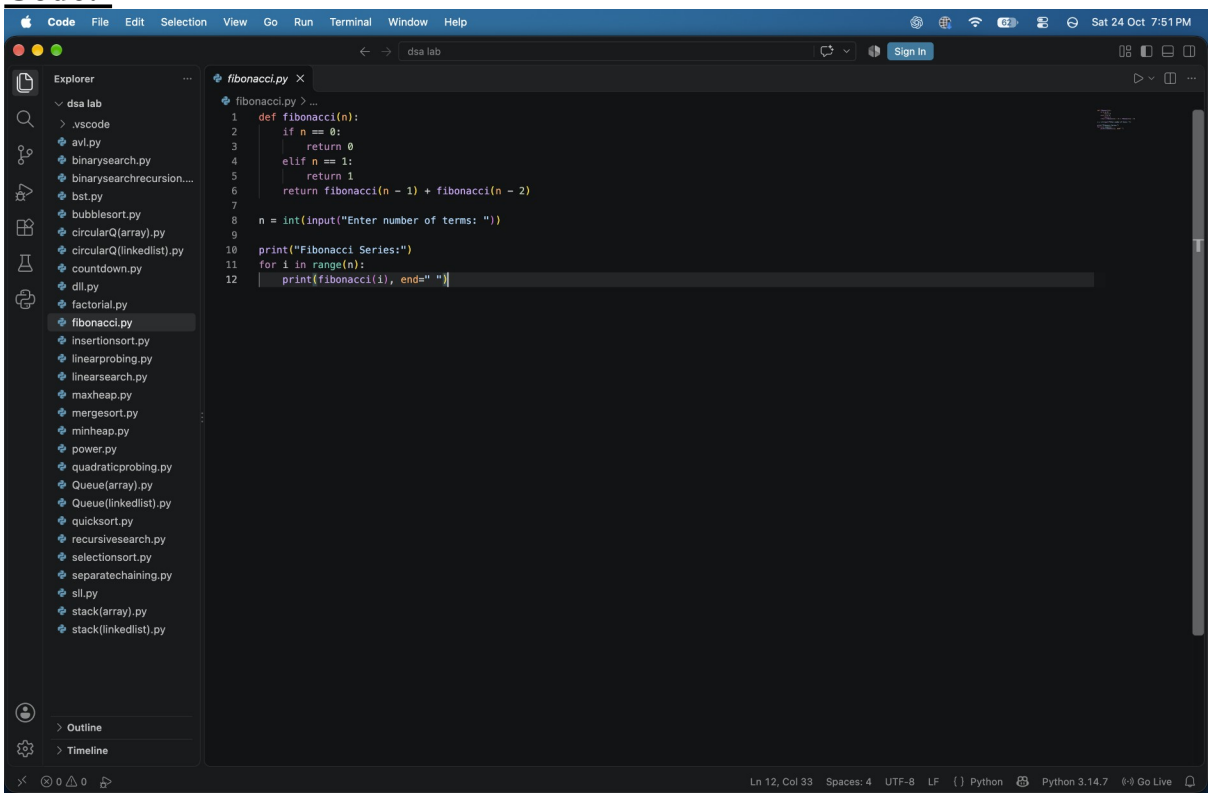


Output:-

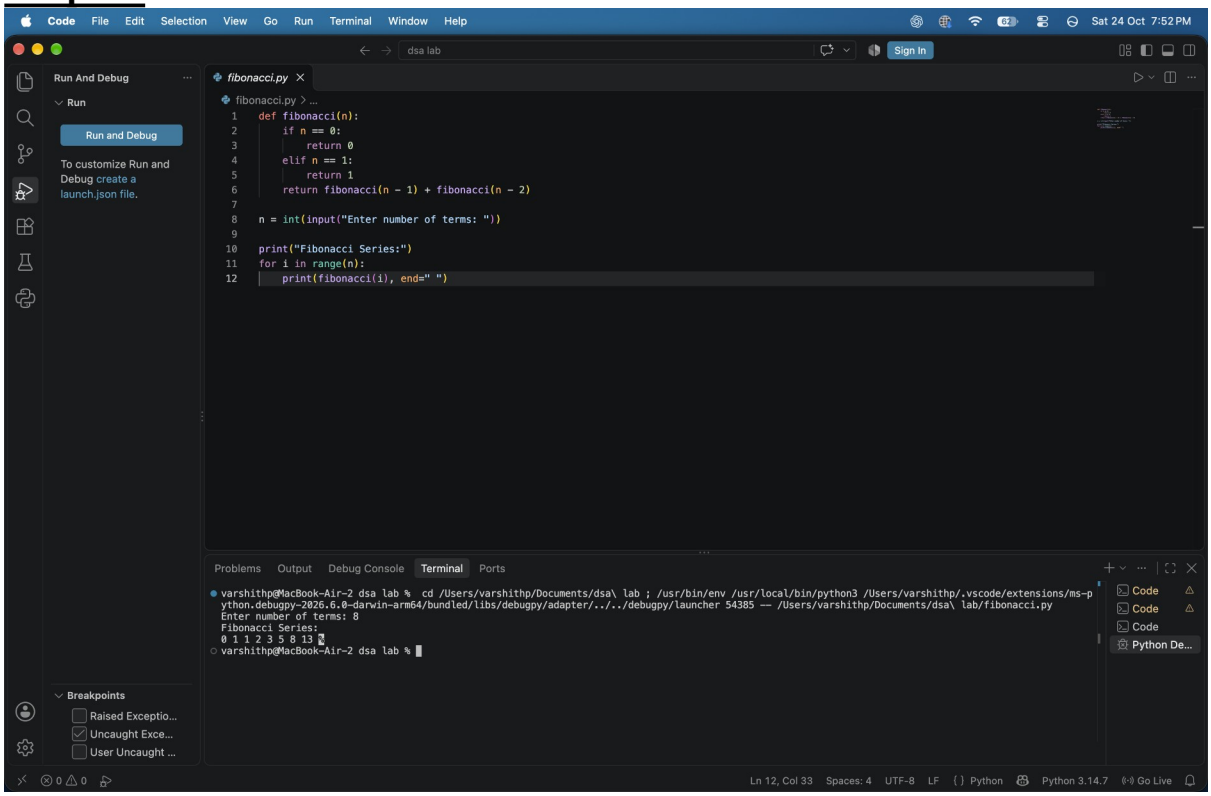

Code File Edit Selection View Go Run Terminal Window Help
Sat 24 Oct 7:49 PM
dsa lab
Sign In
Run And Debug
Run
Run and Debug
To customize Run and Debug create a launch.json file.
factorial.py
def factorial(n):
if n == 0 or n == 1:
return 1
return n * factorial(n - 1)
n = int(input("Enter a number: "))
print("Factorial =", factorial(n))
Problems Output Debug Console Terminal Ports
varshithp@MacBook-Air-2 dsa lab % cd /Users/varshithp/Documents/dsa\ lab ; /usr/bin/env /usr/local/bin/python3 /Users/varshithp/.vscode/extensions/ms-python.debugpy-2026.6.0-darwin-arm64/bundled/libs/debugpy/adapter/../../debugpy/launcher 54353 -- /Users/varshithp/Documents/dsa\ lab/factorial.py
Enter a number: 7
Factorial = 5040
varshithp@MacBook-Air-2 dsa lab %
Breakpoints
Raised Exceptio...
Uncaught Exce...
User Uncaught ...
Ln 7, Col 35 Spaces: 4 UTF-8 LF Python Python 3.14.7 Go Live

1(E). Write a recursive python program to print the first n terms of the Fibonacci series.

Code:-



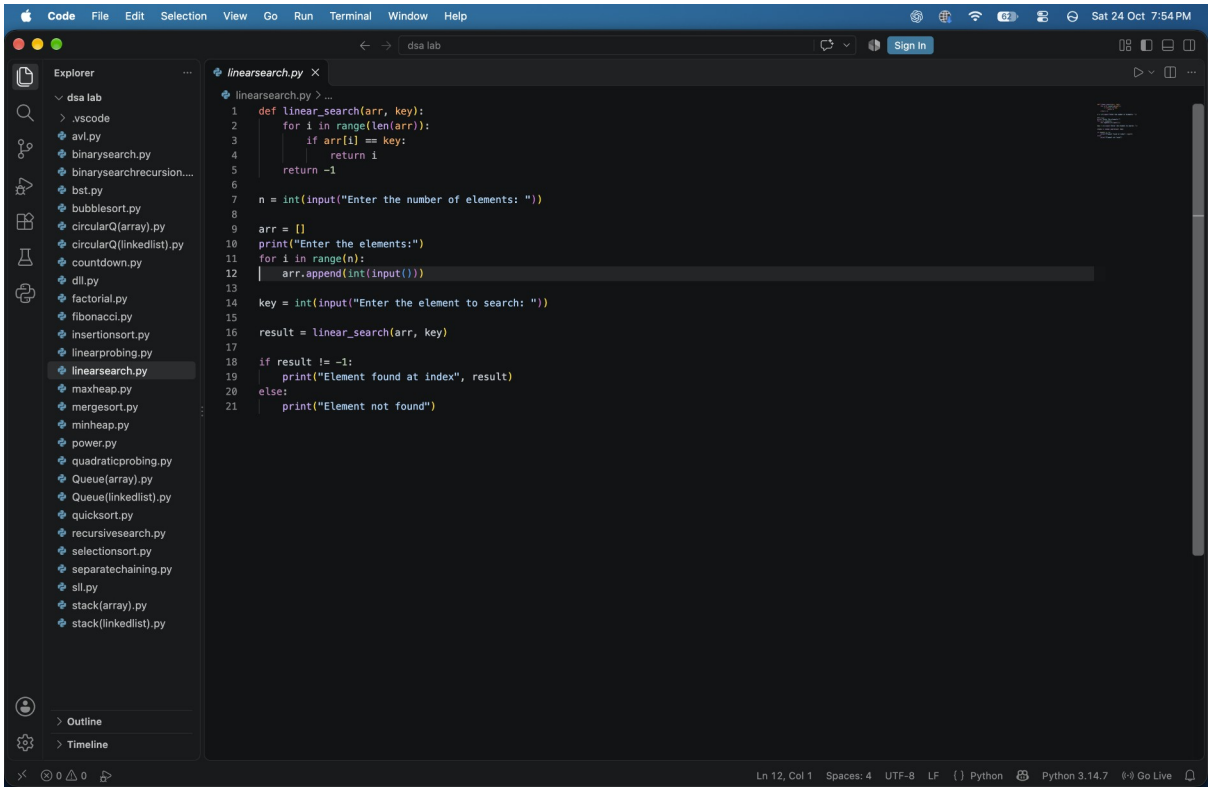
Output:-


Code File Edit Selection View Go Run Terminal Window Help
Sat 24 Oct 7:52 PM
dsa lab
Sign In
Run And Debug
Run
Run and Debug
To customize Run and Debug create a launch.json file.
fibonacci.py
def fibonacci(n):
if n == 0:
return 0
elif n == 1:
return 1
return fibonacci(n - 1) + fibonacci(n - 2)
n = int(input("Enter number of terms: "))
print("Fibonacci Series:")
for i in range(n):
print(fibonacci(i), end=" ")
Problems Output Debug Console Terminal Ports
varshithp@MacBook-Air-2 dsa lab % cd /Users/varshithp/Documents/dsa\ lab ; /usr/bin/env /usr/local/bin/python3 /Users/varshithp/.vscode/extensions/ms-python.debugpy-2026.6.0-darwin-arm64/bundled/libs/debugpy/adapter/../../debugpy/launcher 54385 -- /Users/varshithp/Documents/dsa\ lab/fibonacci.py
Enter number of terms: 8
Fibonacci Series:
0 1 1 2 3 5 8 13
varshithp@MacBook-Air-2 dsa lab %
Breakpoints
Raised Exceptio...
Uncaught Exce...
User Uncaught ...
Ln 12, Col 33 Spaces: 4 UTF-8 LF Python Python 3.14.7 Go Live

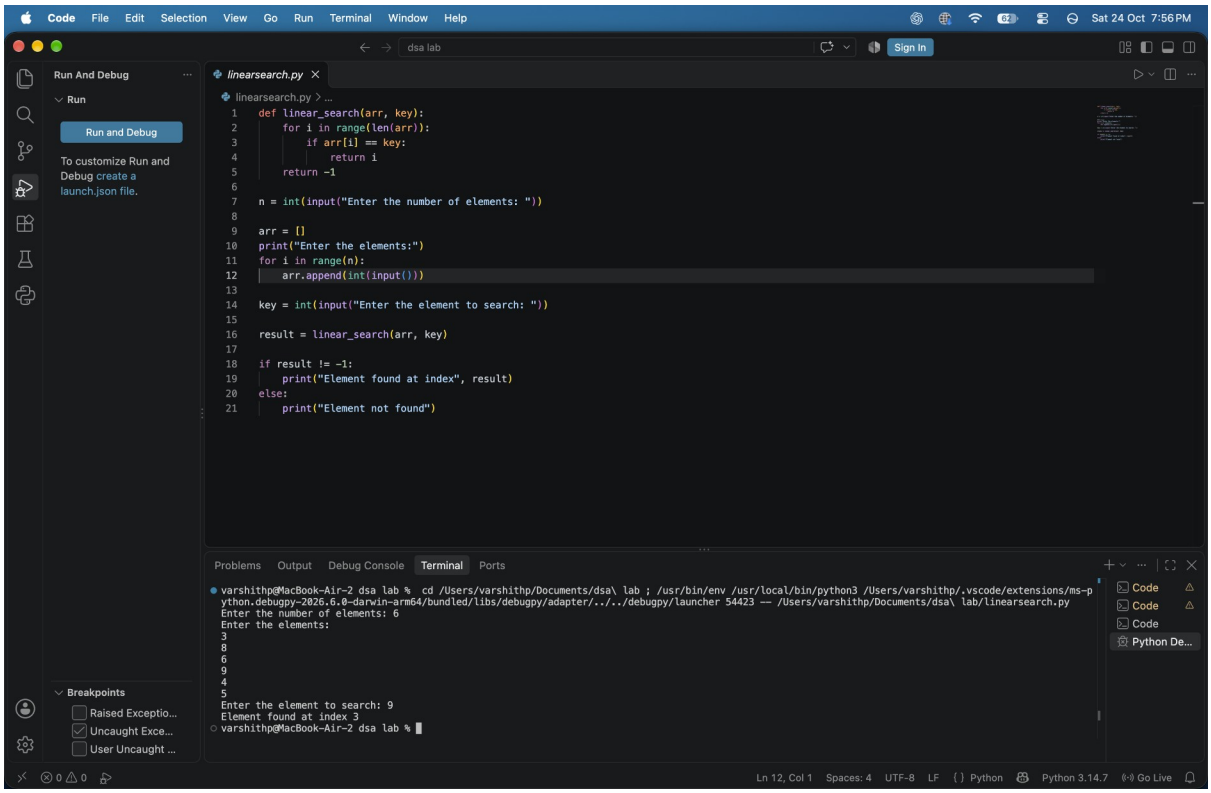
WEEK-2

2(A). Write a python program to implement linear search.

Code:-



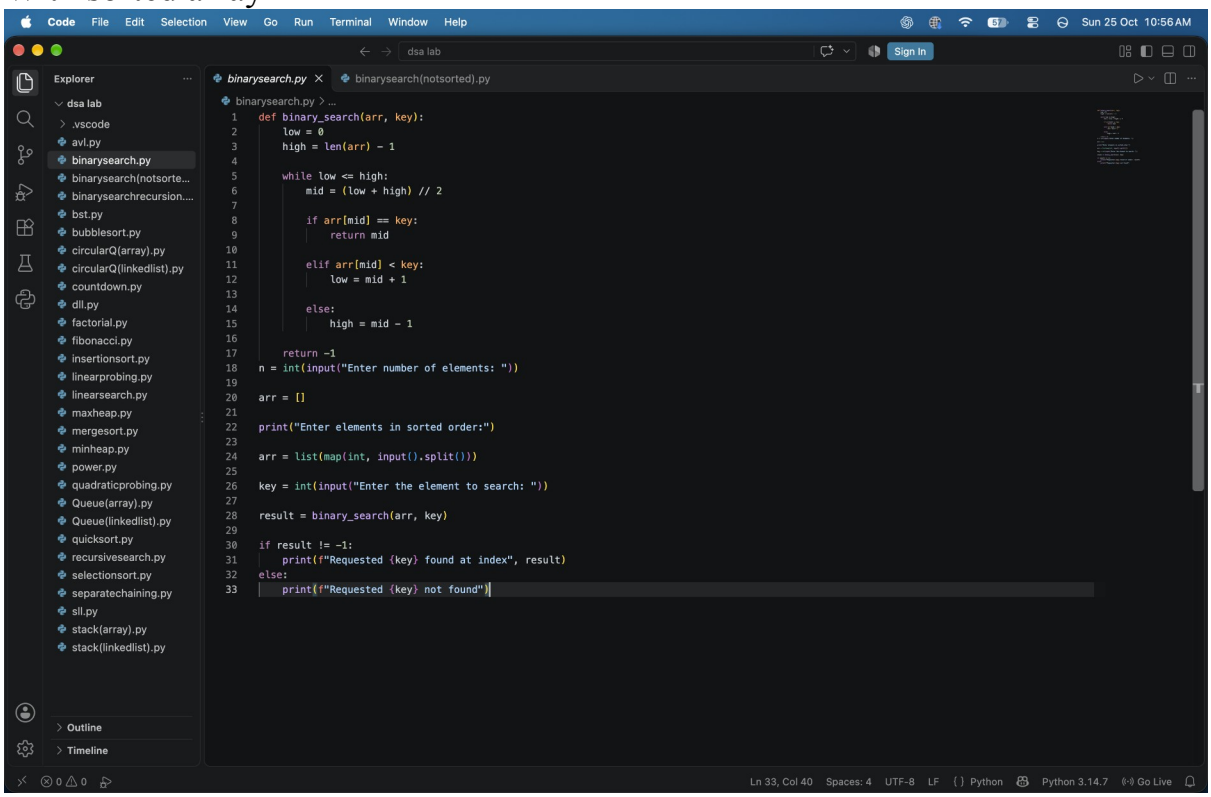
Output:-


def linear_search(arr, key):
for i in range(len(arr)):
if arr[i] == key:
return i
return -1
n = int(input("Enter the number of elements: "))
arr = []
print("Enter the elements:")
for i in range(n):
arr.append(int(input()))
key = int(input("Enter the element to search: "))
result = linear_search(arr, key)
if result != -1:
print("Element found at index", result)
else:
print("Element not found")
Enter the number of elements: 6
Enter the elements:
3
8
6
9
4
5
Enter the element to search: 9
Element found at index 3

2(B). Write a python program to implement binary search with and without sorted array as an input.

Code:-

With sorted array



```
def binary_search(arr, key):
    low = 0
    high = len(arr) - 1

    while low <= high:
        mid = (low + high) // 2

        if arr[mid] == key:
            return mid

        elif arr[mid] < key:
            low = mid + 1

        else:
            high = mid - 1

    return -1
n = int(input("Enter number of elements: "))

arr = []

print("Enter elements in sorted order:")

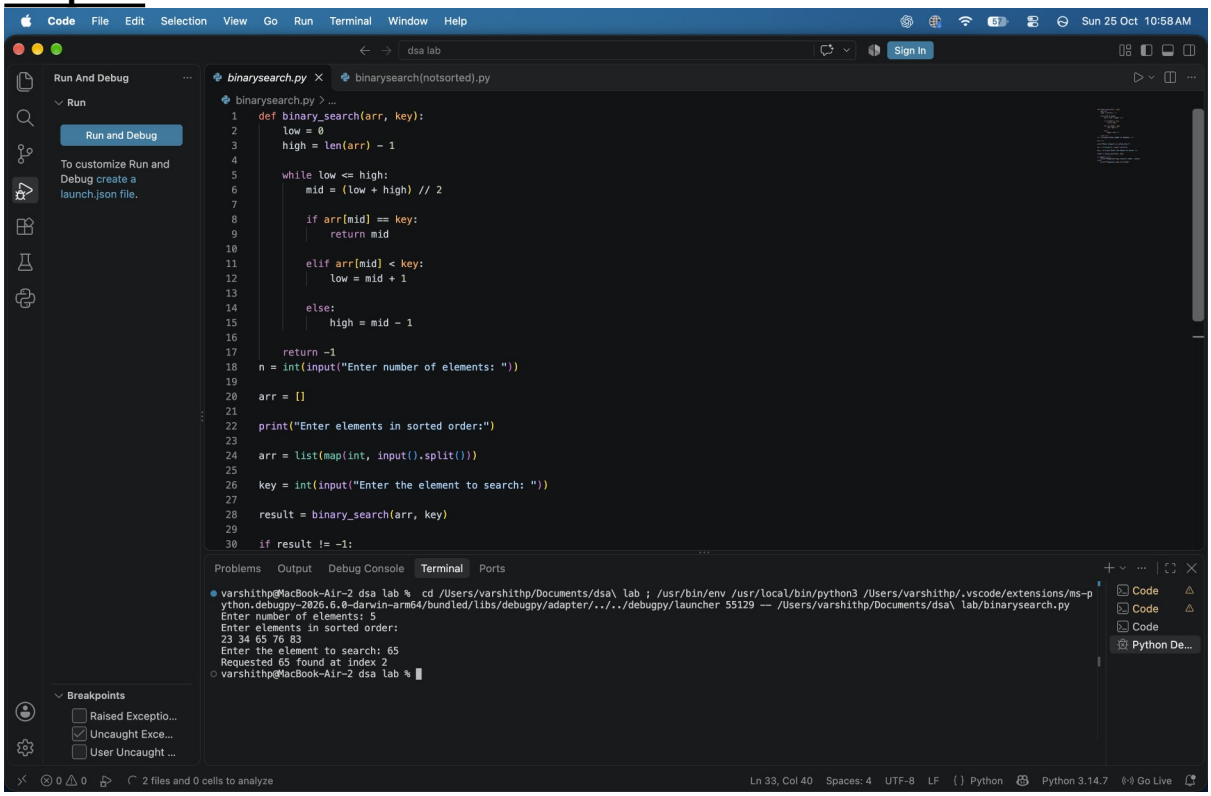
arr = list(map(int, input().split()))

key = int(input("Enter the element to search: "))

result = binary_search(arr, key)

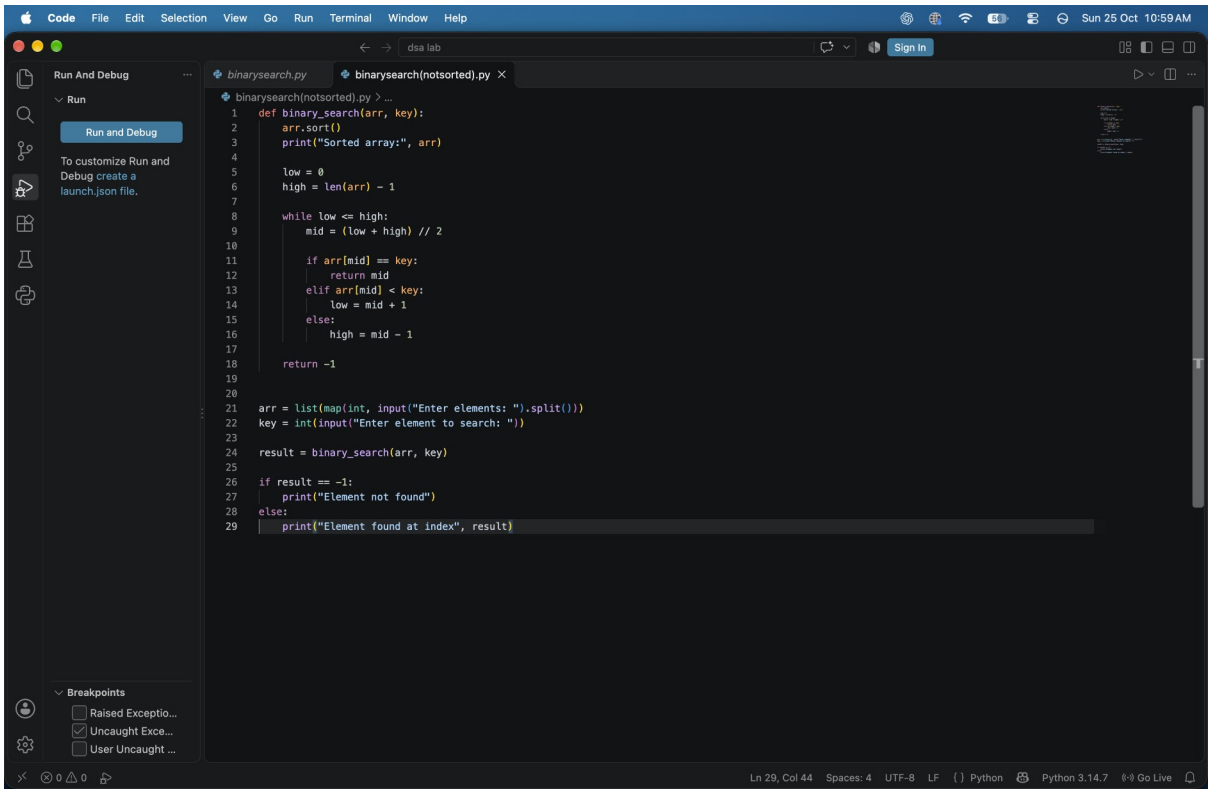
if result != -1:
    print(f"Requested {key} found at index", result)
else:
    print(f"Requested {key} not found")
```

Output:-


Code File Edit Selection View Go Run Terminal Window Help
Sun 25 Oct 10:58 AM
dsa lab
Sign In
Run And Debug
Run
Run and Debug
To customize Run and Debug create a launch.json file.
Breakpoints
Raised Exceptio...
Uncaught Exce...
User Uncaught ...
binarysearch.py
binarysearch(notsorted).py
def binary_search(arr, key):
low = 0
high = len(arr) - 1
while low <= high:
mid = (low + high) // 2
if arr[mid] == key:
return mid
elif arr[mid] < key:
low = mid + 1
else:
high = mid - 1
return -1
n = int(input("Enter number of elements: "))
arr = []
print("Enter elements in sorted order:")
arr = list(map(int, input().split()))
key = int(input("Enter the element to search: "))
result = binary_search(arr, key)
if result != -1:
Problems Output Debug Console Terminal Ports
varshithp@MacBook-Air-2 dsa lab % cd /Users/varshithp/Documents/dsa\ lab ; /usr/bin/env /usr/local/bin/python3 /Users/varshithp/.vscode/extensions/ms-python.debugpy-2026.6.0-darwin-arm64/bundled/libs/debugpy/adapter/../../debugpy/launcher 55129 -- /Users/varshithp/Documents/dsa\ lab/binarysearch.py
Enter number of elements: 5
Enter elements in sorted order:
23 34 65 76 83
Enter the element to search: 65
Requested 65 found at index 2
varshithp@MacBook-Air-2 dsa lab %
Code
Code
Code
Python De...
2 files and 0 cells to analyze
Ln 33, Col 40 Spaces: 4 UTF-8 LF Python Python 3.14.7 Go Live

Without Sorted array

Code:-



```
def binary_search(arr, key):
    arr.sort()
    print("Sorted array:", arr)

    low = 0
    high = len(arr) - 1

    while low <= high:
        mid = (low + high) // 2

        if arr[mid] == key:
            return mid
        elif arr[mid] < key:
            low = mid + 1
        else:
            high = mid - 1

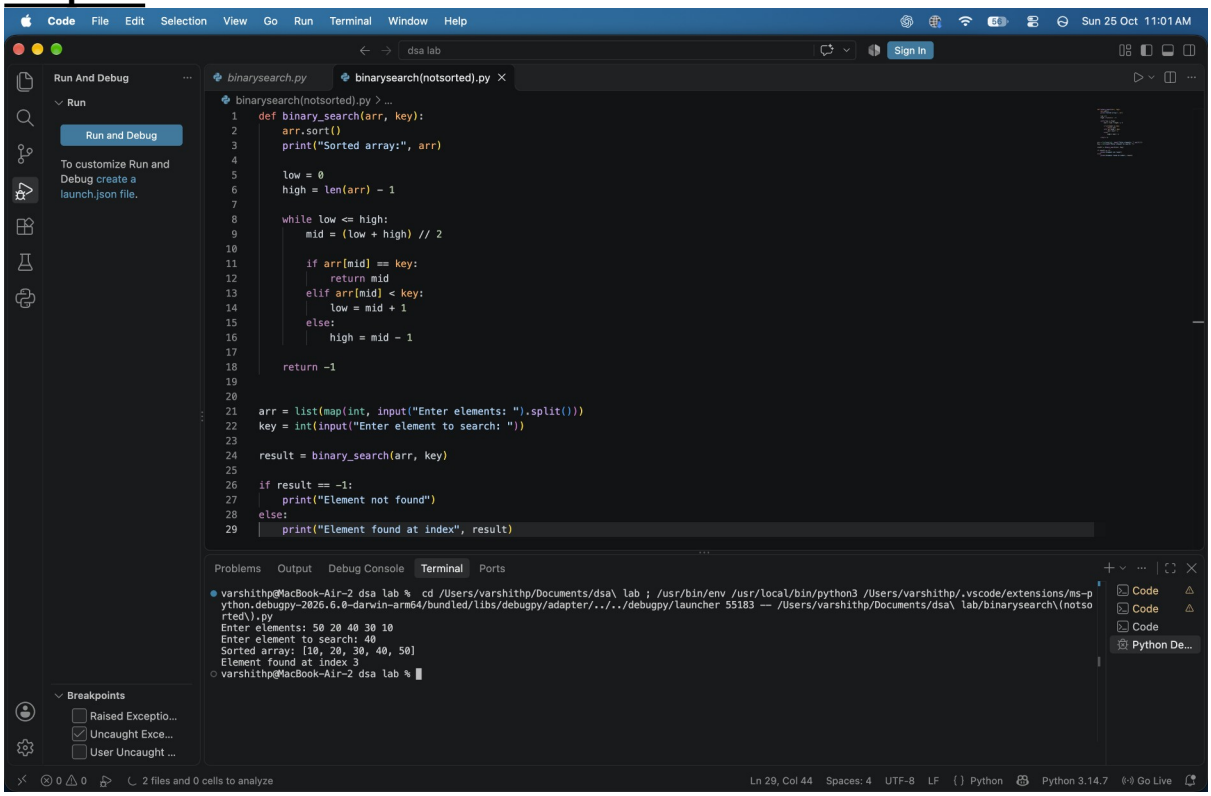
    return -1


arr = list(map(int, input("Enter elements: ").split()))
key = int(input("Enter element to search: "))

result = binary_search(arr, key)

if result == -1:
    print("Element not found")
else:
    print("Element found at index", result)
```

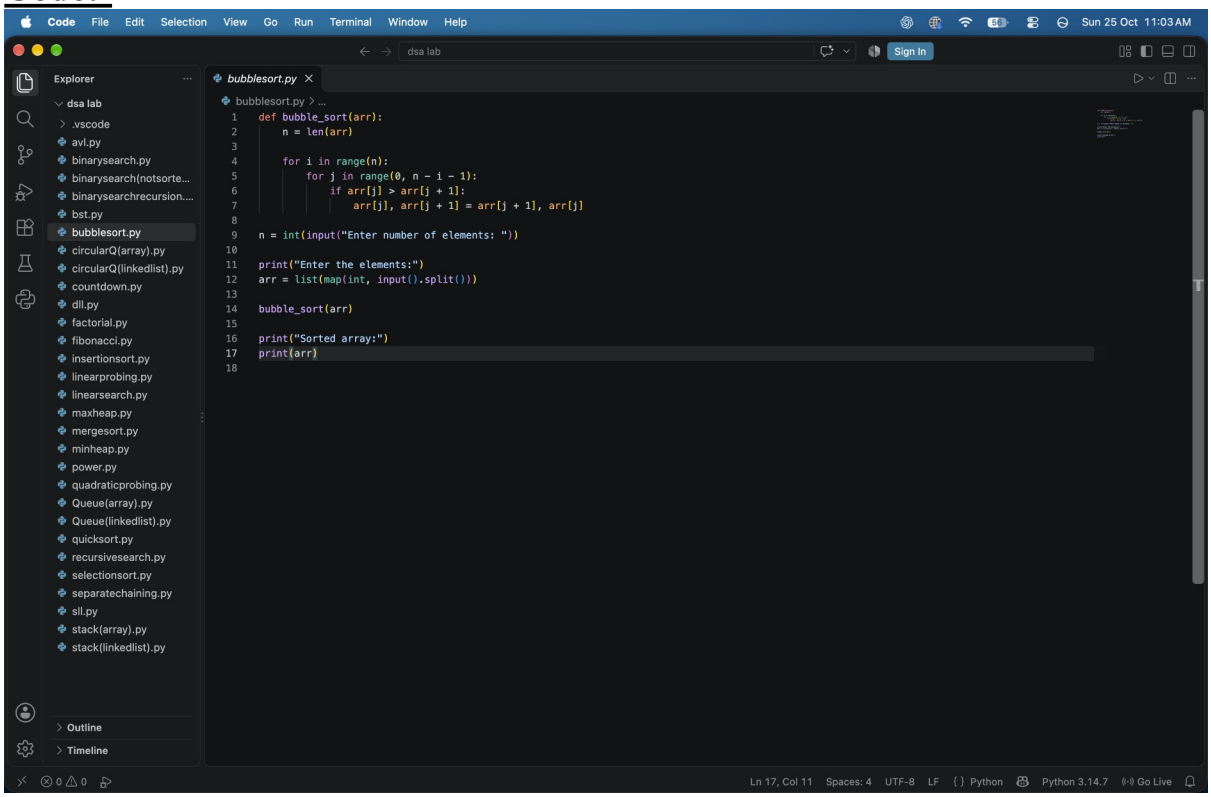
Output:-


Code File Edit Selection View Go Run Terminal Window Help
Sun 25 Oct 11:01 AM
dsa lab
Sign In
Run And Debug
Run
Run and Debug
To customize Run and Debug create a launch.json file.
binarysearch.py
binarysearch(notsorted).py
def binary_search(arr, key):
arr.sort()
print("Sorted array:", arr)
low = 0
high = len(arr) - 1
while low <= high:
mid = (low + high) // 2
if arr[mid] == key:
return mid
elif arr[mid] < key:
low = mid + 1
else:
high = mid - 1
return -1
arr = list(map(int, input("Enter elements: ").split()))
key = int(input("Enter element to search: "))
result = binary_search(arr, key)
if result == -1:
print("Element not found")
else:
print("Element found at index", result)
Problems Output Debug Console Terminal Ports
varshithp@MacBook-Air-2 dsa lab % cd /Users/varshithp/Documents/dsa\ lab ; /usr/bin/env /usr/local/bin/python3 /Users/varshithp/.vscode/extensions/ms-python.debugpy-2026.6.0-darwin-arm64/bundled/libs/debugpy/adapter/../../debugpy/launcher 55183 -- /Users/varshithp/Documents/dsa\ lab/binarysearch\(notsorted\).py
Enter elements: 50 20 40 30 10
Enter element to search: 40
Sorted array: [10, 20, 30, 40, 50]
Element found at index 3
varshithp@MacBook-Air-2 dsa lab %
Code
Code
Code
Python De...
Breakpoints
Raised Exceptio...
Uncaught Exce...
User Uncaught ...
2 files and 0 cells to analyze
Ln 29, Col 44 Spaces: 4 UTF-8 LF Python Python 3.14.7 Go Live

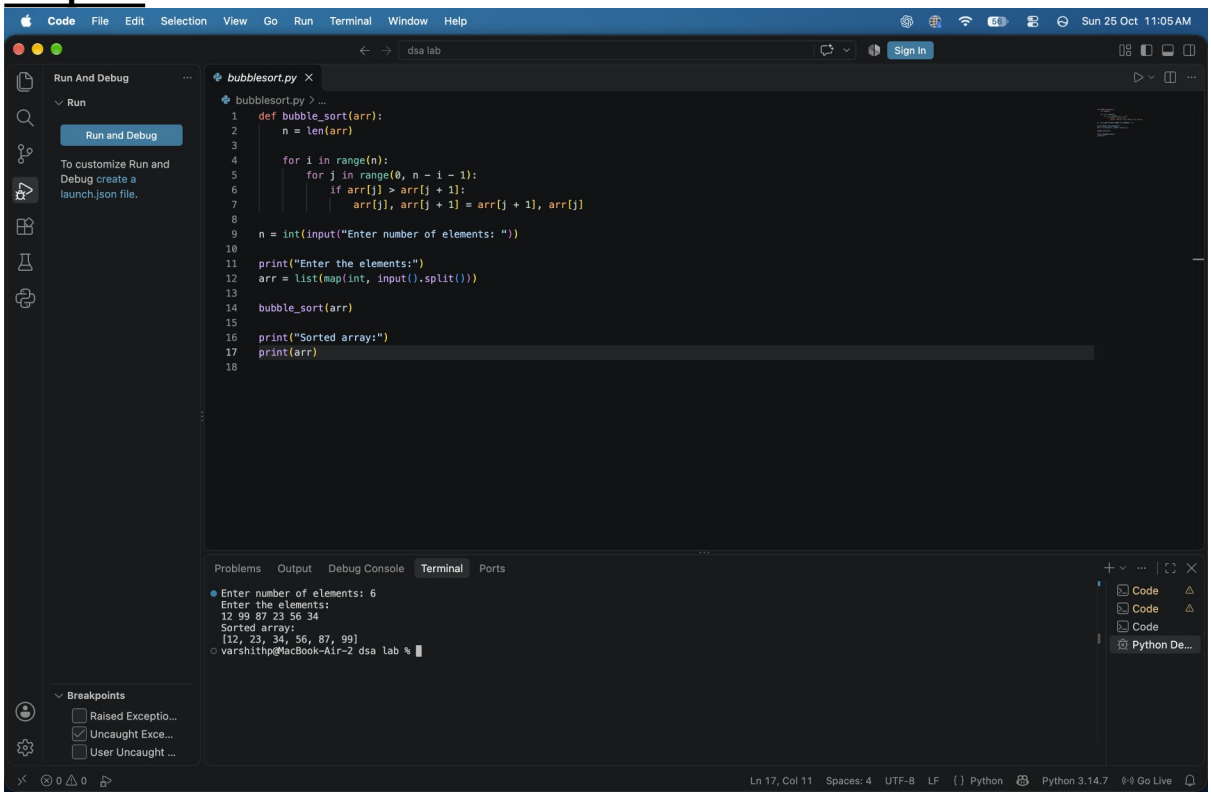
WEEK-3

3(A). Write a python program to implement bubble sort.

Code:-

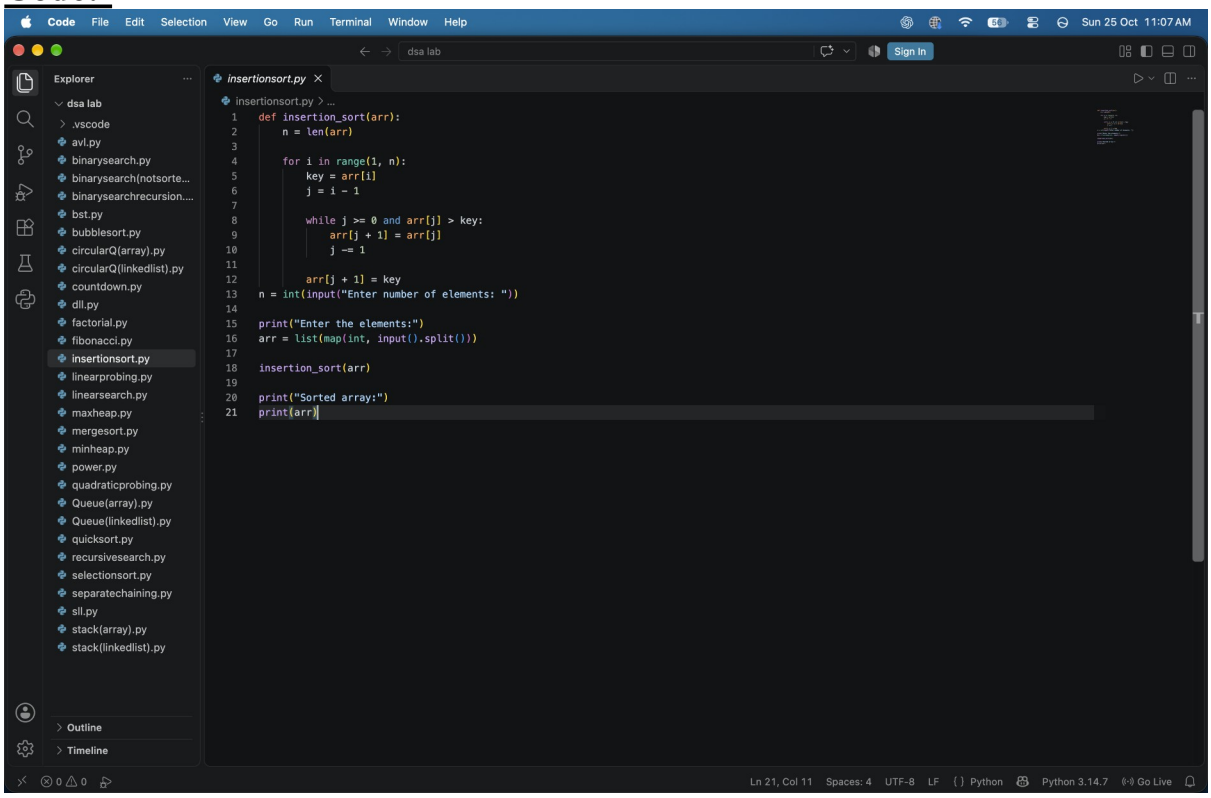


Output:-

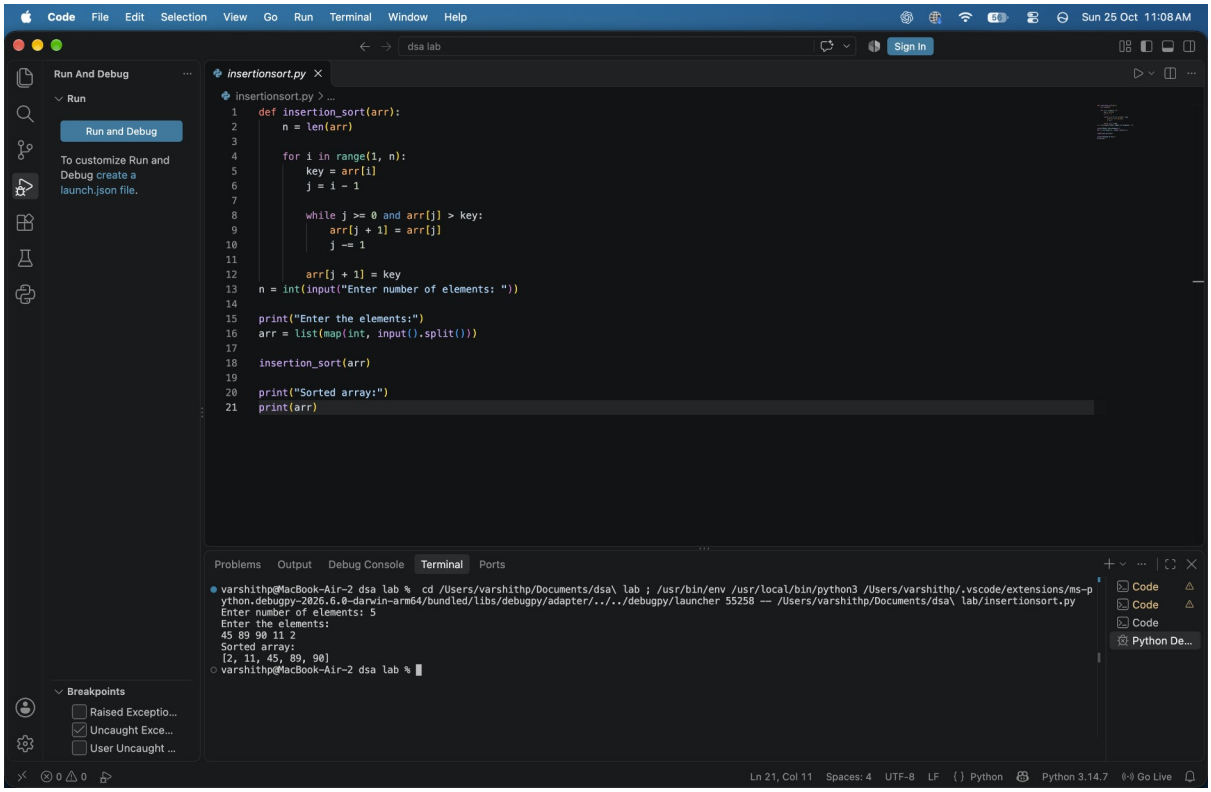

Code File Edit Selection View Go Run Terminal Window Help
Sun 25 Oct 11:05 AM
dsa lab
Sign In
Run And Debug
Run
Run and Debug
To customize Run and Debug create a launch.json file.
bubblesort.py
def bubble_sort(arr):
n = len(arr)
for i in range(n):
for j in range(0, n - i - 1):
if arr[j] > arr[j + 1]:
arr[j], arr[j + 1] = arr[j + 1], arr[j]
n = int(input("Enter number of elements: "))
print("Enter the elements:")
arr = list(map(int, input().split()))
bubble_sort(arr)
print("Sorted array:")
print(arr)
Problems Output Debug Console Terminal Ports
Enter number of elements: 6
Enter the elements:
12 99 87 23 56 34
Sorted array:
[12, 23, 34, 56, 87, 99]
varshithp@MacBook-Air-2 dsa lab %
Code
Code
Code
Python De...
Breakpoints
Raised Exceptio...
Uncaught Exce...
User Uncaught ...
Ln 17, Col 11 Spaces: 4 UTF-8 LF Python Python 3.14.7 Go Live

3(B). Write a python program to implement insertion sort.

Code:-

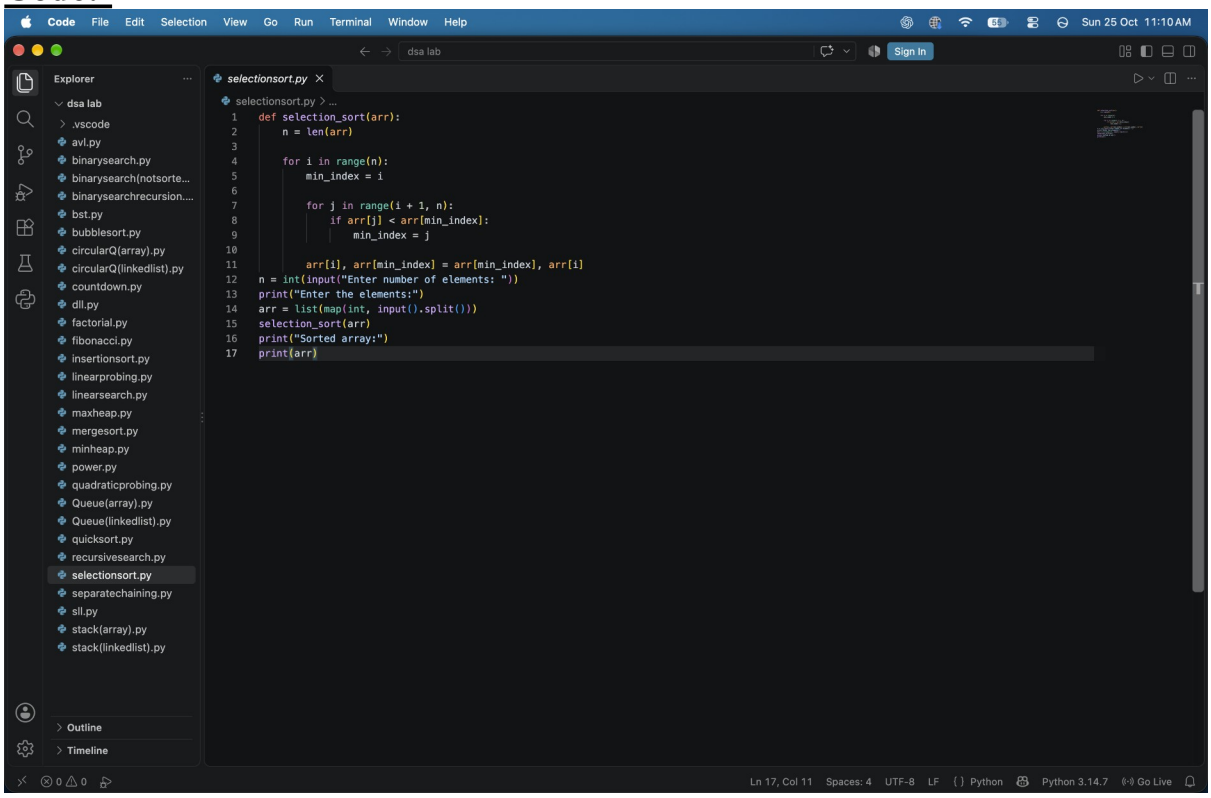


Output:-

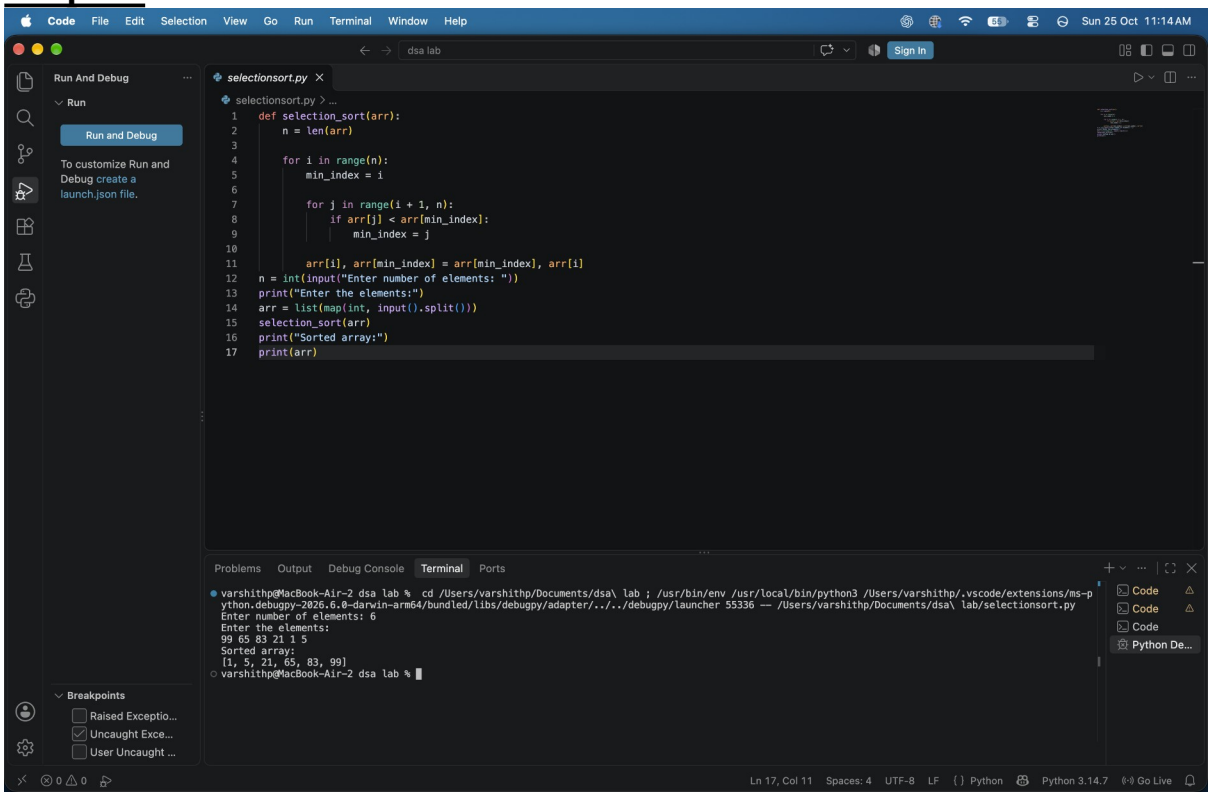

Code File Edit Selection View Go Run Terminal Window Help
Sun 25 Oct 11:08 AM
dsa lab
Sign In
Run And Debug
Run
Run and Debug
To customize Run and Debug create a launch.json file.
insertionsort.py
insertionsort.py > ...
def insertion_sort(arr):
n = len(arr)
for i in range(1, n):
key = arr[i]
j = i - 1
while j >= 0 and arr[j] > key:
arr[j + 1] = arr[j]
j -= 1
arr[j + 1] = key
n = int(input("Enter number of elements: "))
print("Enter the elements:")
arr = list(map(int, input().split()))
insertion_sort(arr)
print("Sorted array:")
print(arr)
Problems Output Debug Console Terminal Ports
varshithp@MacBook-Air-2 dsa lab % cd /Users/varshithp/Documents/dsa\ lab ; /usr/bin/env /usr/local/bin/python3 /Users/varshithp/.vscode/extensions/ms-python.debugpy-2026.6.0-darwin-arm64/bundled/libs/debugpy/adapter/../../debugpy/launcher 55258 -- /Users/varshithp/Documents/dsa\ lab/insertionsort.py
Enter number of elements: 5
Enter the elements:
45 89 90 11 2
Sorted array:
[2, 11, 45, 89, 90]
varshithp@MacBook-Air-2 dsa lab %
Code
Code
Code
Python De...
Breakpoints
Raised Exceptio...
Uncaught Exce...
User Uncaught ...
Ln 21, Col 11 Spaces: 4 UTF-8 LF Python Python 3.14.7 Go Live

3(C). Write a python program to implement selection sort.

Code:-



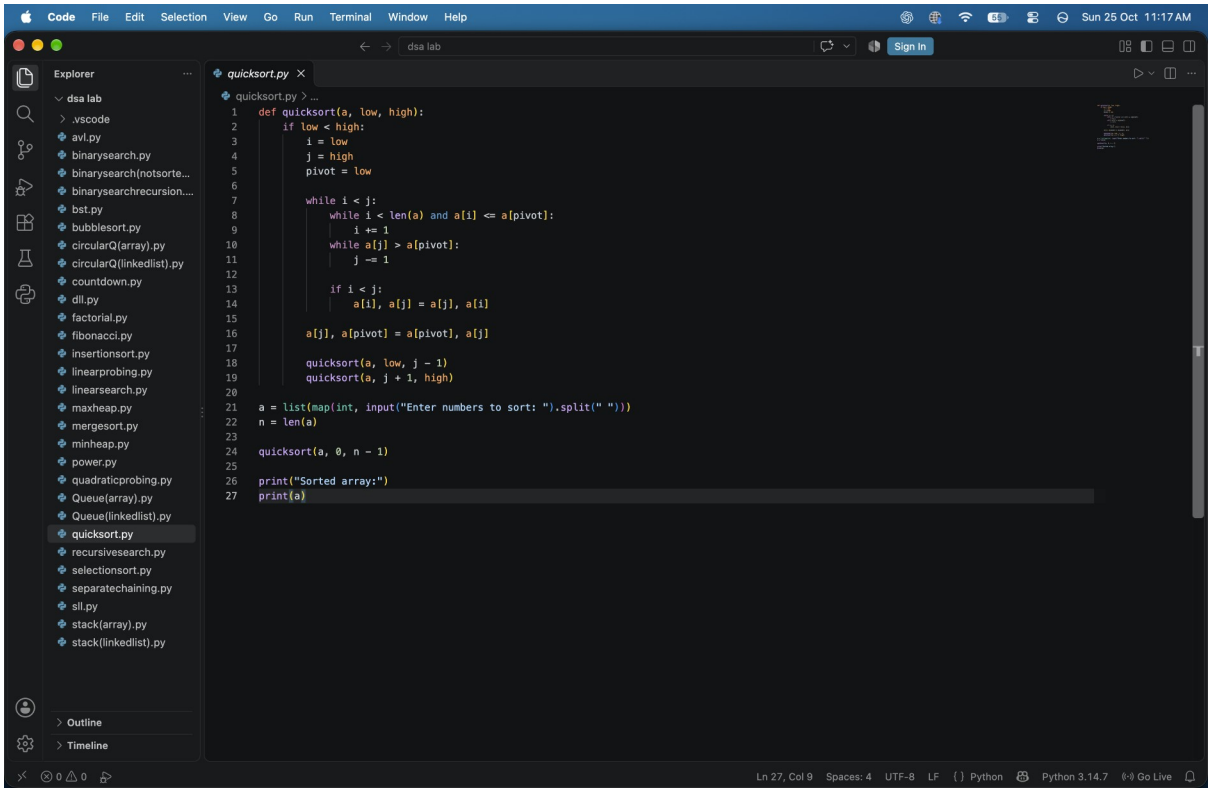
Output:-


Code File Edit Selection View Go Run Terminal Window Help
Sun 25 Oct 11:14 AM
dsa lab
Sign In
Run And Debug
Run
Run and Debug
To customize Run and Debug create a launch.json file.
selectionsort.py
def selection_sort(arr):
n = len(arr)
for i in range(n):
min_index = i
for j in range(i + 1, n):
if arr[j] < arr[min_index]:
min_index = j
arr[i], arr[min_index] = arr[min_index], arr[i]
n = int(input("Enter number of elements: "))
print("Enter the elements:")
arr = list(map(int, input().split()))
selection_sort(arr)
print("Sorted array:")
print(arr)
Problems Output Debug Console Terminal Ports
varshithp@MacBook-Air-2 dsa lab % cd /Users/varshithp/Documents/dsa\ lab ; /usr/bin/env /usr/local/bin/python3 /Users/varshithp/.vscode/extensions/ms-python.debugpy-2026.6.0-darwin-arm64/bundled/libs/debugpy/adapter/../../debugpy/launcher 55336 -- /Users/varshithp/Documents/dsa\ lab/selectionsort.py
Enter number of elements: 6
Enter the elements:
99 65 83 21 1 5
Sorted array:
[1, 5, 21, 65, 83, 99]
varshithp@MacBook-Air-2 dsa lab %
Breakpoints
Raised Exceptio...
Uncaught Exce...
User Uncaught ...
Ln 17, Col 11 Spaces: 4 UTF-8 LF Python Python 3.14.7 Go Live

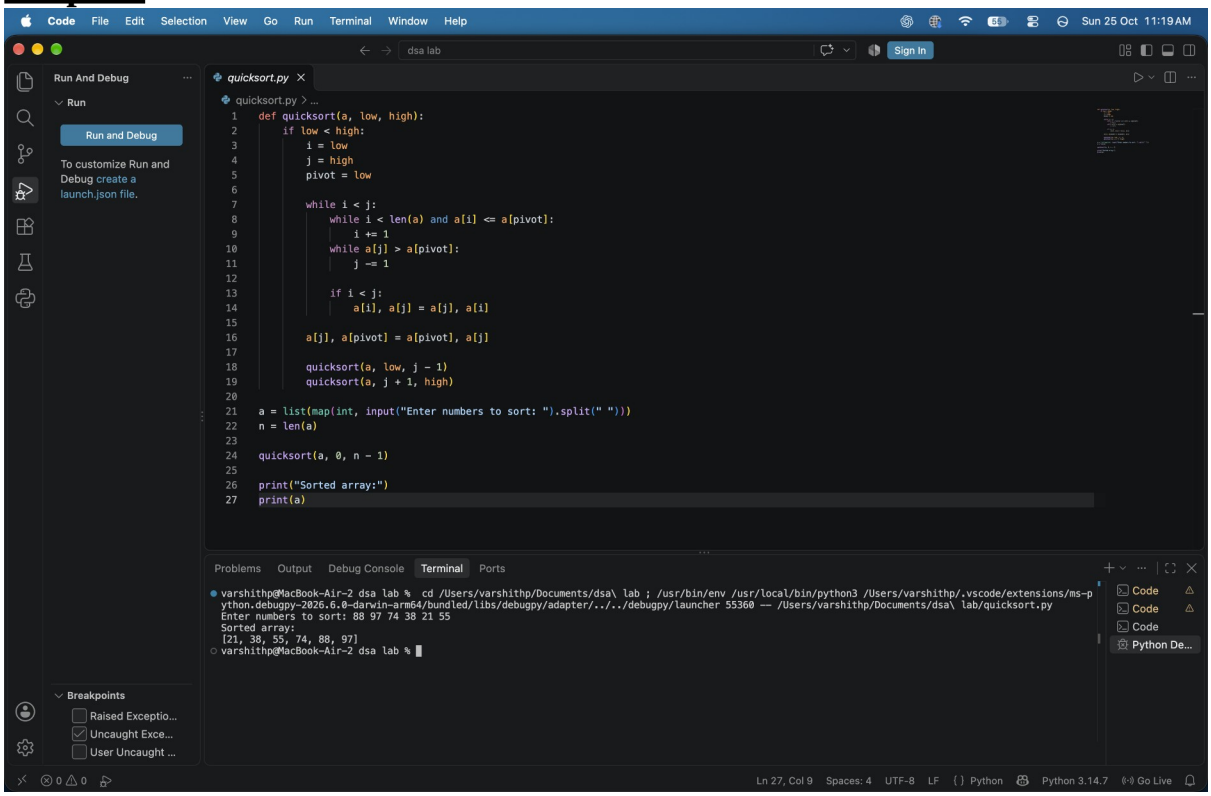
WEEK-4

4(A). Write a python program to implement quick sort.

Code:-

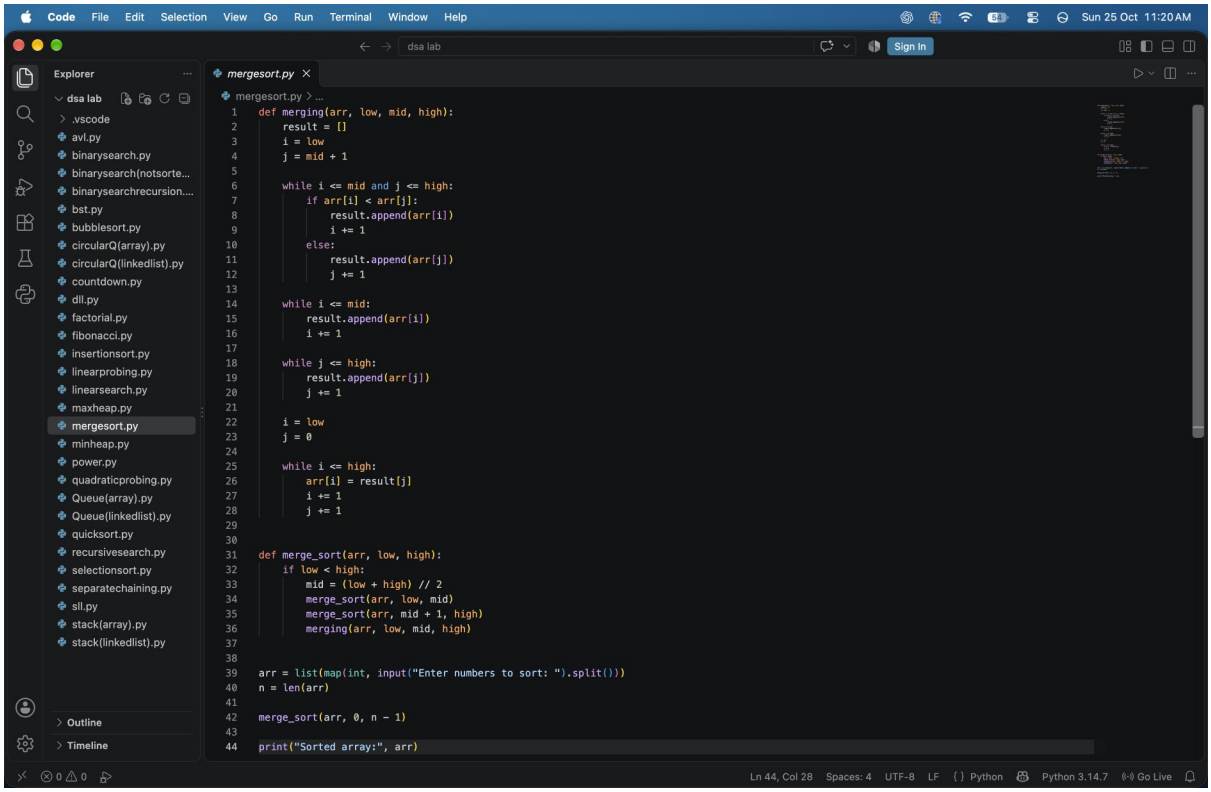


Output:-


Code File Edit Selection View Go Run Terminal Window Help
Sun 25 Oct 11:19 AM
dsa lab
Sign In
Run And Debug
Run
Run and Debug
To customize Run and Debug create a launch.json file.
quicksort.py
quicksort.py > ...
def quicksort(a, low, high):
if low < high:
i = low
j = high
pivot = low
while i < j:
while i < len(a) and a[i] <= a[pivot]:
i += 1
while a[j] > a[pivot]:
j -= 1
if i < j:
a[i], a[j] = a[j], a[i]
a[j], a[pivot] = a[pivot], a[j]
quicksort(a, low, j - 1)
quicksort(a, j + 1, high)
a = list(map(int, input("Enter numbers to sort: ").split(" ")))
n = len(a)
quicksort(a, 0, n - 1)
print("Sorted array:")
print(a)
Problems Output Debug Console Terminal Ports
varshithp@MacBook-Air-2 dsa lab % cd /Users/varshithp/Documents/dsa\ lab ; /usr/bin/env /usr/local/bin/python3 /Users/varshithp/.vscode/extensions/ms-python.debugpy-2026.6.0-darwin-arm64/bundled/libs/debugpy/adapter/../../debugpy/launcher 55360 -- /Users/varshithp/Documents/dsa\ lab/quicksort.py
Enter numbers to sort: 88 97 74 38 21 55
Sorted array:
[21, 38, 55, 74, 88, 97]
varshithp@MacBook-Air-2 dsa lab %
Code
Code
Code
Python De...
Breakpoints
Raised Exceptio...
Uncaught Exce...
User Uncaught ...
Ln 27, Col 9 Spaces: 4 UTF-8 LF Python Python 3.14.7 Go Live

4(B). Write a python program to implement merge sort.

Code:-



```
def merging(arr, low, mid, high):
    result = []
    i = low
    j = mid + 1

    while i <= mid and j <= high:
        if arr[i] < arr[j]:
            result.append(arr[i])
            i += 1
        else:
            result.append(arr[j])
            j += 1

    while i <= mid:
        result.append(arr[i])
        i += 1

    while j <= high:
        result.append(arr[j])
        j += 1

    i = low
    j = 0

    while i <= high:
        arr[i] = result[j]
        i += 1
        j += 1

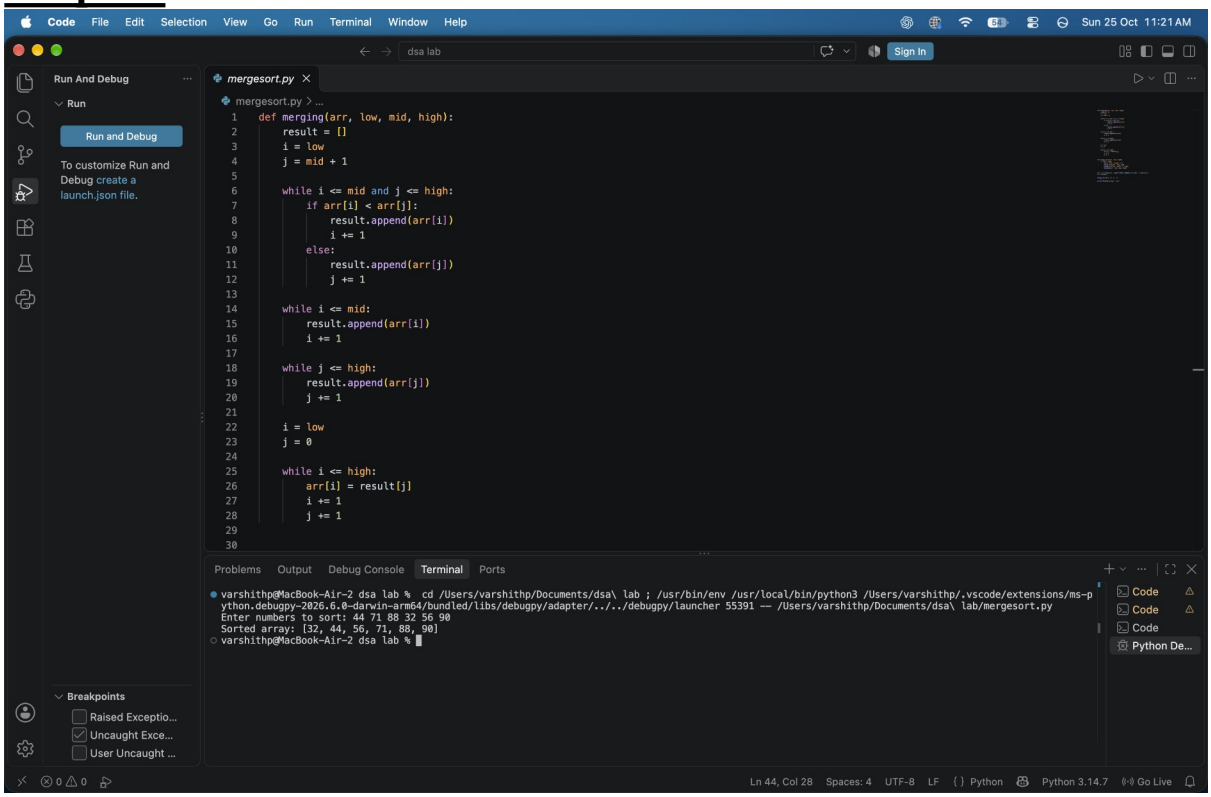

def merge_sort(arr, low, high):
    if low < high:
        mid = (low + high) // 2
        merge_sort(arr, low, mid)
        merge_sort(arr, mid + 1, high)
        merging(arr, low, mid, high)


arr = list(map(int, input("Enter numbers to sort: ").split()))
n = len(arr)

merge_sort(arr, 0, n - 1)

print("Sorted array:", arr)
```

Output:-

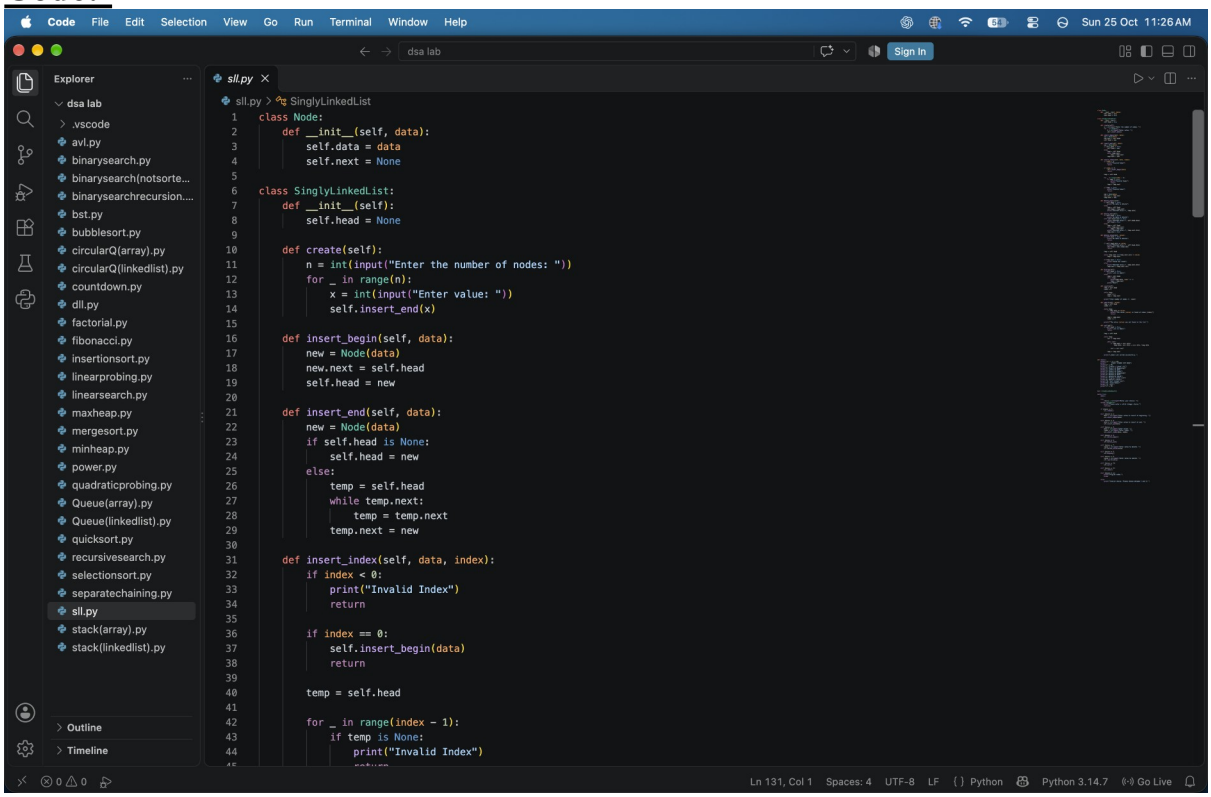

Code File Edit Selection View Go Run Terminal Window Help
Sun 25 Oct 11:21 AM
dsa lab
Sign In
Run And Debug
Run
Run and Debug
To customize Run and Debug create a launch.json file.
mergesort.py
mergesort.py > ...
def merging(arr, low, mid, high):
result = []
i = low
j = mid + 1
while i <= mid and j <= high:
if arr[i] < arr[j]:
result.append(arr[i])
i += 1
else:
result.append(arr[j])
j += 1
while i <= mid:
result.append(arr[i])
i += 1
while j <= high:
result.append(arr[j])
j += 1
i = low
j = 0
while i <= high:
arr[i] = result[j]
i += 1
j += 1
Problems Output Debug Console Terminal Ports
varshithp@MacBook-Air-2 dsa lab % cd /Users/varshithp/Documents/dsa\ lab ; /usr/bin/env /usr/local/bin/python3 /Users/varshithp/.vscode/extensions/ms-python.debugpy-2026.6.0-darwin-arm64/bundled/libs/debugpy/adapter/../../debugpy/launcher 55391 -- /Users/varshithp/Documents/dsa\ lab/mergesort.py
Enter numbers to sort: 44 71 88 32 56 90
Sorted array: [32, 44, 56, 71, 88, 90]
varshithp@MacBook-Air-2 dsa lab %
Code
Code
Code
Python De...
Breakpoints
Raised Exceptio...
Uncaught Exce...
User Uncaught ...
Ln 44, Col 28 Spaces: 4 UTF-8 LF Python Python 3.14.7 Go Live

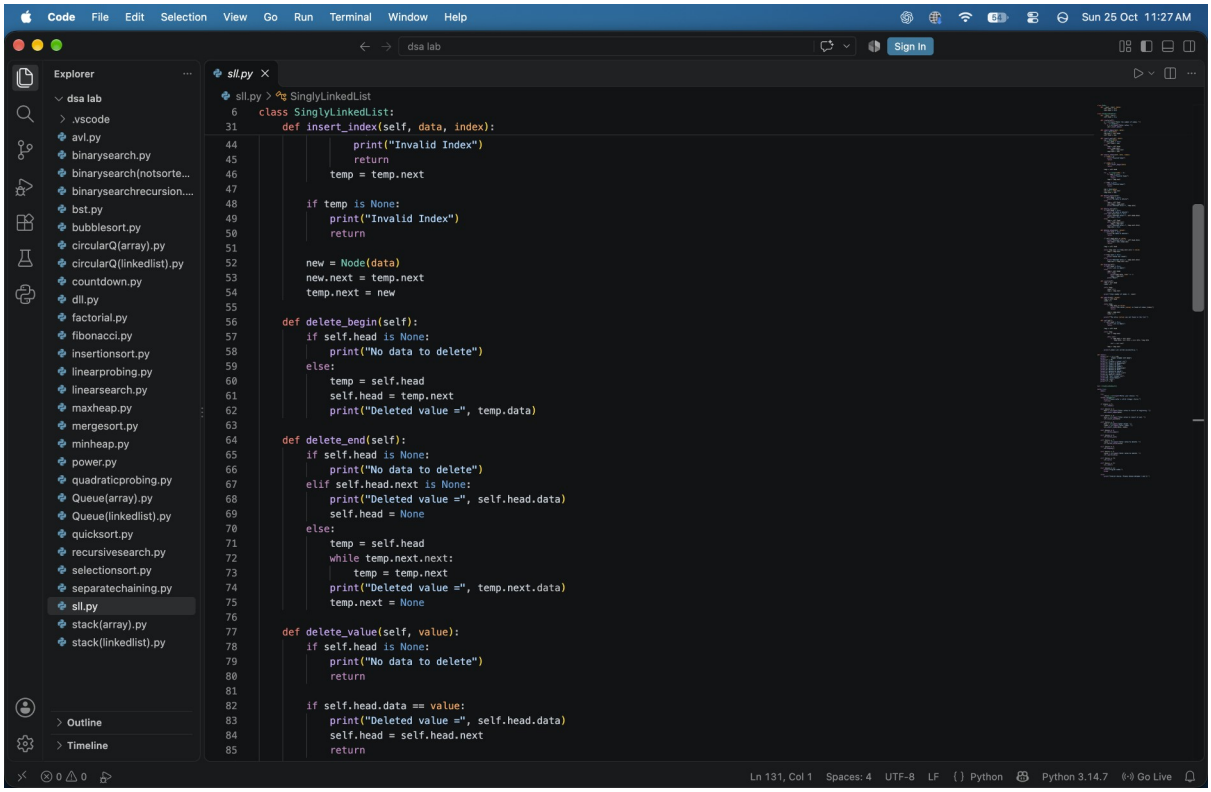
WEEK-5

5(A). Write a python program to implement a Singly Linked List with the below operations.

1)Create Linked List
2)Insert Beginning
3)Insert End
4)Insert at Index
5)Delete by Value
6)Delete First Node
7)Delete Last Node
8)Count number of nodes
9)Sort the data of SLL
10)Search data in SLL
11)Display or Traverse
12)Exit

Code:-





```
class SinglyLinkedList:
    def insert_index(self, data, index):
                print("Invalid Index")
                return
            temp = temp.next

        if temp is None:
            print("Invalid Index")
            return

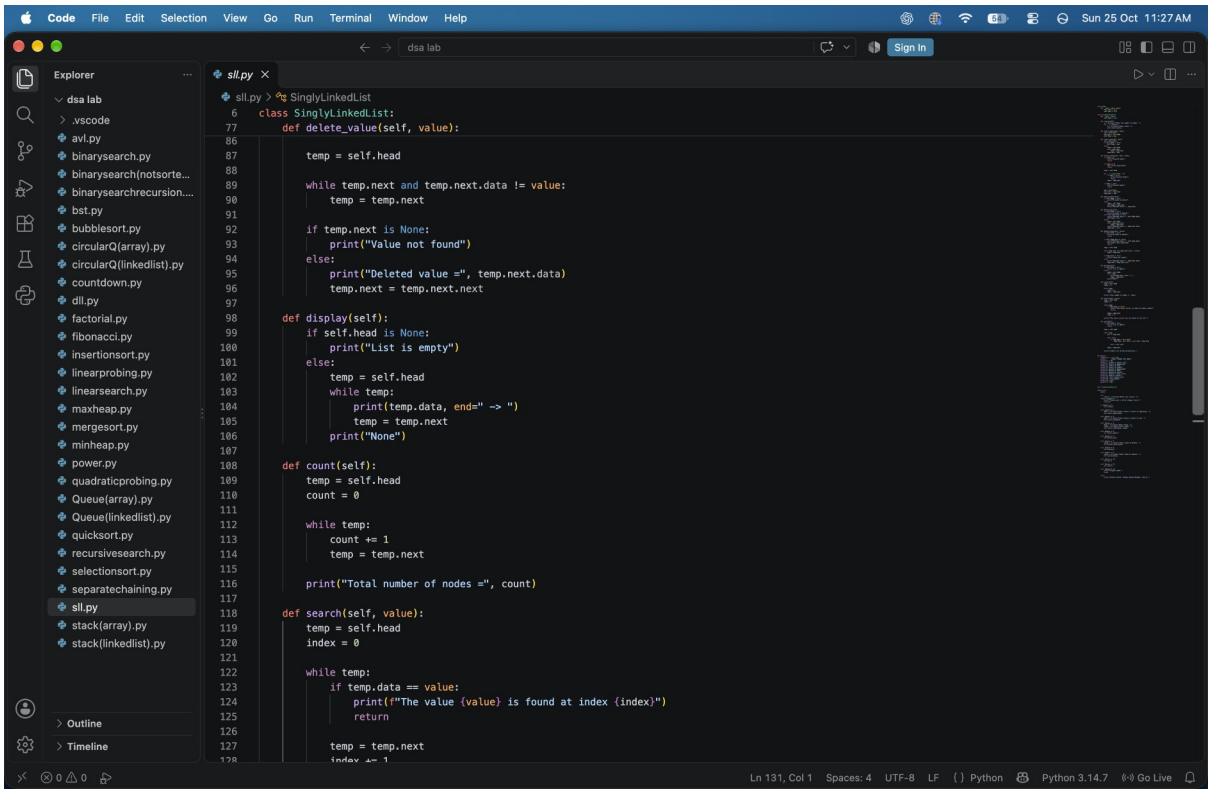
        new = Node(data)
        new.next = temp.next
        temp.next = new

    def delete_begin(self):
        if self.head is None:
            print("No data to delete")
        else:
            temp = self.head
            self.head = temp.next
            print("Deleted value =", temp.data)

    def delete_end(self):
        if self.head is None:
            print("No data to delete")
        elif self.head.next is None:
            print("Deleted value =", self.head.data)
            self.head = None
        else:
            temp = self.head
            while temp.next.next:
                temp = temp.next
            print("Deleted value =", temp.next.data)
            temp.next = None

    def delete_value(self, value):
        if self.head is None:
            print("No data to delete")
            return

        if self.head.data == value:
            print("Deleted value =", self.head.data)
            self.head = self.head.next
            return
```



```
class SinglyLinkedList:
    def delete_value(self, value):

        temp = self.head

        while temp.next and temp.next.data != value:
            temp = temp.next

        if temp.next is None:
            print("Value not found")
        else:
            print("Deleted value =", temp.next.data)
            temp.next = temp.next.next

    def display(self):
        if self.head is None:
            print("List is empty")
        else:
            temp = self.head
            while temp:
                print(temp.data, end=" -> ")
                temp = temp.next
            print("None")

    def count(self):
        temp = self.head
        count = 0

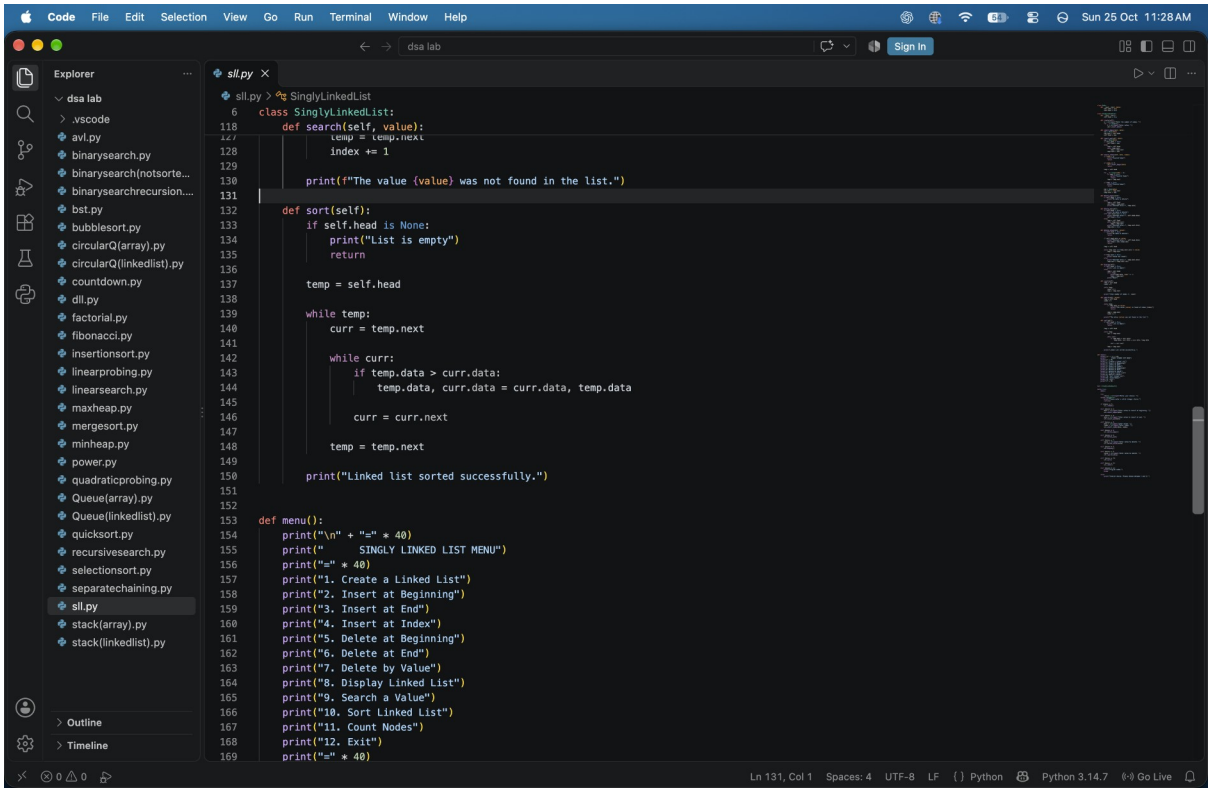
        while temp:
            count += 1
            temp = temp.next

        print("Total number of nodes =", count)

    def search(self, value):
        temp = self.head
        index = 0

        while temp:
            if temp.data == value:
                print(f"The value {value} is found at index {index}")
                return

            temp = temp.next
```



```
class SinglyLinkedList:
    def search(self, value):
            temp = temp.next
            index += 1

        print(f"The value {value} was not found in the list.")

    def sort(self):
        if self.head is None:
            print("List is empty")
            return

        temp = self.head

        while temp:
            curr = temp.next

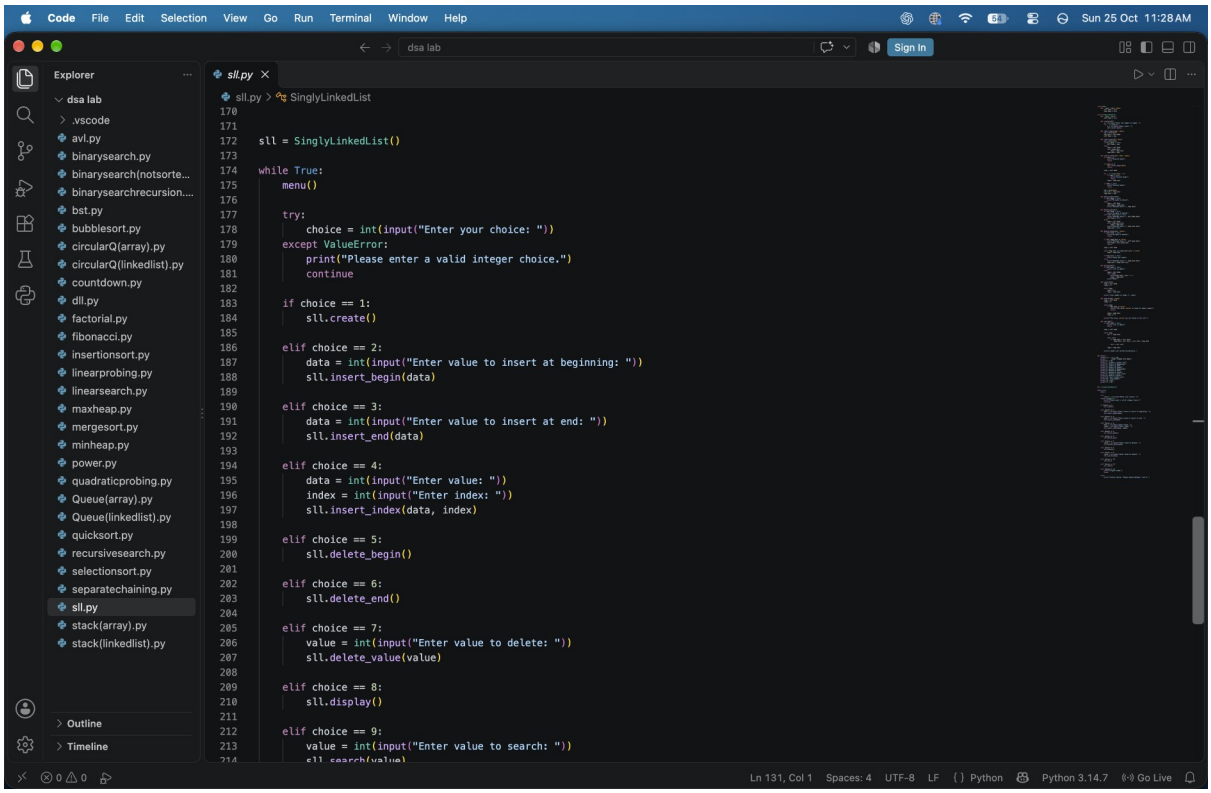
            while curr:
                if temp.data > curr.data:
                    temp.data, curr.data = curr.data, temp.data


                curr = curr.next

            temp = temp.next

        print("Linked list sorted successfully.")


def menu():
    print("\n" + "=" * 40)
    print("      SINGLY LINKED LIST MENU")
    print("=" * 40)
    print("1. Create a Linked List")
    print("2. Insert at Beginning")
    print("3. Insert at End")
    print("4. Insert at Index")
    print("5. Delete at Beginning")
    print("6. Delete at End")
    print("7. Delete by Value")
    print("8. Display Linked List")
    print("9. Search a Value")
    print("10. Sort Linked List")
    print("11. Count Nodes")
    print("12. Exit")
    print("=" * 40)
```



```
sll = SinglyLinkedList()

while True:
    menu()

    try:
        choice = int(input("Enter your choice: "))
    except ValueError:
        print("Please enter a valid integer choice.")
        continue

    if choice == 1:
        sll.create()

    elif choice == 2:
        data = int(input("Enter value to insert at beginning: "))
        sll.insert_begin(data)

    elif choice == 3:
        data = int(input("Enter value to insert at end: "))
        sll.insert_end(data)

    elif choice == 4:
        data = int(input("Enter value: "))
        index = int(input("Enter index: "))
        sll.insert_index(data, index)

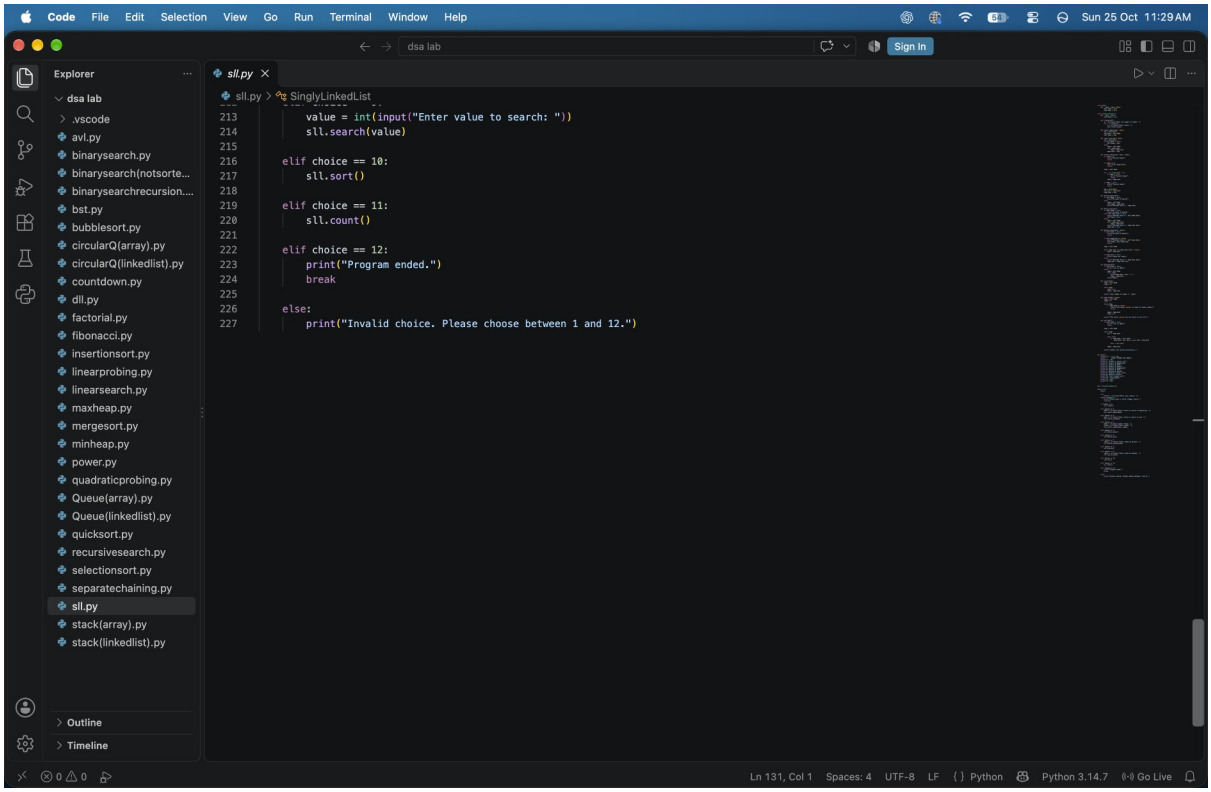
    elif choice == 5:
        sll.delete_begin()

    elif choice == 6:
        sll.delete_end()

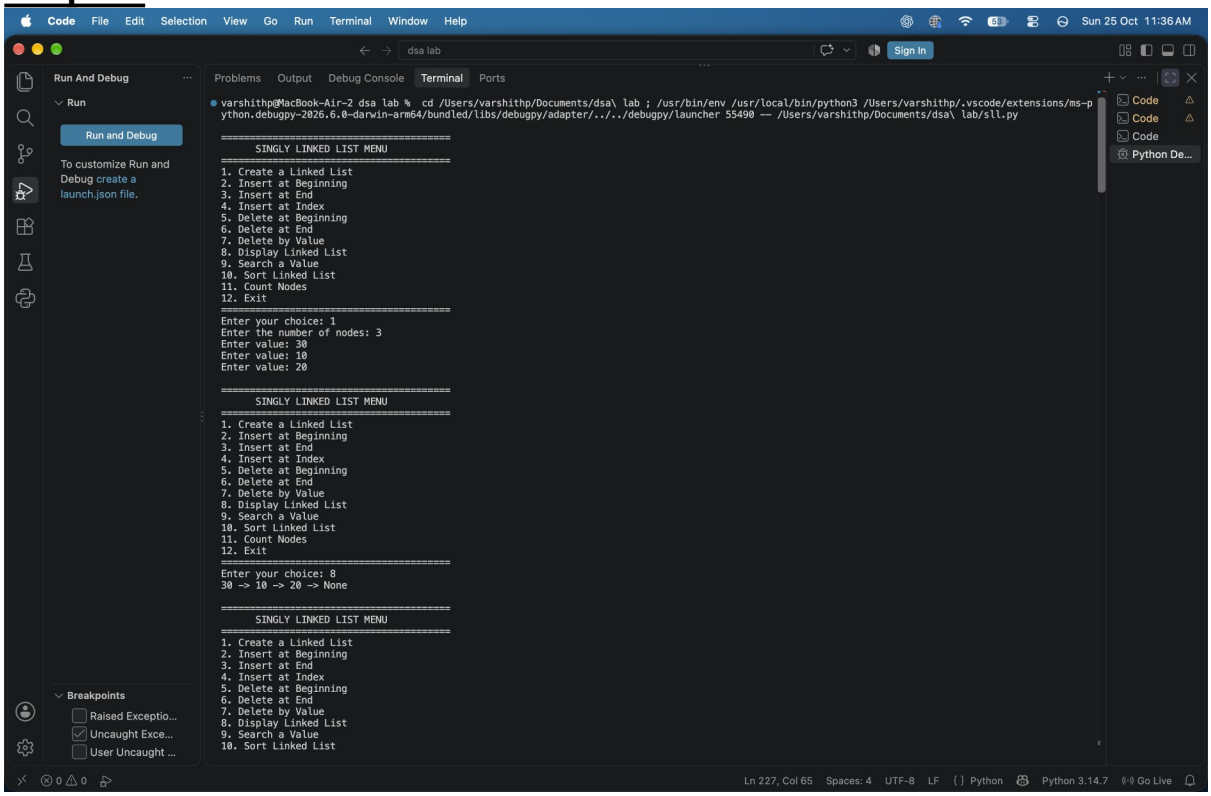
    elif choice == 7:
        value = int(input("Enter value to delete: "))
        sll.delete_value(value)

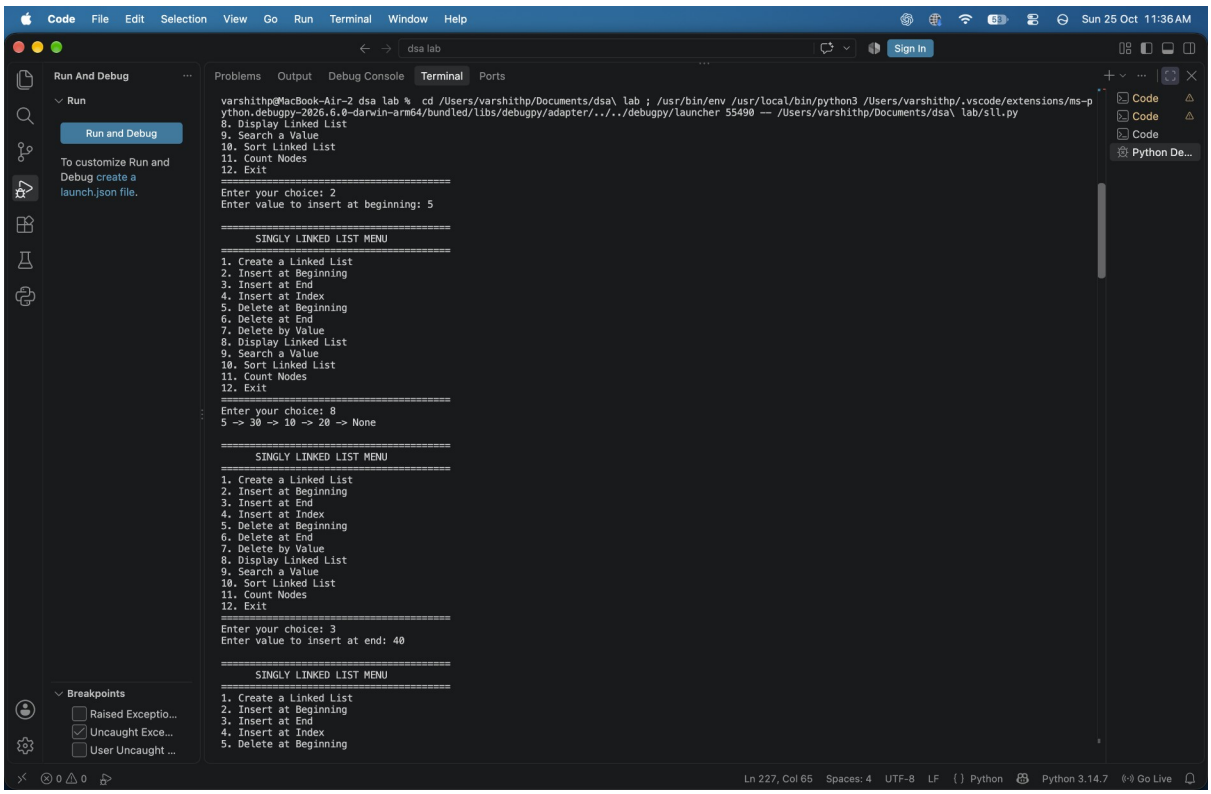
    elif choice == 8:
        sll.display()

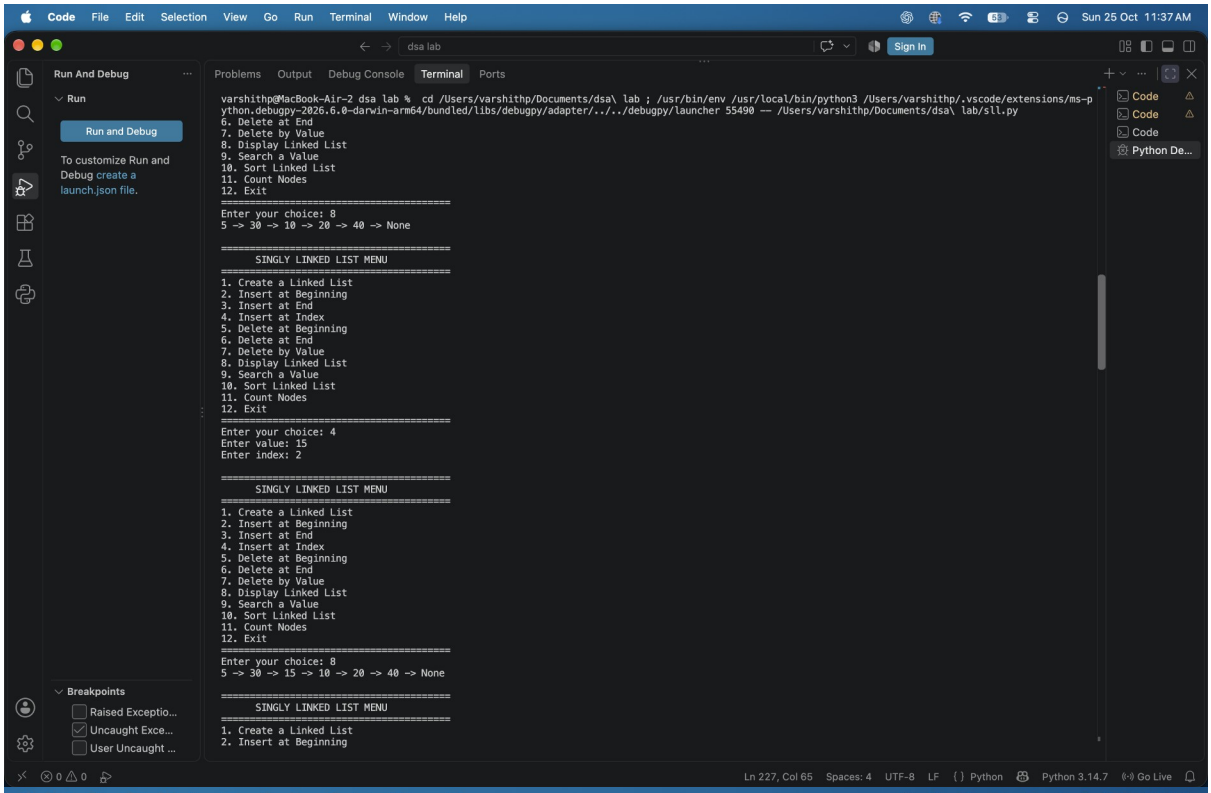
    elif choice == 9:
        value = int(input("Enter value to search: "))
```

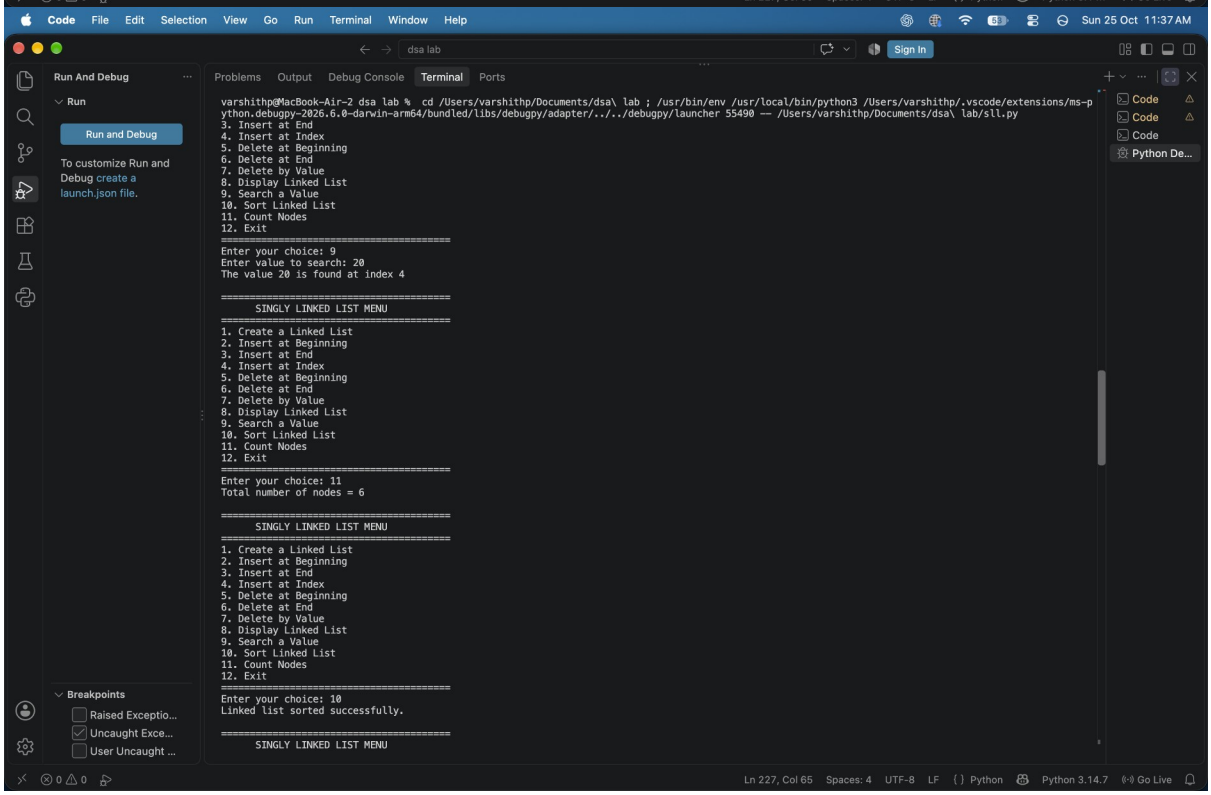

Code File Edit Selection View Go Run Terminal Window Help
Sun 25 Oct 11:29 AM
dsa lab
Sign in
Explorer
dsa lab
.vscode
avl.py
binarysearch.py
binarysearch(notsorte...
binarysearchrecursion....
bst.py
bubblesort.py
circularQ(array).py
circularQ(linkedlist).py
countdown.py
dll.py
factorial.py
fibonacci.py
insertionsort.py
linearprobing.py
linearsearch.py
maxheap.py
mergesort.py
minheap.py
power.py
quadraticprobing.py
Queue(array).py
Queue(linkedlist).py
quicksort.py
recursivesearch.py
selectionsort.py
separatechaining.py
sll.py
stack(array).py
stack(linkedlist).py
Outline
Timeline
sll.py
sll.py > SinglyLinkedList
213 value = int(input("Enter value to search: "))
214 sll.search(value)
215
216 elif choice == 10:
217 sll.sort()
218
219 elif choice == 11:
220 sll.count()
221
222 elif choice == 12:
223 print("Program ended.")
224 break
225
226 else:
227 print("Invalid choice. Please choose between 1 and 12.")
Ln 131, Col 1 Spaces: 4 UTF-8 LF Python Python 3.14.7 Go Live

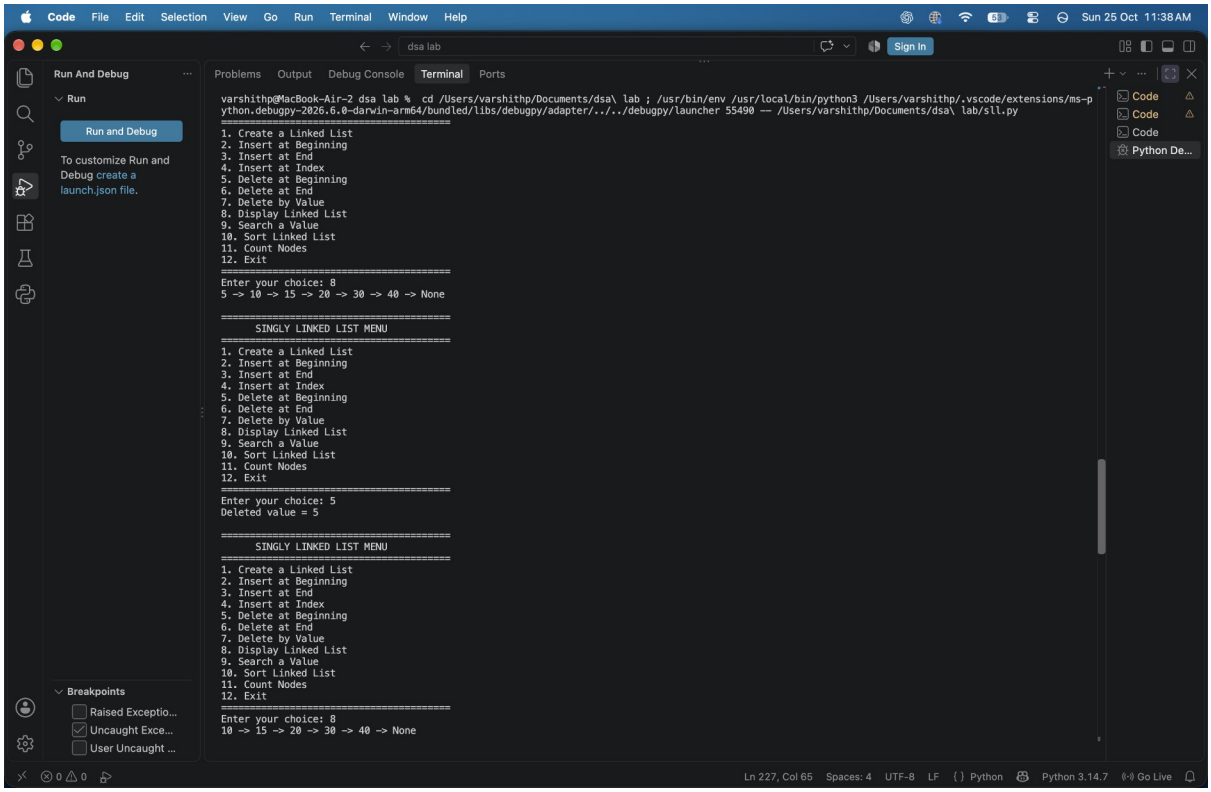
Output:-

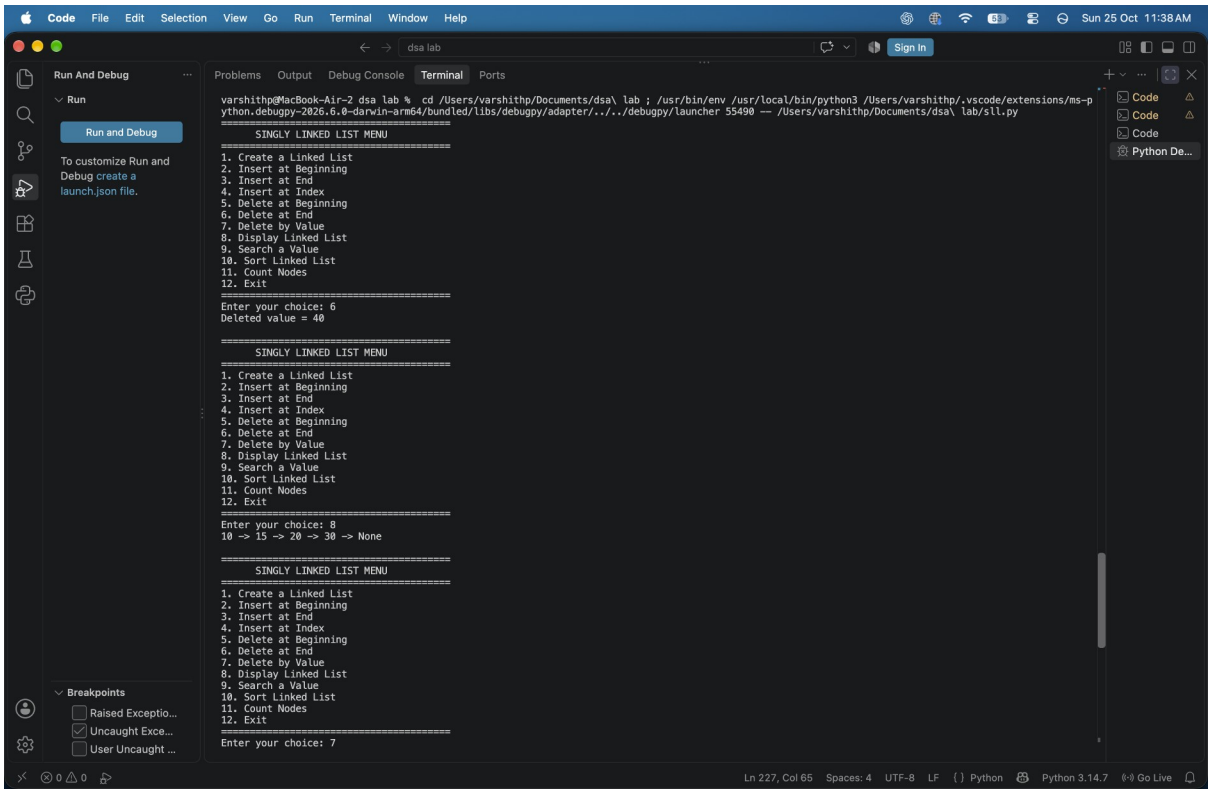

Code File Edit Selection View Go Run Terminal Window Help
Sun 25 Oct 11:36 AM
dsa lab
Sign In
Run And Debug
Run
Run and Debug
To customize Run and Debug create a launch.json file.
Problems Output Debug Console Terminal Ports
varshithp@MacBook-Air-2 dsa lab % cd /Users/varshithp/Documents/dsa\ lab ; /usr/bin/env /usr/local/bin/python3 /Users/varshithp/.vscode/extensions/ms-python.debugpy-2026.6.0-darwin-arm64/bundled/libs/debugpy/adapter/../../debugpy/launcher 55490 -- /Users/varshithp/Documents/dsa\ lab/sll.py
SINGLY LINKED LIST MENU
1. Create a Linked List
2. Insert at Beginning
3. Insert at End
4. Insert at Index
5. Delete at Beginning
6. Delete at End
7. Delete by Value
8. Display Linked List
9. Search a Value
10. Sort Linked List
11. Count Nodes
12. Exit
Enter your choice: 1
Enter the number of nodes: 3
Enter value: 30
Enter value: 10
Enter value: 20
SINGLY LINKED LIST MENU
1. Create a Linked List
2. Insert at Beginning
3. Insert at End
4. Insert at Index
5. Delete at Beginning
6. Delete at End
7. Delete by Value
8. Display Linked List
9. Search a Value
10. Sort Linked List
11. Count Nodes
12. Exit
Enter your choice: 8
30 -> 10 -> 20 -> None
SINGLY LINKED LIST MENU
1. Create a Linked List
2. Insert at Beginning
3. Insert at End
4. Insert at Index
5. Delete at Beginning
6. Delete at End
7. Delete by Value
8. Display Linked List
9. Search a Value
10. Sort Linked List
Breakpoints
Raised Exceptio...
Uncaught Exce...
User Uncaught ...
Code
Code
Code
Python De...
Ln 227, Col 65 Spaces: 4 UTF-8 LF Python Python 3.14.7 Go Live

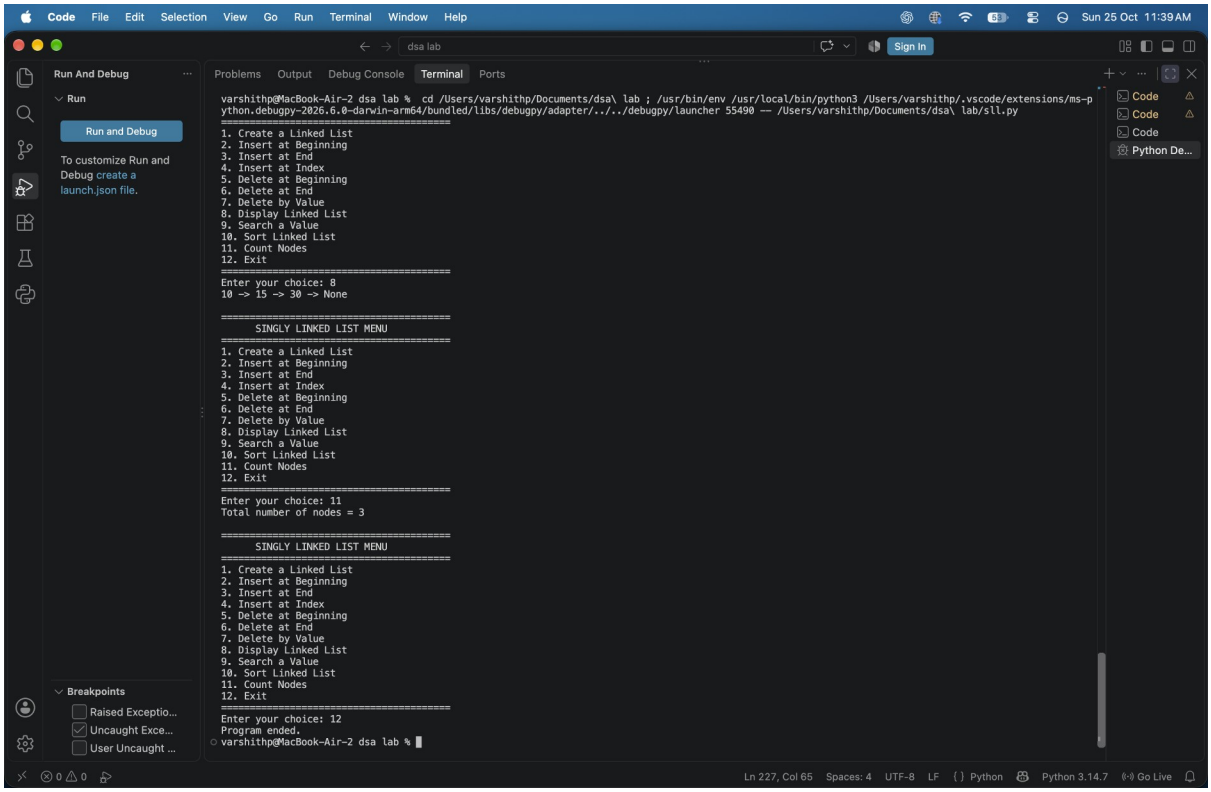

Code File Edit Selection View Go Run Terminal Window Help
Sun 25 Oct 11:36 AM
dsa lab
Sign In
Run And Debug
Run
Run and Debug
To customize Run and Debug create a launch.json file.
Problems Output Debug Console Terminal Ports
varshithp@MacBook-Air-2 dsa lab % cd /Users/varshithp/Documents/dsa\ lab ; /usr/bin/env /usr/local/bin/python3 /Users/varshithp/.vscode/extensions/ms-python.debugpy-2026.6.0-darwin-arm64/bundled/libs/debugpy/adapter/../../debugpy/launcher 55490 -- /Users/varshithp/Documents/dsa\ lab/sll.py
8. Display Linked List
9. Search a Value
10. Sort Linked List
11. Count Nodes
12. Exit
Enter your choice: 2
Enter value to insert at beginning: 5
SINGLY LINKED LIST MENU
1. Create a Linked List
2. Insert at Beginning
3. Insert at End
4. Insert at Index
5. Delete at Beginning
6. Delete at End
7. Delete by Value
8. Display Linked List
9. Search a Value
10. Sort Linked List
11. Count Nodes
12. Exit
Enter your choice: 8
5 -> 30 -> 10 -> 20 -> None
SINGLY LINKED LIST MENU
1. Create a Linked List
2. Insert at Beginning
3. Insert at End
4. Insert at Index
5. Delete at Beginning
6. Delete at End
7. Delete by Value
8. Display Linked List
9. Search a Value
10. Sort Linked List
11. Count Nodes
12. Exit
Enter your choice: 3
Enter value to insert at end: 40
SINGLY LINKED LIST MENU
1. Create a Linked List
2. Insert at Beginning
3. Insert at End
4. Insert at Index
5. Delete at Beginning
Breakpoints
Raised Exceptio...
Uncaught Exce...
User Uncaught ...
Code
Code
Code
Python De...
Ln 227, Col 65 Spaces: 4 UTF-8 LF Python Python 3.14.7 Go Live


Code File Edit Selection View Go Run Terminal Window Help
Sun 25 Oct 11:37 AM
dsa lab
Sign in
Run And Debug
Run
Run and Debug
To customize Run and Debug create a launch.json file.
Problems Output Debug Console Terminal Ports
varshithp@MacBook-Air-2 dsa lab % cd /Users/varshithp/Documents/dsa\ lab ; /usr/bin/env /usr/local/bin/python3 /Users/varshithp/.vscode/extensions/ms-python.debugpy-2026.6.0-darwin-arm64/bundled/libs/debugpy/adapter/../../debugpy/launcher 55490 -- /Users/varshithp/Documents/dsa\ lab/sll.py
6. Delete at End
7. Delete by Value
8. Display Linked List
9. Search a Value
10. Sort Linked List
11. Count Nodes
12. Exit
Enter your choice: 8
5 -> 30 -> 10 -> 20 -> 40 -> None
SINGLY LINKED LIST MENU
Enter your choice: 4
Enter value: 15
Enter index: 2
SINGLY LINKED LIST MENU
Enter your choice: 8
5 -> 30 -> 15 -> 10 -> 20 -> 40 -> None
SINGLY LINKED LIST MENU
Breakpoints
Raised Exceptio...
Uncaught Exce...
User Uncaught ...
Code
Code
Code
Python De...
Ln 227, Col 65 Spaces: 4 UTF-8 LF Python Python 3.14.7 Go Live


Code File Edit Selection View Go Run Terminal Window Help
Sun 25 Oct 11:37 AM
dsa lab
Sign in
Run And Debug
Run
Run and Debug
To customize Run and Debug create a launch.json file.
Problems Output Debug Console Terminal Ports
varshithp@MacBook-Air-2 dsa lab % cd /Users/varshithp/Documents/dsa\ lab ; /usr/bin/env /usr/local/bin/python3 /Users/varshithp/.vscode/extensions/ms-python.debugpy-2026.6.0-darwin-arm64/bundled/libs/debugpy/adapter/../../debugpy/launcher 55490 -- /Users/varshithp/Documents/dsa\ lab/sll.py
3. Insert at End
4. Insert at Index
5. Delete at Beginning
6. Delete at End
7. Delete by Value
8. Display Linked List
9. Search a Value
10. Sort Linked List
11. Count Nodes
12. Exit
Enter your choice: 9
Enter value to search: 20
The value 20 is found at index 4
SINGLY LINKED LIST MENU
Enter your choice: 11
Total number of nodes = 6
SINGLY LINKED LIST MENU
Enter your choice: 10
Linked list sorted successfully.
SINGLY LINKED LIST MENU
Breakpoints
Raised Exceptio...
Uncaught Exce...
User Uncaught ...
Code
Code
Code
Python De...
Ln 227, Col 65 Spaces: 4 UTF-8 LF Python Python 3.14.7 Go Live


Code File Edit Selection View Go Run Terminal Window Help
Sun 25 Oct 11:38 AM
dsa lab
Sign In
Run And Debug
Run
Run and Debug
To customize Run and Debug create a launch.json file.
Problems Output Debug Console Terminal Ports
varshithp@MacBook-Air-2 dsa lab % cd /Users/varshithp/Documents/dsa\ lab ; /usr/bin/env /usr/local/bin/python3 /Users/varshithp/.vscode/extensions/ms-python.debugpy-2026.6.0-darwin-arm64/bundled/libs/debugpy/adapter/../../debugpy/launcher 55490 -- /Users/varshithp/Documents/dsa\ lab/sll.py
==========
1. Create a Linked List
2. Insert at Beginning
3. Insert at End
4. Insert at Index
5. Delete at Beginning
6. Delete at End
7. Delete by Value
8. Display Linked List
9. Search a Value
10. Sort Linked List
11. Count Nodes
12. Exit
==========
Enter your choice: 8
5 -> 10 -> 15 -> 20 -> 30 -> 40 -> None
==========
SINGLY LINKED LIST MENU
==========
1. Create a Linked List
2. Insert at Beginning
3. Insert at End
4. Insert at Index
5. Delete at Beginning
6. Delete at End
7. Delete by Value
8. Display Linked List
9. Search a Value
10. Sort Linked List
11. Count Nodes
12. Exit
==========
Enter your choice: 5
Deleted value = 5
==========
SINGLY LINKED LIST MENU
==========
1. Create a Linked List
2. Insert at Beginning
3. Insert at End
4. Insert at Index
5. Delete at Beginning
6. Delete at End
7. Delete by Value
8. Display Linked List
9. Search a Value
10. Sort Linked List
11. Count Nodes
12. Exit
==========
Enter your choice: 8
10 -> 15 -> 20 -> 30 -> 40 -> None
Code
Code
Code
Python De...
Breakpoints
Raised Exceptio...
Uncaught Exce...
User Uncaught ...
Ln 227, Col 65 Spaces: 4 UTF-8 LF { } Python Python 3.14.7 Go Live


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varshithp@MacBook-Air-2 dsa lab % cd /Users/varshithp/Documents/dsa\ lab ; /usr/bin/env /usr/local/bin/python3 /Users/varshithp/.vscode/extensions/ms-python.debugpy-2026.6.0-darwin-arm64/bundled/libs/debugpy/adapter/../../debugpy/launcher 55490 -- /Users/varshithp/Documents/dsa\ lab/sll.py
==========
SINGLY LINKED LIST MENU
==========
1. Create a Linked List
2. Insert at Beginning
3. Insert at End
4. Insert at Index
5. Delete at Beginning
6. Delete at End
7. Delete by Value
8. Display Linked List
9. Search a Value
10. Sort Linked List
11. Count Nodes
12. Exit
==========
Enter your choice: 6
Deleted value = 40
==========
SINGLY LINKED LIST MENU
==========
1. Create a Linked List
2. Insert at Beginning
3. Insert at End
4. Insert at Index
5. Delete at Beginning
6. Delete at End
7. Delete by Value
8. Display Linked List
9. Search a Value
10. Sort Linked List
11. Count Nodes
12. Exit
==========
Enter your choice: 8
10 -> 15 -> 20 -> 30 -> None
==========
SINGLY LINKED LIST MENU
==========
1. Create a Linked List
2. Insert at Beginning
3. Insert at End
4. Insert at Index
5. Delete at Beginning
6. Delete at End
7. Delete by Value
8. Display Linked List
9. Search a Value
10. Sort Linked List
11. Count Nodes
12. Exit
==========
Enter your choice: 7
Code
Code
Code
Python De...
Breakpoints
Raised Exceptio...
Uncaught Exce...
User Uncaught ...
Ln 227, Col 65 Spaces: 4 UTF-8 LF { } Python Python 3.14.7 Go Live


Code File Edit Selection View Go Run Terminal Window Help
Sun 25 Oct 11:39 AM
dsa lab
Sign In
Run And Debug
Run
Run and Debug
To customize Run and Debug create a launch.json file.
Breakpoints
Raised Exceptio...
Uncaught Exce...
User Uncaught ...
Problems Output Debug Console Terminal Ports
varshithp@MacBook-Air-2 dsa lab % cd /Users/varshithp/Documents/dsa\ lab ; /usr/bin/env /usr/local/bin/python3 /Users/varshithp/.vscode/extensions/ms-python.debugpy-2026.6.0-darwin-arm64/bundled/libs/debugpy/adapter/../../debugpy/launcher 55490 -- /Users/varshithp/Documents/dsa\ lab/sll.py
==========
1. Create a Linked List
2. Insert at Beginning
3. Insert at End
4. Insert at Index
5. Delete at Beginning
6. Delete at End
7. Delete by Value
8. Display Linked List
9. Search a Value
10. Sort Linked List
11. Count Nodes
12. Exit
==========
Enter your choice: 8
10 -> 15 -> 30 -> None
==========
SINGLY LINKED LIST MENU
==========
1. Create a Linked List
2. Insert at Beginning
3. Insert at End
4. Insert at Index
5. Delete at Beginning
6. Delete at End
7. Delete by Value
8. Display Linked List
9. Search a Value
10. Sort Linked List
11. Count Nodes
12. Exit
==========
Enter your choice: 11
Total number of nodes = 3
==========
SINGLY LINKED LIST MENU
==========
1. Create a Linked List
2. Insert at Beginning
3. Insert at End
4. Insert at Index
5. Delete at Beginning
6. Delete at End
7. Delete by Value
8. Display Linked List
9. Search a Value
10. Sort Linked List
11. Count Nodes
12. Exit
==========
Enter your choice: 12
Program ended.
varshithp@MacBook-Air-2 dsa lab %
Code
Code
Code
Python De...
Ln 227, Col 65 Spaces: 4 UTF-8 LF Python Python 3.14.7 Go Live

WEEK-6

6. Write a python program to implement a Double Linked List with the below operations.

1)Create Linked List

2)Insert Beginning

3)Insert End

4)Insert at Index

5)Delete by Value

6)Delete First Node

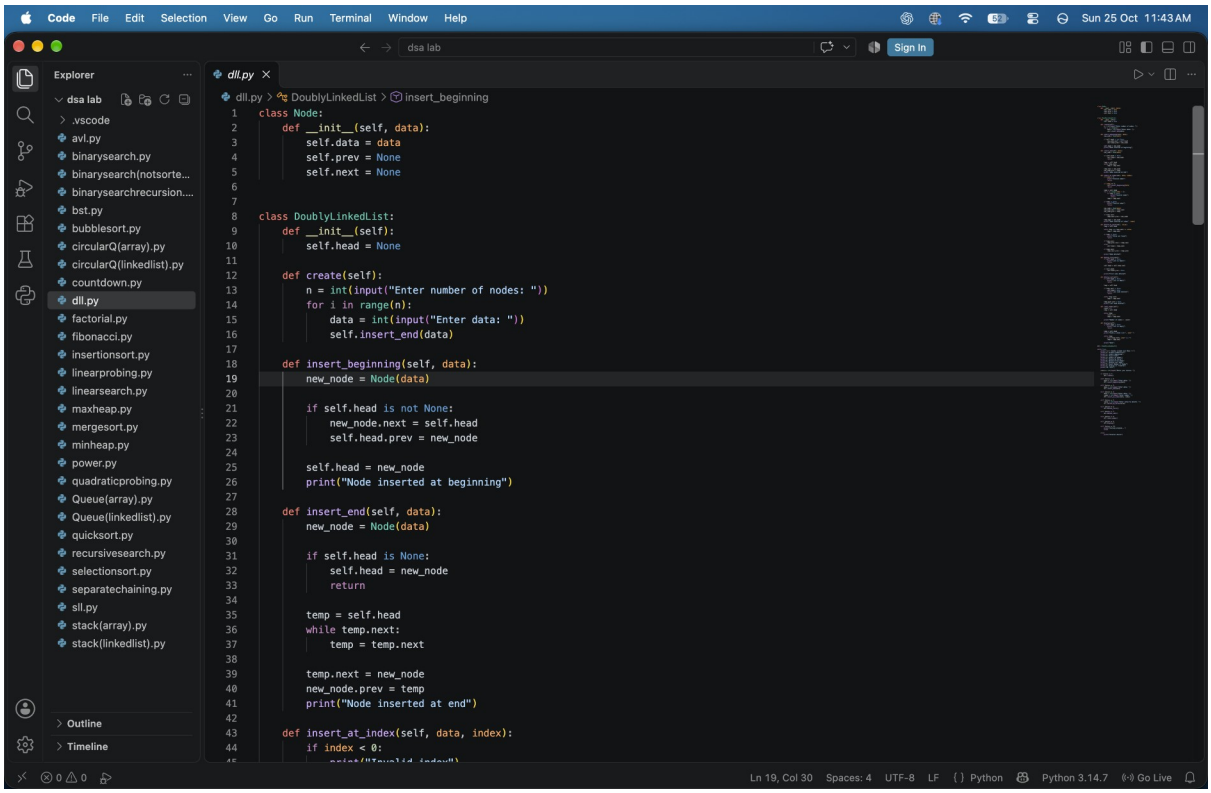
7)Delete Last Node

8)Count number of nodes

9)Display or Traverse

10)Exit

Code:-



```python
class Node:
    def __init__(self, data):
        self.data = data
        self.prev = None
        self.next = None


class DoublyLinkedList:
    def __init__(self):
        self.head = None

    def create(self):
        n = int(input("Enter number of nodes: "))
        for i in range(n):
            data = int(input("Enter data: "))
            self.insert_end(data)

    def insert_beginning(self, data):
        new_node = Node(data)

        if self.head is not None:
            new_node.next = self.head
            self.head.prev = new_node

        self.head = new_node
        print("Node inserted at beginning")

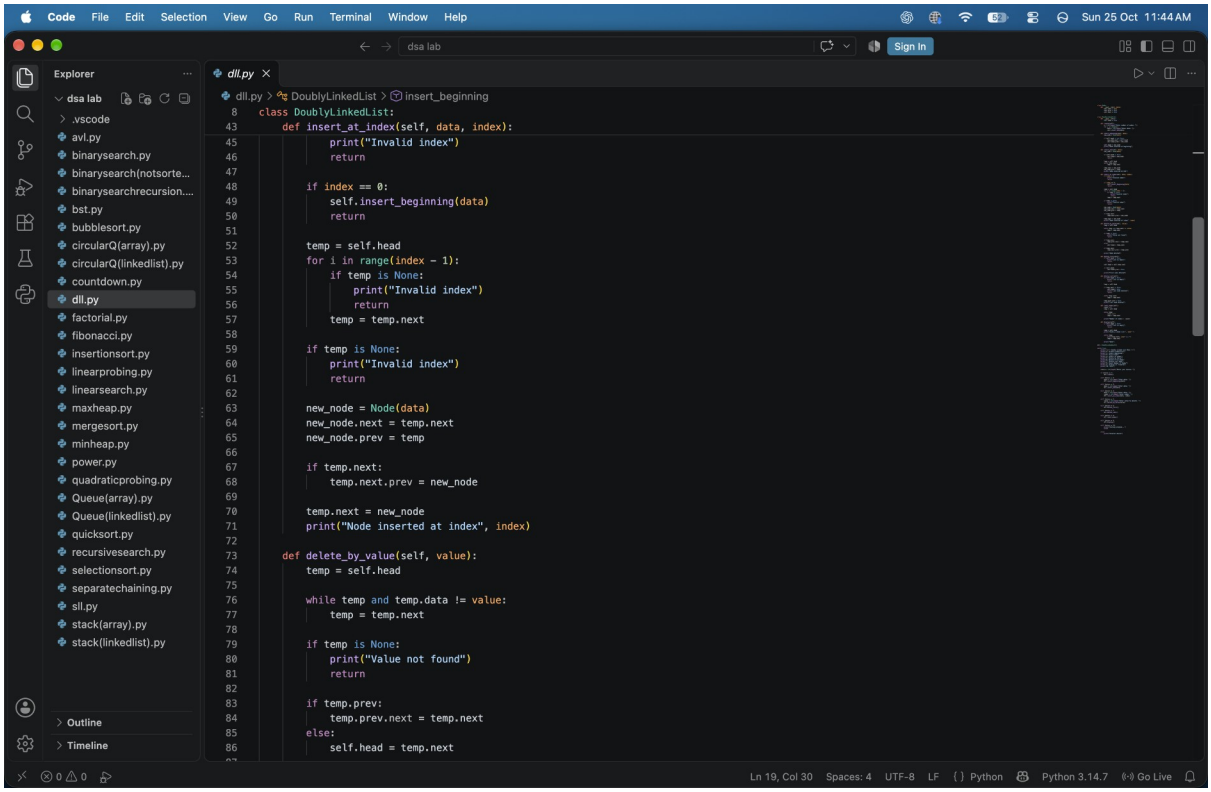
    def insert_end(self, data):
        new_node = Node(data)

        if self.head is None:
            self.head = new_node
            return

        temp = self.head
        while temp.next:
            temp = temp.next

        temp.next = new_node
        new_node.prev = temp
        print("Node inserted at end")

    def insert_at_index(self, data, index):
        if index < 0:
```



```
class DoublyLinkedList:
    def insert_at_index(self, data, index):
            print("Invalid index")
            return

        if index == 0:
            self.insert_beginning(data)
            return

        temp = self.head
        for i in range(index - 1):
            if temp is None:
                print("Invalid index")
                return
            temp = temp.next

        if temp is None:
            print("Invalid index")
            return

        new_node = Node(data)
        new_node.next = temp.next
        new_node.prev = temp

        if temp.next:
            temp.next.prev = new_node

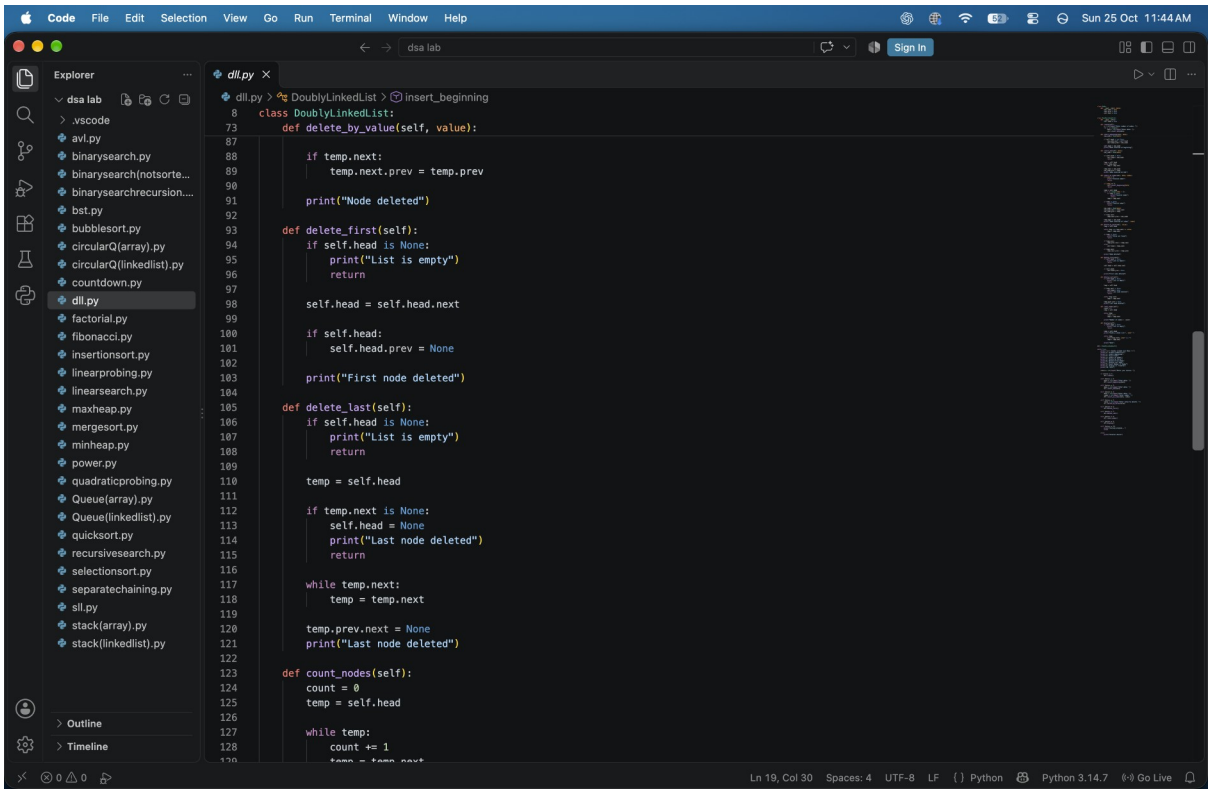
        temp.next = new_node
        print("Node inserted at index", index)

    def delete_by_value(self, value):
        temp = self.head

        while temp and temp.data != value:
            temp = temp.next

        if temp is None:
            print("Value not found")
            return

        if temp.prev:
            temp.prev.next = temp.next
        else:
            self.head = temp.next
```



```
class DoublyLinkedList:
    def delete_by_value(self, value):

        if temp.next:
            temp.next.prev = temp.prev

        print("Node deleted")

    def delete_first(self):
        if self.head is None:
            print("List is empty")
            return

        self.head = self.head.next

        if self.head:
            self.head.prev = None

        print("First node deleted")

    def delete_last(self):
        if self.head is None:
            print("List is empty")
            return

        temp = self.head

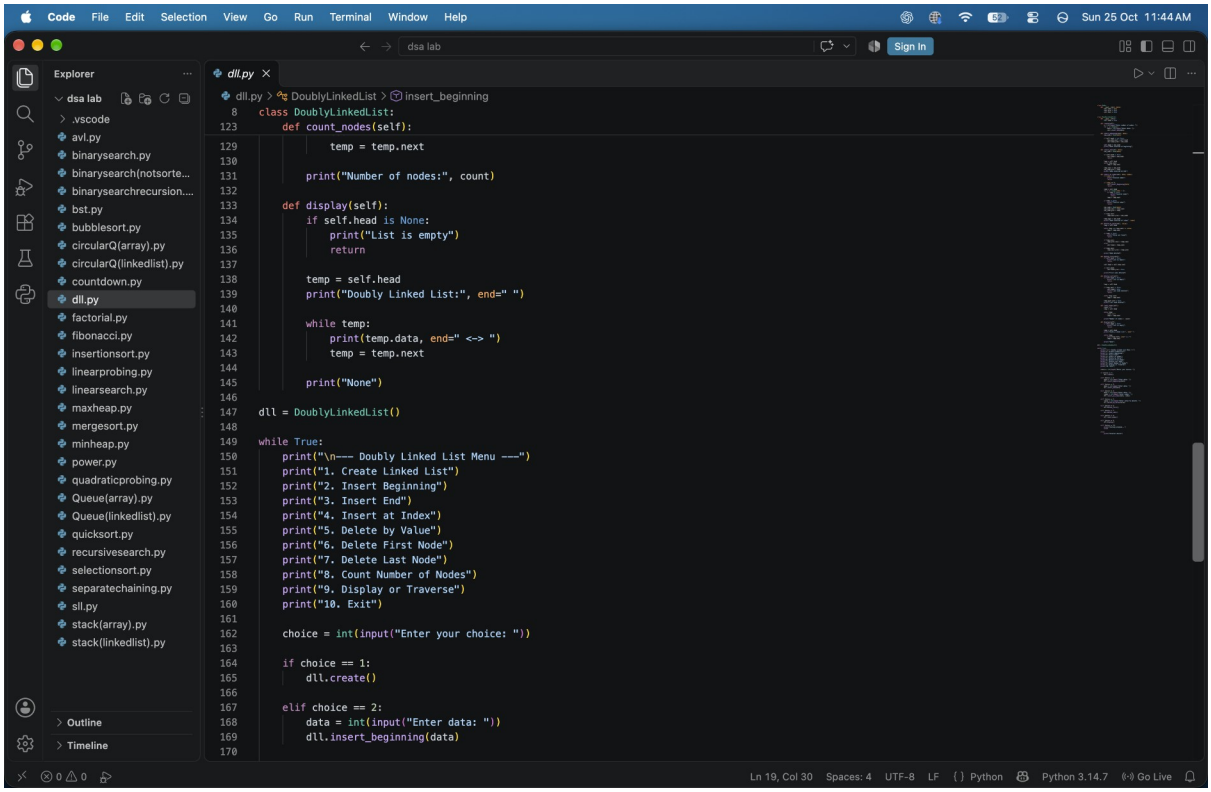
        if temp.next is None:
            self.head = None
            print("Last node deleted")
            return

        while temp.next:
            temp = temp.next

        temp.prev.next = None
        print("Last node deleted")

    def count_nodes(self):
        count = 0
        temp = self.head

        while temp:
            count += 1
```



```
class DoublyLinkedList:
    def count_nodes(self):
                temp = temp.next

        print("Number of nodes:", count)

    def display(self):
        if self.head is None:
            print("List is empty")
            return

        temp = self.head
        print("Doubly Linked List:", end=" ")

        while temp:
            print(temp.data, end=" <-> ")
            temp = temp.next

        print("None")

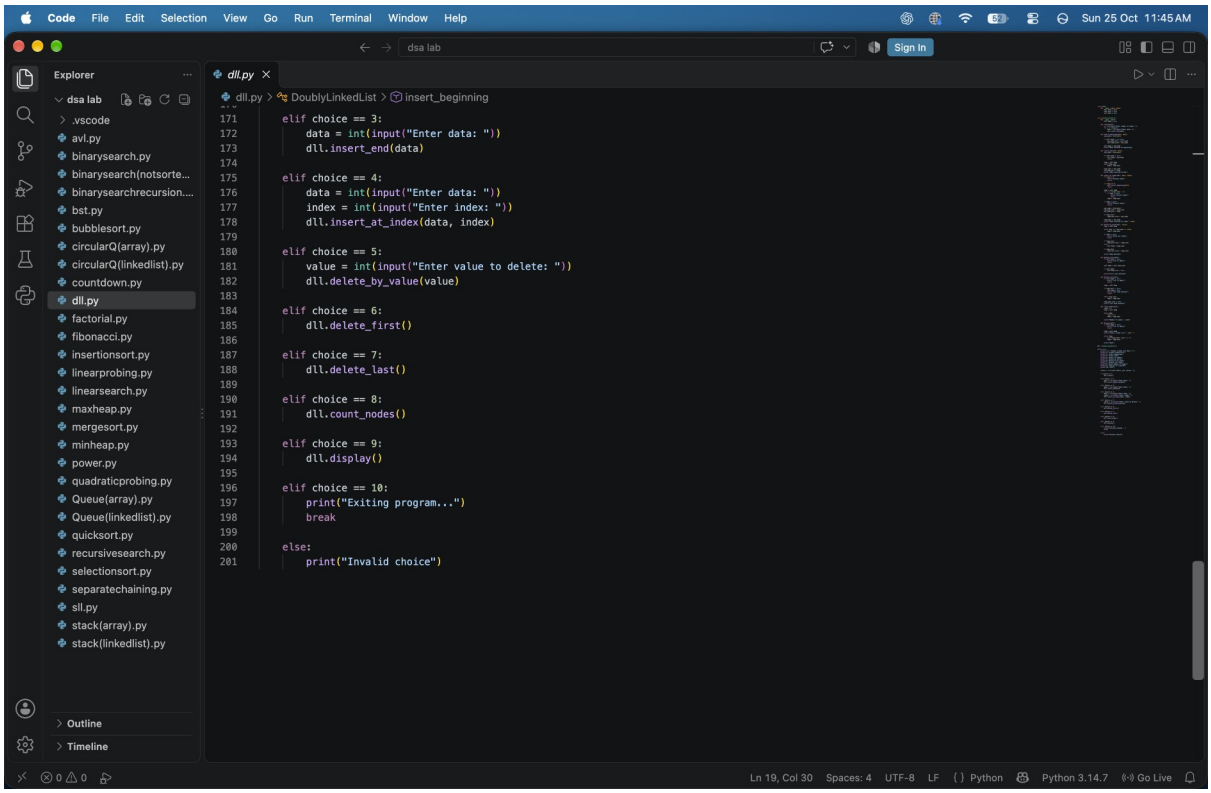
dll = DoublyLinkedList()

while True:
    print("\n--- Doubly Linked List Menu ---")
    print("1. Create Linked List")
    print("2. Insert Beginning")
    print("3. Insert End")
    print("4. Insert at Index")
    print("5. Delete by Value")
    print("6. Delete First Node")
    print("7. Delete Last Node")
    print("8. Count Number of Nodes")
    print("9. Display or Traverse")
    print("10. Exit")

    choice = int(input("Enter your choice: "))

    if choice == 1:
        dll.create()

    elif choice == 2:
        data = int(input("Enter data: "))
        dll.insert_beginning(data)
```



```
    elif choice == 3:
        data = int(input("Enter data: "))
        dll.insert_end(data)

    elif choice == 4:
        data = int(input("Enter data: "))
        index = int(input("Enter index: "))
        dll.insert_at_index(data, index)

    elif choice == 5:
        value = int(input("Enter value to delete: "))
        dll.delete_by_value(value)

    elif choice == 6:
        dll.delete_first()

    elif choice == 7:
        dll.delete_last()

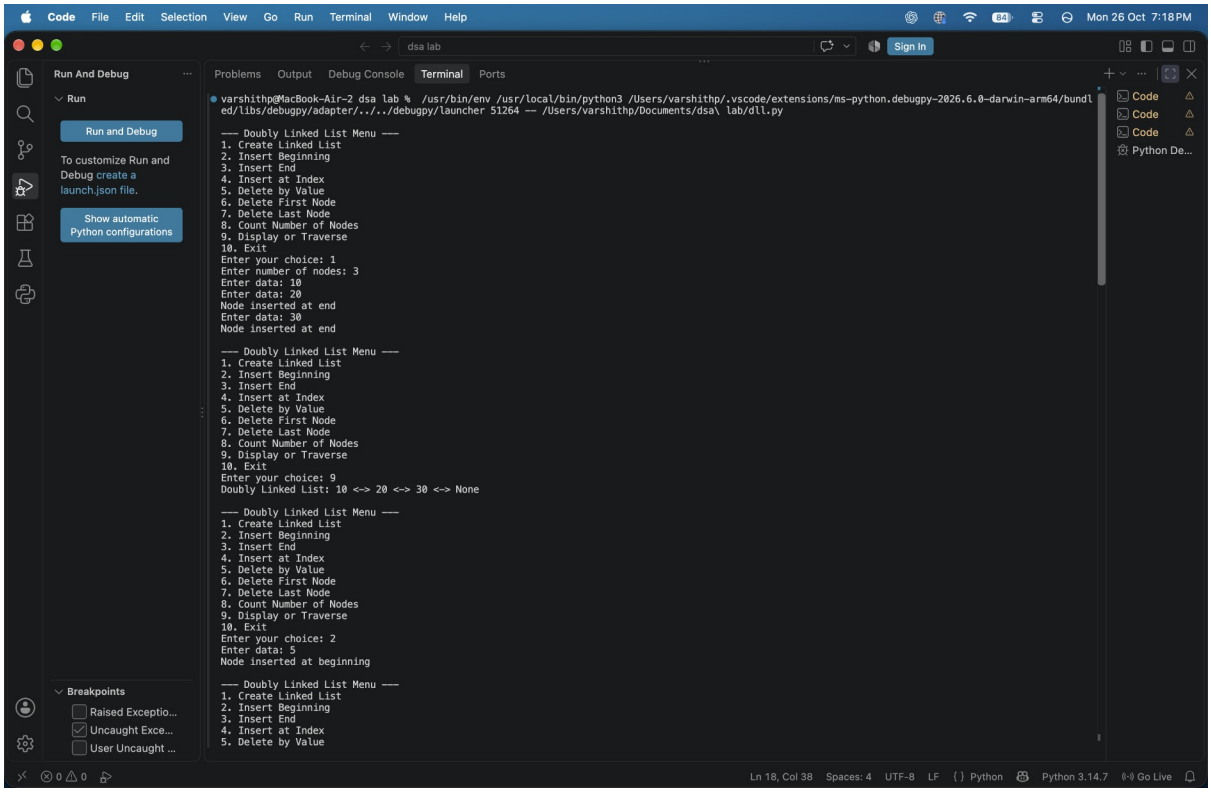
    elif choice == 8:
        dll.count_nodes()

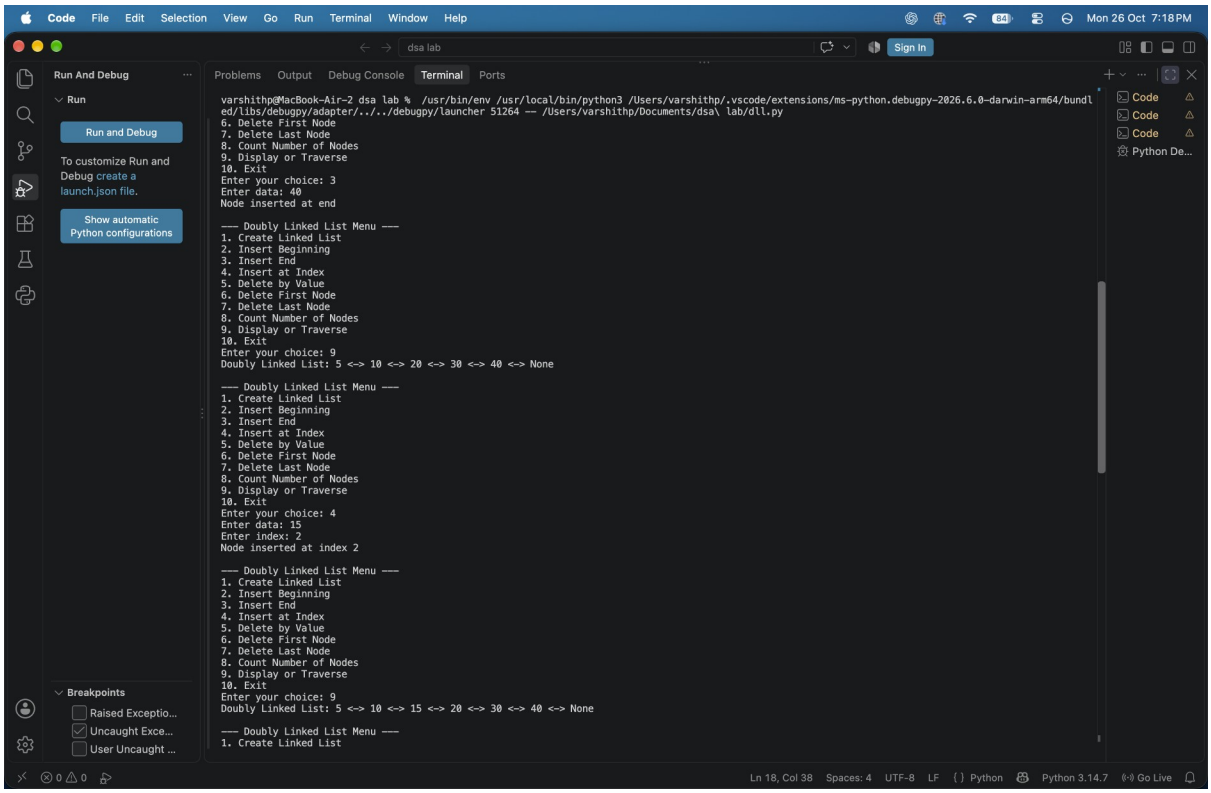
    elif choice == 9:
        dll.display()

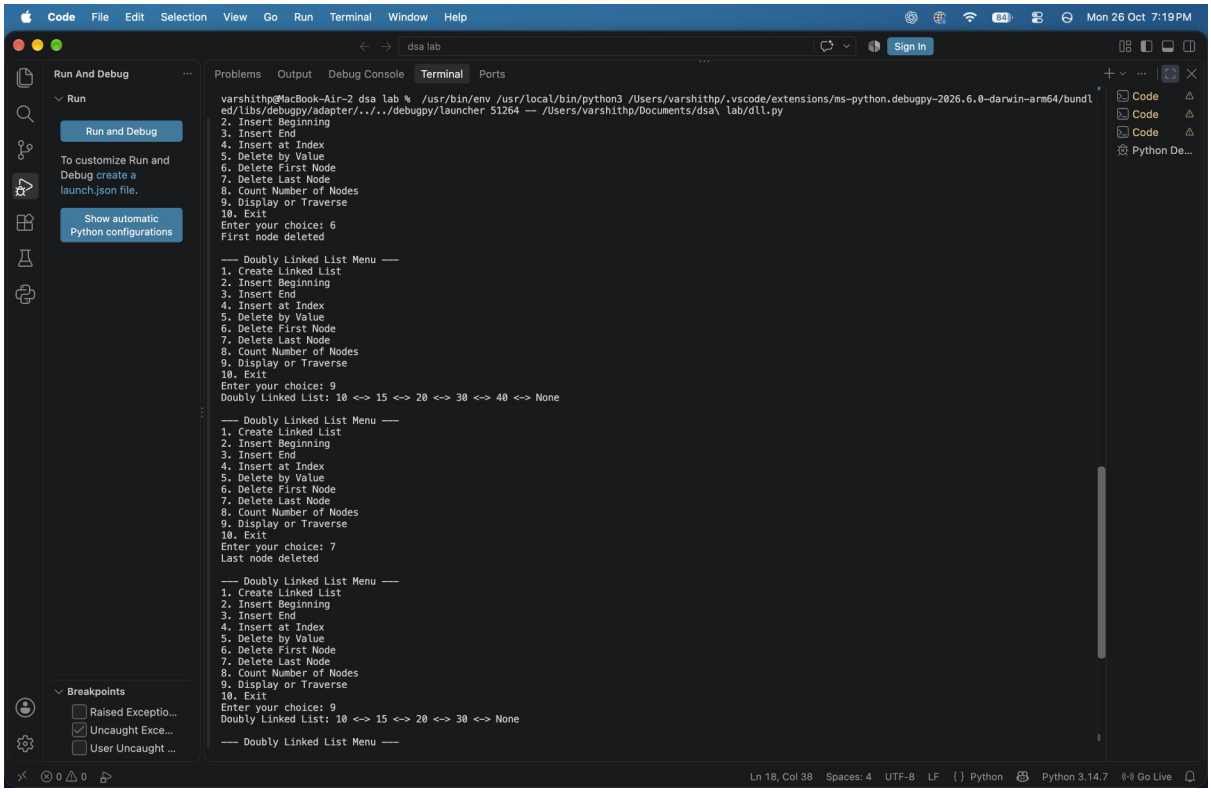
    elif choice == 10:
        print("Exiting program...")
        break

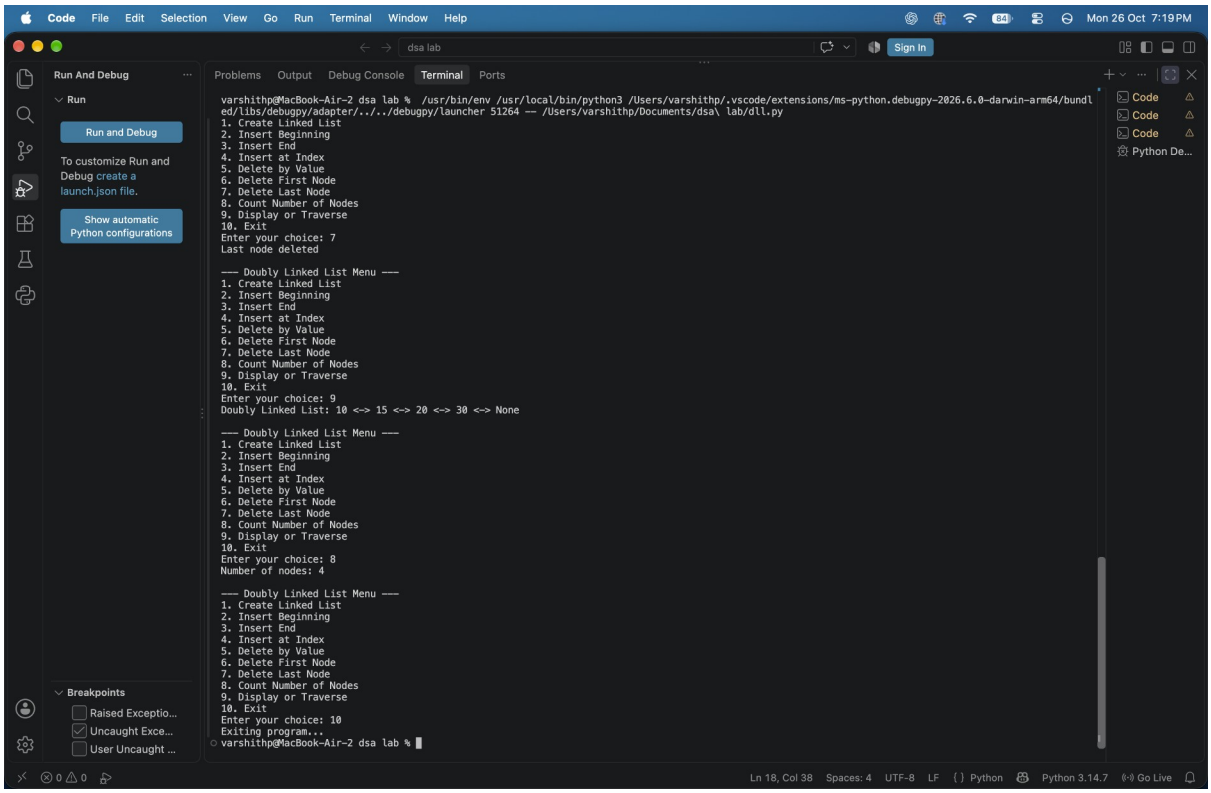
    else:
        print("Invalid choice")
```

Output:-


varshithp@MacBook-Air-2 dsa lab % /usr/bin/env /usr/local/bin/python3 /Users/varshithp/.vscode/extensions/ms-python.debugpy-2026.6.0-darwin-arm64/bundled/libs/debugpy/adapter/../../debugpy/launcher 51264 -- /Users/varshithp/Documents/dsa\ lab/dll.py
--- Doubly Linked List Menu ---
1. Create Linked List
2. Insert Beginning
3. Insert End
4. Insert at Index
5. Delete by Value
6. Delete First Node
7. Delete Last Node
8. Count Number of Nodes
9. Display or Traverse
10. Exit
Enter your choice: 1
Enter number of nodes: 3
Enter data: 10
Enter data: 20
Node inserted at end
Enter data: 30
Node inserted at end
--- Doubly Linked List Menu ---
1. Create Linked List
2. Insert Beginning
3. Insert End
4. Insert at Index
5. Delete by Value
6. Delete First Node
7. Delete Last Node
8. Count Number of Nodes
9. Display or Traverse
10. Exit
Enter your choice: 9
Doubly Linked List: 10 <-> 20 <-> 30 <-> None
--- Doubly Linked List Menu ---
1. Create Linked List
2. Insert Beginning
3. Insert End
4. Insert at Index
5. Delete by Value
6. Delete First Node
7. Delete Last Node
8. Count Number of Nodes
9. Display or Traverse
10. Exit
Enter your choice: 2
Enter data: 5
Node inserted at beginning
--- Doubly Linked List Menu ---
1. Create Linked List
2. Insert Beginning
3. Insert End
4. Insert at Index
5. Delete by Value


6. Delete First Node
7. Delete Last Node
8. Count Number of Nodes
9. Display or Traverse
10. Exit
Enter your choice: 3
Enter data: 40
Node inserted at end
--- Doubly Linked List Menu ---
1. Create Linked List
2. Insert Beginning
3. Insert End
4. Insert at Index
5. Delete by Value
6. Delete First Node
7. Delete Last Node
8. Count Number of Nodes
9. Display or Traverse
10. Exit
Enter your choice: 9
Doubly Linked List: 5 <-> 10 <-> 20 <-> 30 <-> 40 <-> None
--- Doubly Linked List Menu ---
1. Create Linked List
2. Insert Beginning
3. Insert End
4. Insert at Index
5. Delete by Value
6. Delete First Node
7. Delete Last Node
8. Count Number of Nodes
9. Display or Traverse
10. Exit
Enter your choice: 4
Enter data: 15
Enter index: 2
Node inserted at index 2
--- Doubly Linked List Menu ---
1. Create Linked List
2. Insert Beginning
3. Insert End
4. Insert at Index
5. Delete by Value
6. Delete First Node
7. Delete Last Node
8. Count Number of Nodes
9. Display or Traverse
10. Exit
Enter your choice: 9
Doubly Linked List: 5 <-> 10 <-> 15 <-> 20 <-> 30 <-> 40 <-> None
--- Doubly Linked List Menu ---
1. Create Linked List

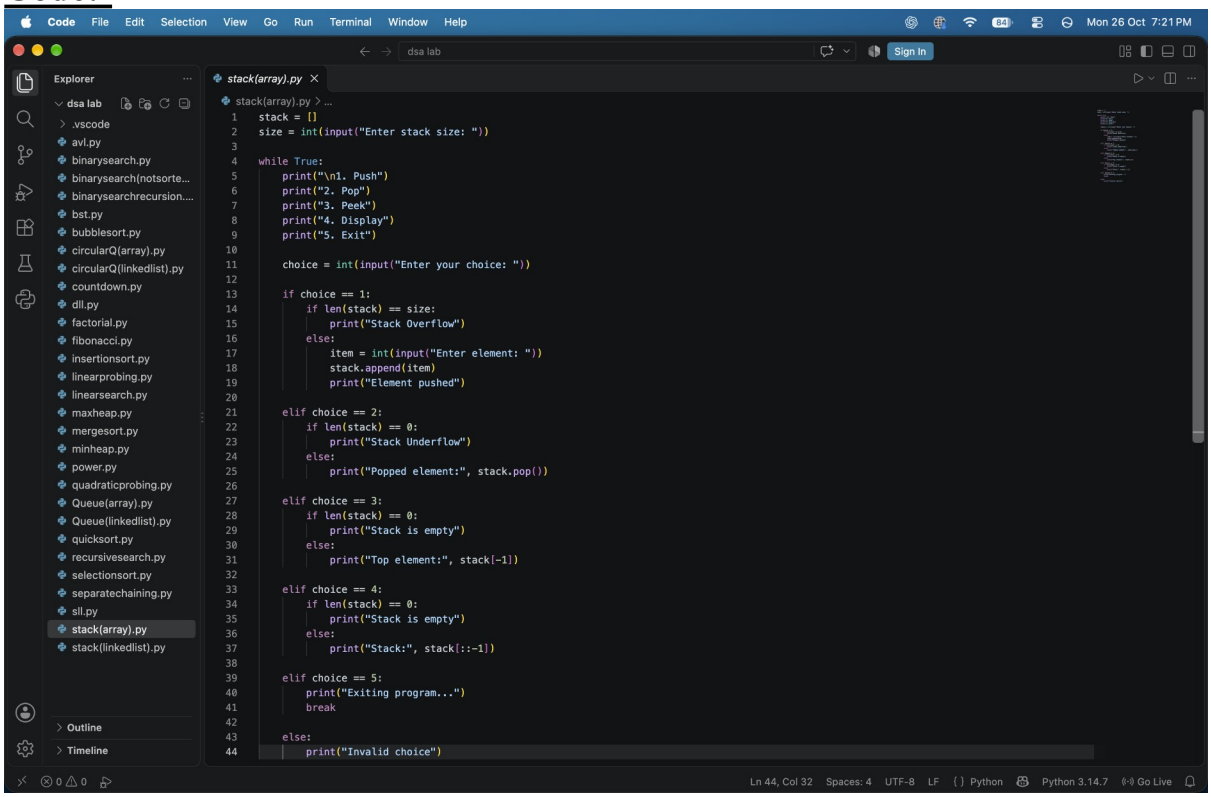

Code File Edit Selection View Go Run Terminal Window Help
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Run
Run and Debug
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Raised Exceptio...
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Problems Output Debug Console Terminal Ports
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2. Insert Beginning
3. Insert End
4. Insert at Index
5. Delete by Value
6. Delete First Node
7. Delete Last Node
8. Count Number of Nodes
9. Display or Traverse
10. Exit
Enter your choice: 6
First node deleted
--- Doubly Linked List Menu ---
1. Create Linked List
2. Insert Beginning
3. Insert End
4. Insert at Index
5. Delete by Value
6. Delete First Node
7. Delete Last Node
8. Count Number of Nodes
9. Display or Traverse
10. Exit
Enter your choice: 9
Doubly Linked List: 10 <-> 15 <-> 20 <-> 30 <-> 40 <-> None
--- Doubly Linked List Menu ---
1. Create Linked List
2. Insert Beginning
3. Insert End
4. Insert at Index
5. Delete by Value
6. Delete First Node
7. Delete Last Node
8. Count Number of Nodes
9. Display or Traverse
10. Exit
Enter your choice: 7
Last node deleted
--- Doubly Linked List Menu ---
1. Create Linked List
2. Insert Beginning
3. Insert End
4. Insert at Index
5. Delete by Value
6. Delete First Node
7. Delete Last Node
8. Count Number of Nodes
9. Display or Traverse
10. Exit
Enter your choice: 9
Doubly Linked List: 10 <-> 15 <-> 20 <-> 30 <-> None
--- Doubly Linked List Menu ---
Code
Code
Code
Python De...
Ln 18, Col 38 Spaces: 4 UTF-8 LF Python Python 3.14.7 Go Live


Code File Edit Selection View Go Run Terminal Window Help
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1. Create Linked List
2. Insert Beginning
3. Insert End
4. Insert at Index
5. Delete by Value
6. Delete First Node
7. Delete Last Node
8. Count Number of Nodes
9. Display or Traverse
10. Exit
Enter your choice: 7
Last node deleted
--- Doubly Linked List Menu ---
1. Create Linked List
2. Insert Beginning
3. Insert End
4. Insert at Index
5. Delete by Value
6. Delete First Node
7. Delete Last Node
8. Count Number of Nodes
9. Display or Traverse
10. Exit
Enter your choice: 9
Doubly Linked List: 10 <-> 15 <-> 20 <-> 30 <-> None
--- Doubly Linked List Menu ---
1. Create Linked List
2. Insert Beginning
3. Insert End
4. Insert at Index
5. Delete by Value
6. Delete First Node
7. Delete Last Node
8. Count Number of Nodes
9. Display or Traverse
10. Exit
Enter your choice: 8
Number of nodes: 4
--- Doubly Linked List Menu ---
1. Create Linked List
2. Insert Beginning
3. Insert End
4. Insert at Index
5. Delete by Value
6. Delete First Node
7. Delete Last Node
8. Count Number of Nodes
9. Display or Traverse
10. Exit
Enter your choice: 10
Exiting program...
varshithp@MacBook-Air-2 dsa lab %
Code
Code
Code
Python De...
Ln 18, Col 38 Spaces: 4 UTF-8 LF Python Python 3.14.7 Go Live

WEEK-7

7(A). Write a python program to implement stack operations such as push, pop, peek, display using an array.

Code:-



```
stack = []
size = int(input("Enter stack size: "))

while True:
    print("\n1. Push")
    print("2. Pop")
    print("3. Peek")
    print("4. Display")
    print("5. Exit")

    choice = int(input("Enter your choice: "))

    if choice == 1:
        if len(stack) == size:
            print("Stack Overflow")
        else:
            item = int(input("Enter element: "))
            stack.append(item)
            print("Element pushed")

    elif choice == 2:
        if len(stack) == 0:
            print("Stack Underflow")
        else:
            print("Popped element:", stack.pop())

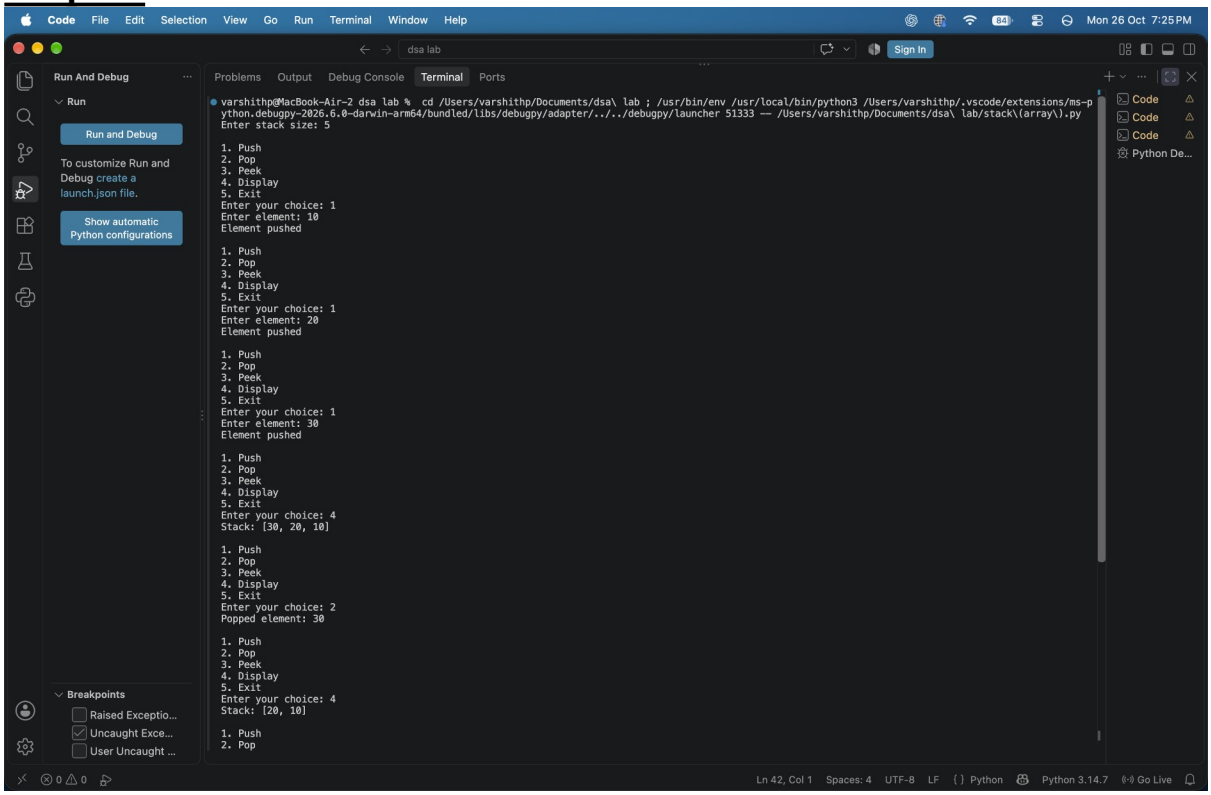
    elif choice == 3:
        if len(stack) == 0:
            print("Stack is empty")
        else:
            print("Top element:", stack[-1])

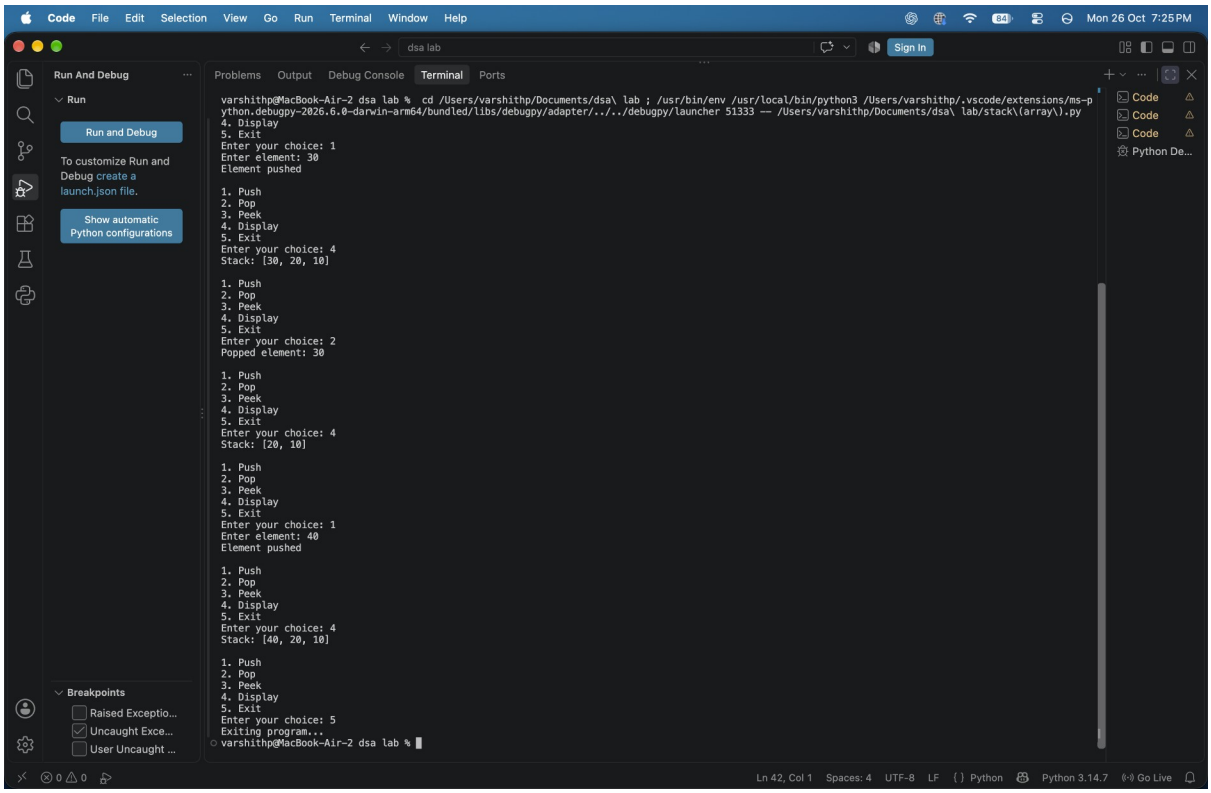
    elif choice == 4:
        if len(stack) == 0:
            print("Stack is empty")
        else:
            print("Stack:", stack[::-1])

    elif choice == 5:
        print("Exiting program...")
        break

    else:
        print("Invalid choice")
```

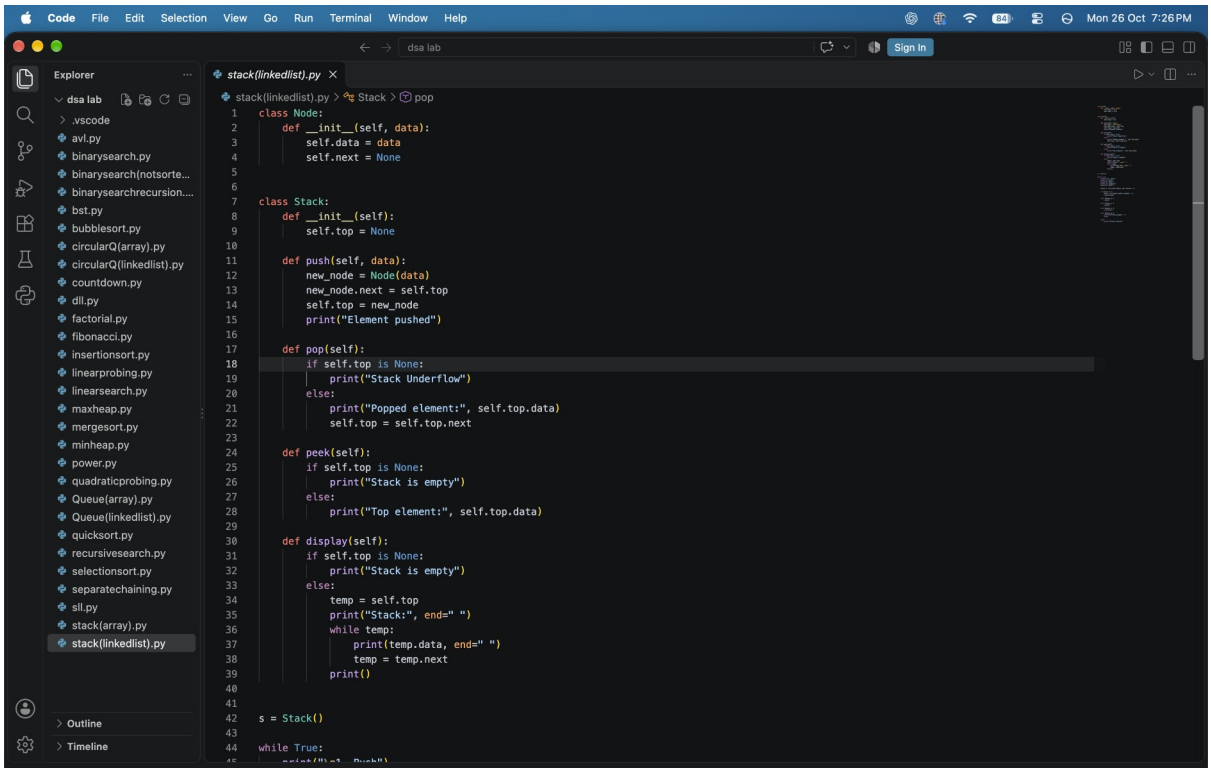
Output:-


varshithp@MacBook-Air-2 dsa lab % cd /Users/varshithp/Documents/dsa\ lab ; /usr/bin/env /usr/local/bin/python3 /Users/varshithp/.vscode/extensions/ms-python.debugpy-2026.6.0-darwin-arm64/bundled/libs/debugpy/adapter/../../debugpy/launcher 51333 -- /Users/varshithp/Documents/dsa\ lab/stack\(array\).py
Enter stack size: 5
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice: 1
Enter element: 10
Element pushed
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice: 1
Enter element: 20
Element pushed
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice: 1
Enter element: 30
Element pushed
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice: 4
Stack: [30, 20, 10]
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice: 2
Popped element: 30
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice: 4
Stack: [20, 10]
1. Push
2. Pop


4. Display
5. Exit
Enter your choice: 1
Enter element: 30
Element pushed
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice: 4
Stack: [30, 20, 10]
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice: 2
Popped element: 30
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice: 4
Stack: [20, 10]
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice: 1
Enter element: 40
Element pushed
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice: 4
Stack: [40, 20, 10]
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice: 5
Exiting program...
varshithp@MacBook-Air-2 dsa lab %

7(B). Write a python program to implement stack operations such as push, pop, peek, display using a linked list.

Code:-



```python
class Node:
    def __init__(self, data):
        self.data = data
        self.next = None


class Stack:
    def __init__(self):
        self.top = None

    def push(self, data):
        new_node = Node(data)
        new_node.next = self.top
        self.top = new_node
        print("Element pushed")

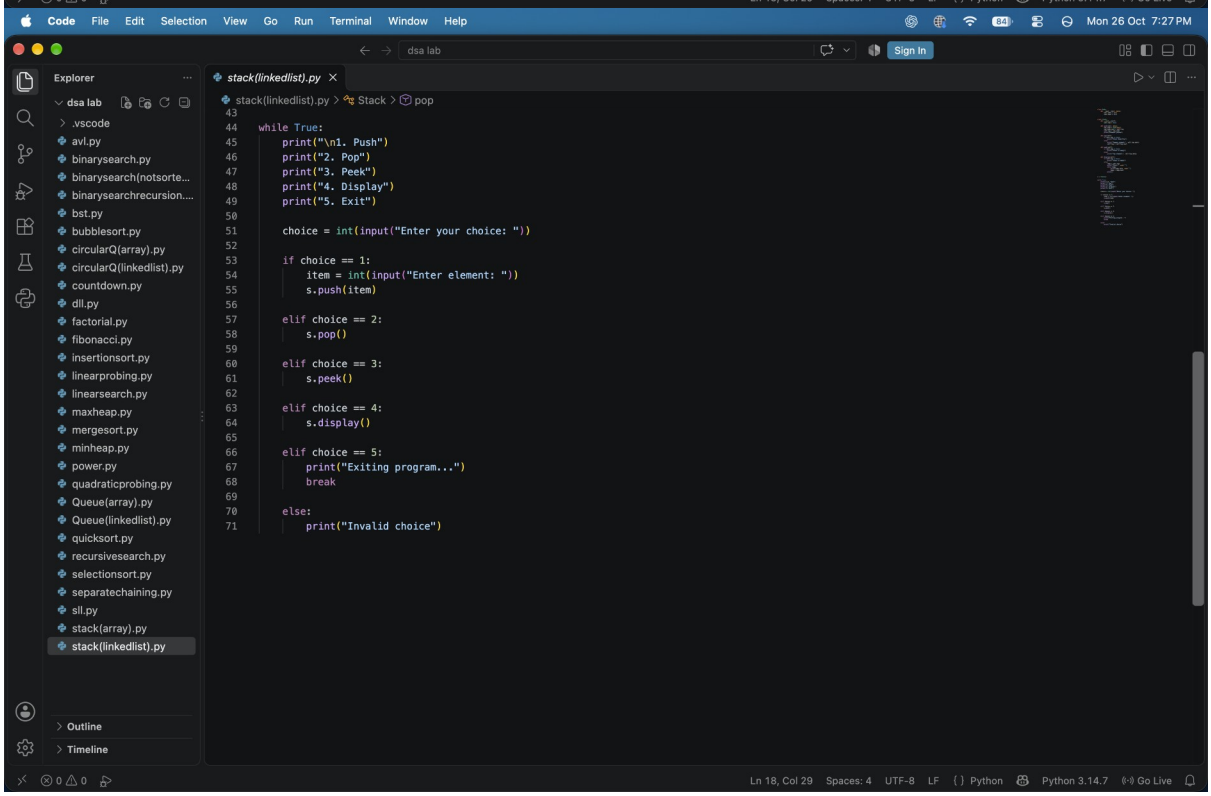
    def pop(self):
        if self.top is None:
            print("Stack Underflow")
        else:
            print("Popped element:", self.top.data)
            self.top = self.top.next

    def peek(self):
        if self.top is None:
            print("Stack is empty")
        else:
            print("Top element:", self.top.data)

    def display(self):
        if self.top is None:
            print("Stack is empty")
        else:
            temp = self.top
            print("Stack:", end=" ")
            while temp:
                print(temp.data, end=" ")
                temp = temp.next
            print()


s = Stack()

while True:
```



```python
while True:
    print("\n1. Push")
    print("2. Pop")
    print("3. Peek")
    print("4. Display")
    print("5. Exit")

    choice = int(input("Enter your choice: "))

    if choice == 1:
        item = int(input("Enter element: "))
        s.push(item)

    elif choice == 2:
        s.pop()

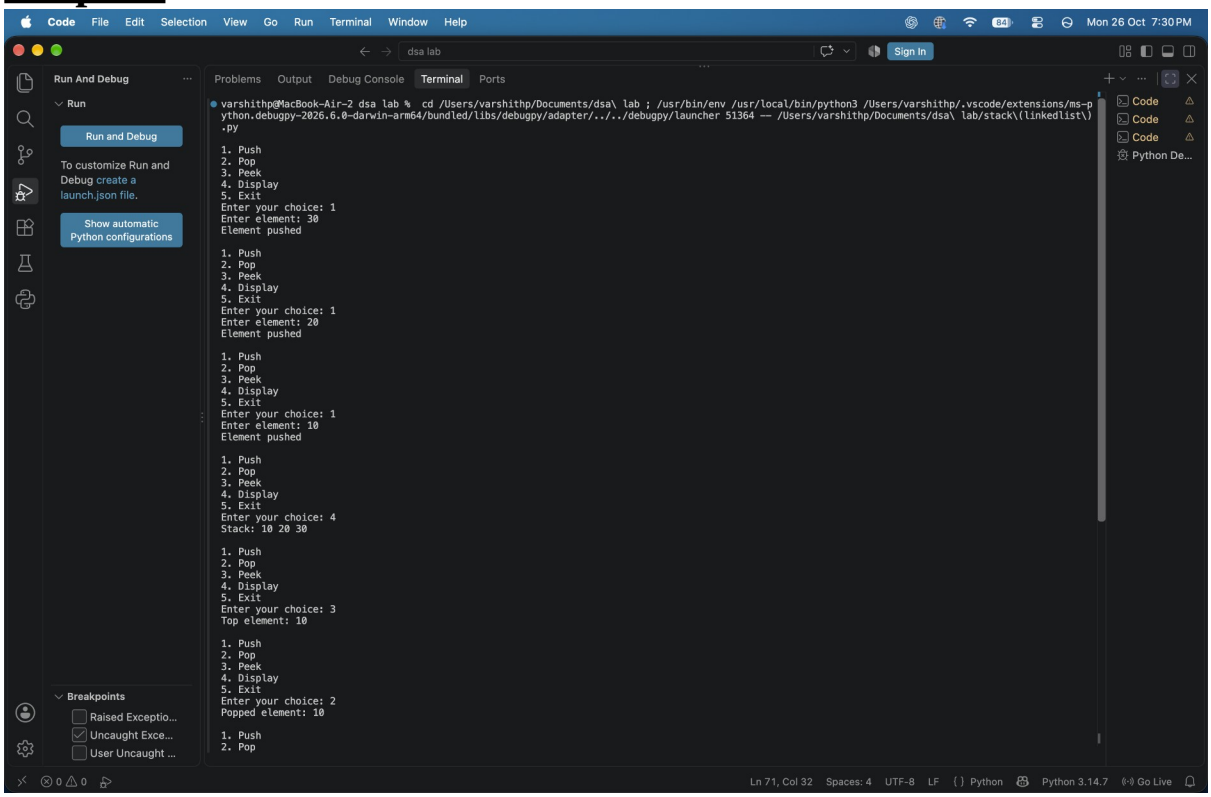
    elif choice == 3:
        s.peek()

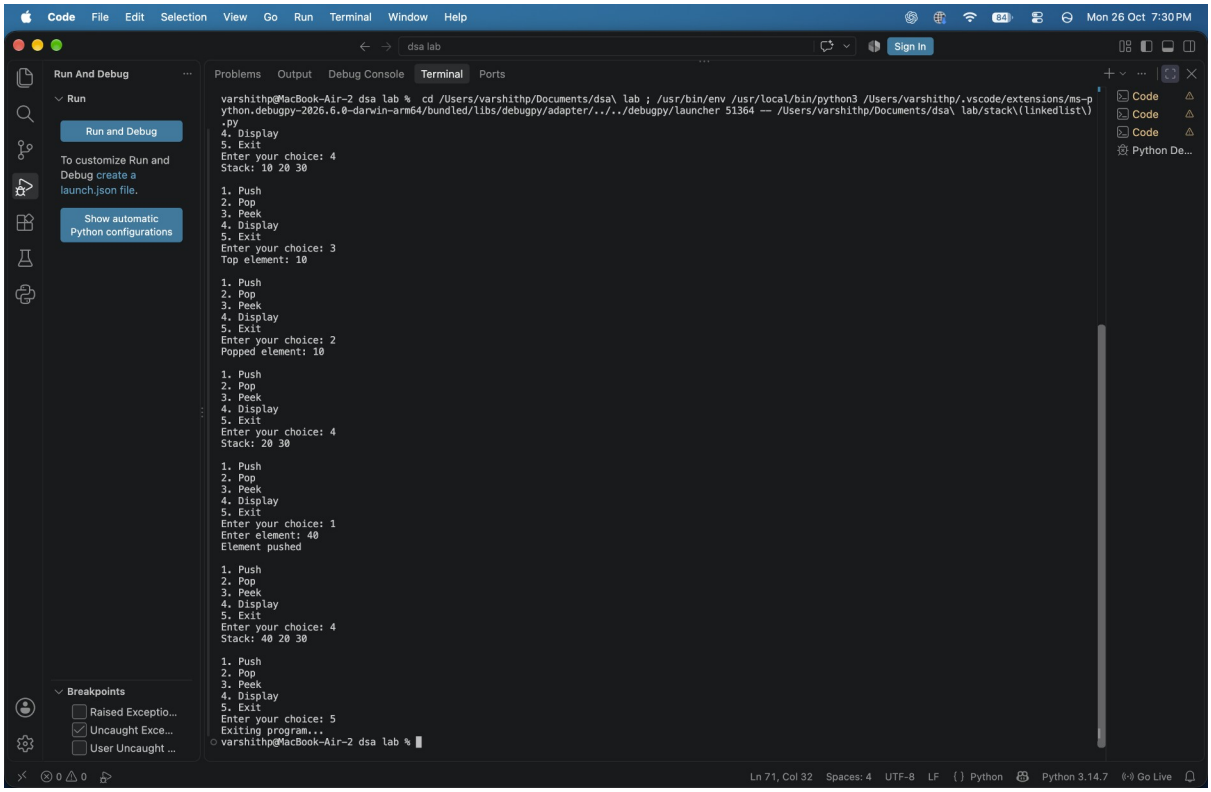
    elif choice == 4:
        s.display()

    elif choice == 5:
        print("Exiting program...")
        break

    else:
        print("Invalid choice")
```

Output:-


Code File Edit Selection View Go Run Terminal Window Help
Mon 26 Oct 7:30 PM
dsa lab
Sign in
Run And Debug
Run
Run and Debug
To customize Run and Debug create a launch.json file.
Show automatic Python configurations
Breakpoints
Raised Exceptio...
Uncaught Exce...
User Uncaught ...
Problems Output Debug Console Terminal Ports
varshithp@MacBook-Air-2 dsa lab % cd /Users/varshithp/Documents/dsa\ lab ; /usr/bin/env /usr/local/bin/python3 /Users/varshithp/.vscode/extensions/ms-python.debugpy-2026.6.0-darwin-arm64/bundled/libs/debugpy/adapter/../../debugpy/launcher 51364 -- /Users/varshithp/Documents/dsa\ lab/stack\(linkedlist\).py
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice: 1
Enter element: 30
Element pushed
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice: 1
Enter element: 20
Element pushed
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice: 1
Enter element: 10
Element pushed
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice: 4
Stack: 10 20 30
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice: 3
Top element: 10
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice: 2
Popped element: 10
1. Push
2. Pop
Code
Code
Code
Python De...
Ln 71, Col 32 Spaces: 4 UTF-8 LF { } Python Python 3.14.7 Go Live

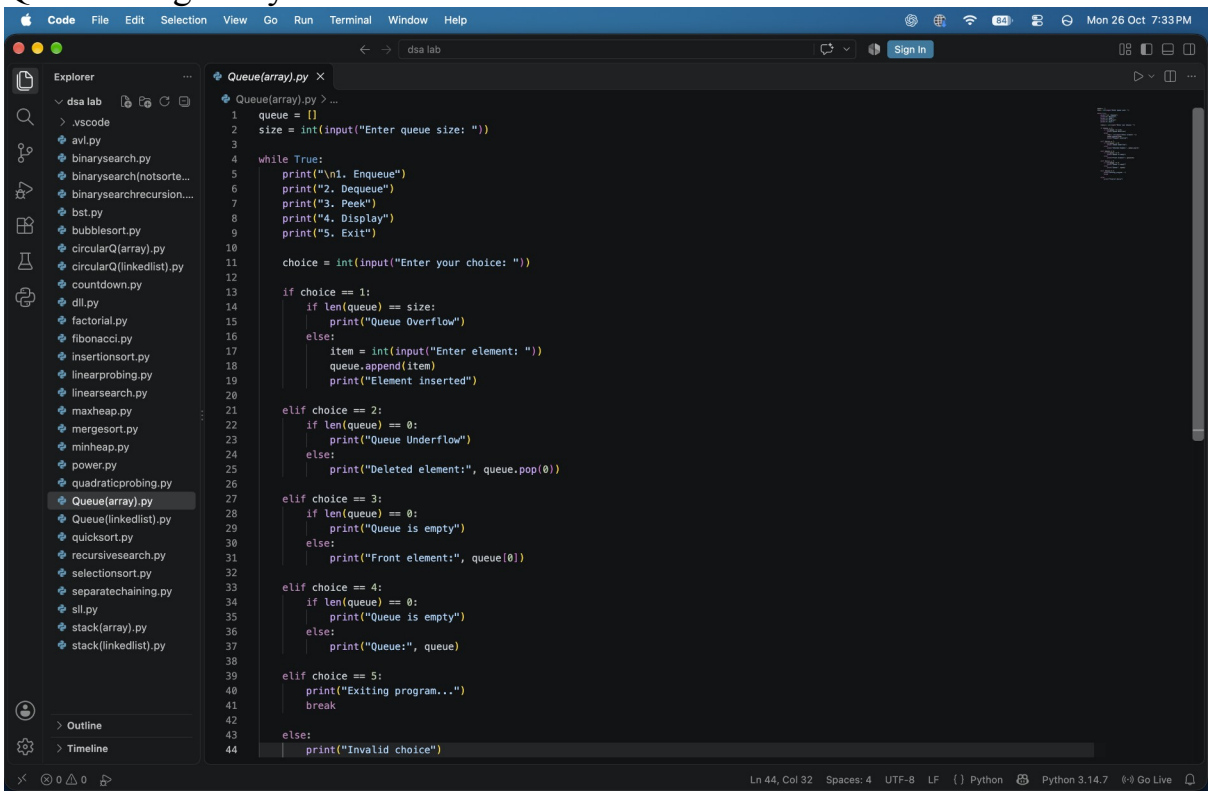

Code File Edit Selection View Go Run Terminal Window Help
Mon 26 Oct 7:30 PM
dsa lab
Sign in
Run And Debug
Run
Run and Debug
To customize Run and Debug create a launch.json file.
Show automatic Python configurations
Breakpoints
Raised Exceptio...
Uncaught Exce...
User Uncaught ...
Problems Output Debug Console Terminal Ports
varshithp@MacBook-Air-2 dsa lab % cd /Users/varshithp/Documents/dsa\ lab ; /usr/bin/env /usr/local/bin/python3 /Users/varshithp/.vscode/extensions/ms-python.debugpy-2026.6.0-darwin-arm64/bundled/libs/debugpy/adapter/../../debugpy/launcher 51364 -- /Users/varshithp/Documents/dsa\ lab/stack\(linkedlist\).py
4. Display
5. Exit
Enter your choice: 4
Stack: 10 20 30
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice: 3
Top element: 10
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice: 2
Popped element: 10
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice: 4
Stack: 20 30
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice: 1
Enter element: 40
Element pushed
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice: 4
Stack: 40 20 30
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice: 5
Exiting program...
varshithp@MacBook-Air-2 dsa lab %
Code
Code
Code
Python De...
Ln 71, Col 32 Spaces: 4 UTF-8 LF { } Python Python 3.14.7 Go Live

WEEK-8

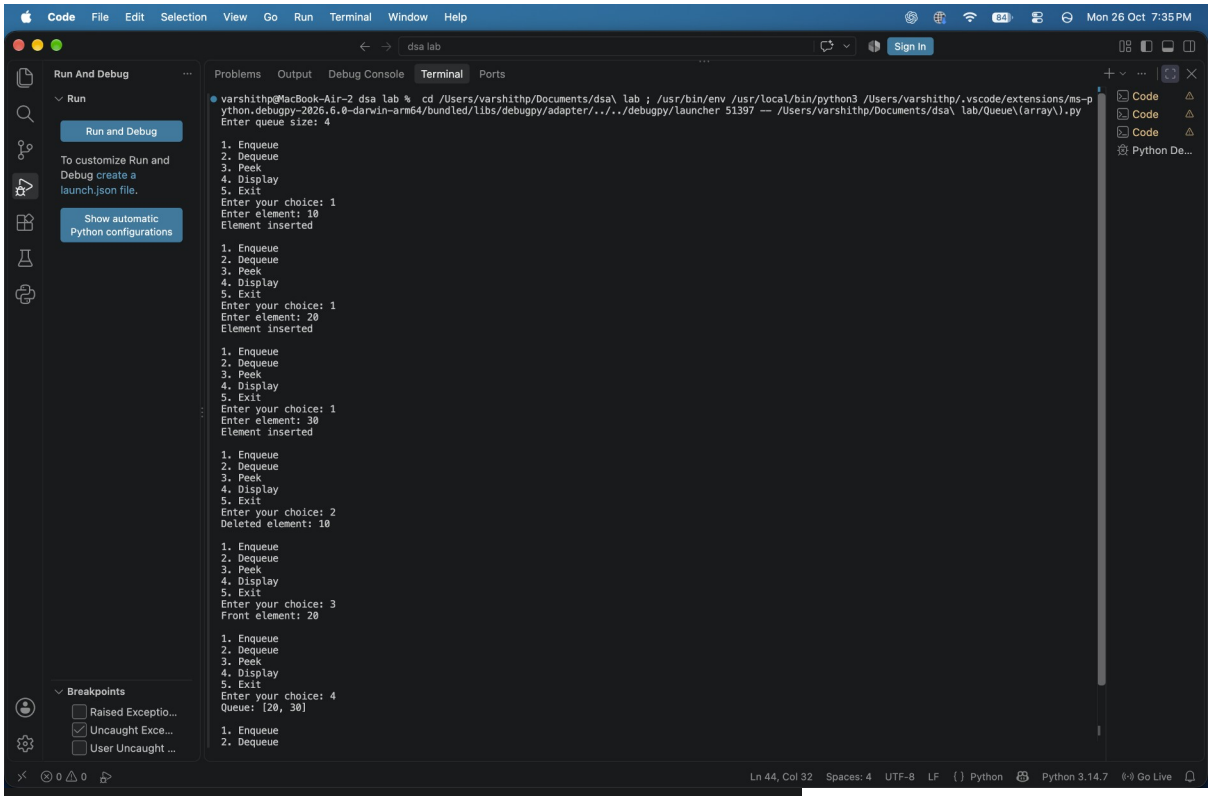
8(A). Write a python program to implement Queue and its operations such as enqueue, dequeue, peek, display with a menu driven approach. Implement using array, and linked list data structures.

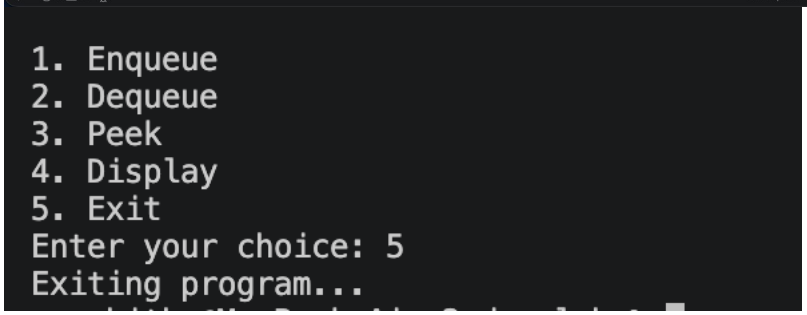
Code:-

Queue using Array



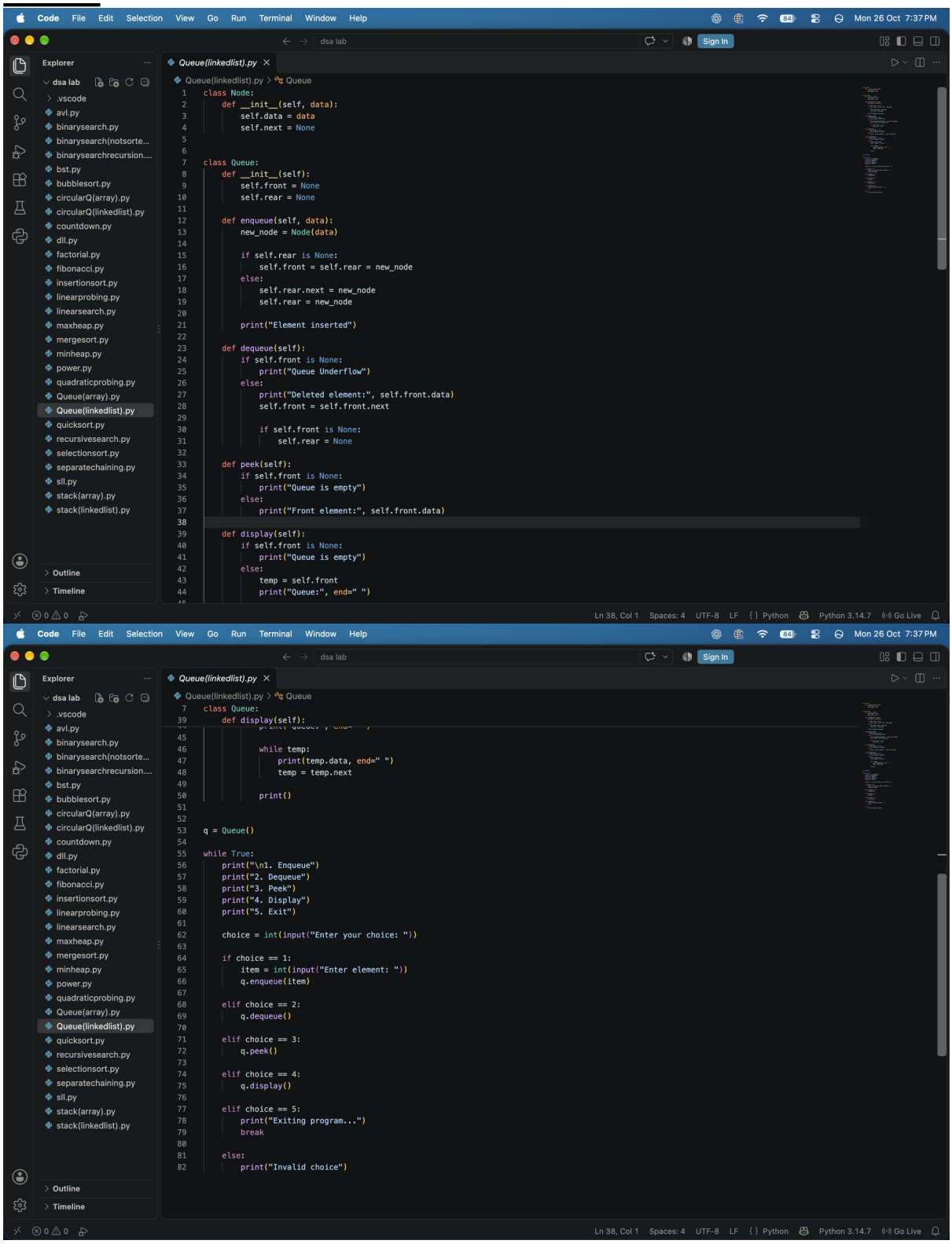
Output:-


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Raised Exceptio...
Uncaught Exce...
User Uncaught ...
Problems Output Debug Console Terminal Ports
varshithp@MacBook-Air-2 dsa lab % cd /Users/varshithp/Documents/dsa\ lab ; /usr/bin/env /usr/local/bin/python3 /Users/varshithp/.vscode/extensions/ms-python.debugpy-2026.6.0-darwin-arm64/bundled/libs/debugpy/adapter/../../debugpy/launcher 51397 -- /Users/varshithp/Documents/dsa\ lab/Queue\(array\).py
Enter queue size: 4
1. Enqueue
2. Dequeue
3. Peek
4. Display
5. Exit
Enter your choice: 1
Enter element: 10
Element inserted
1. Enqueue
2. Dequeue
3. Peek
4. Display
5. Exit
Enter your choice: 1
Enter element: 20
Element inserted
1. Enqueue
2. Dequeue
3. Peek
4. Display
5. Exit
Enter your choice: 1
Enter element: 30
Element inserted
1. Enqueue
2. Dequeue
3. Peek
4. Display
5. Exit
Enter your choice: 2
Deleted element: 10
1. Enqueue
2. Dequeue
3. Peek
4. Display
5. Exit
Enter your choice: 3
Front element: 20
1. Enqueue
2. Dequeue
3. Peek
4. Display
5. Exit
Enter your choice: 4
Queue: [20, 30]
1. Enqueue
2. Dequeue
Code
Code
Code
Python De...
Ln 44, Col 32 Spaces: 4 UTF-8 LF Python Python 3.14.7 Go Live


1. Enqueue
2. Dequeue
3. Peek
4. Display
5. Exit
Enter your choice: 5
Exiting program...

Queue using Linked list

Code:-



```
class Node:
    def __init__(self, data):
        self.data = data
        self.next = None


class Queue:
    def __init__(self):
        self.front = None
        self.rear = None

    def enqueue(self, data):
        new_node = Node(data)

        if self.rear is None:
            self.front = self.rear = new_node
        else:
            self.rear.next = new_node
            self.rear = new_node

        print("Element inserted")

    def dequeue(self):
        if self.front is None:
            print("Queue Underflow")
        else:
            print("Deleted element:", self.front.data)
            self.front = self.front.next

            if self.front is None:
                self.rear = None

    def peek(self):
        if self.front is None:
            print("Queue is empty")
        else:
            print("Front element:", self.front.data)

    def display(self):
        if self.front is None:
            print("Queue is empty")
        else:
            temp = self.front
            print("Queue:", end=" ")
```



```
class Queue:
    def display(self):

            while temp:
                print(temp.data, end=" ")
                temp = temp.next

            print()


q = Queue()

while True:
    print("\n1. Enqueue")
    print("2. Dequeue")
    print("3. Peek")
    print("4. Display")
    print("5. Exit")

    choice = int(input("Enter your choice: "))

    if choice == 1:
        item = int(input("Enter element: "))
        q.enqueue(item)

    elif choice == 2:
        q.dequeue()

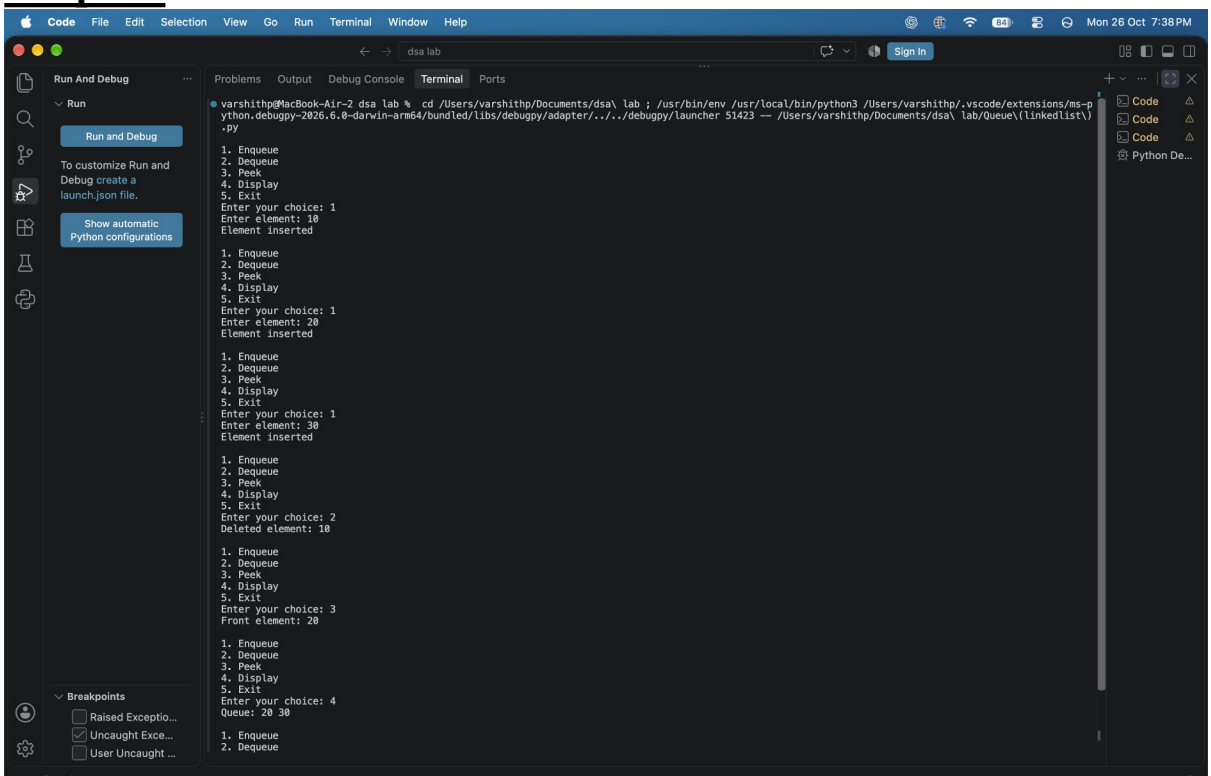
    elif choice == 3:
        q.peek()

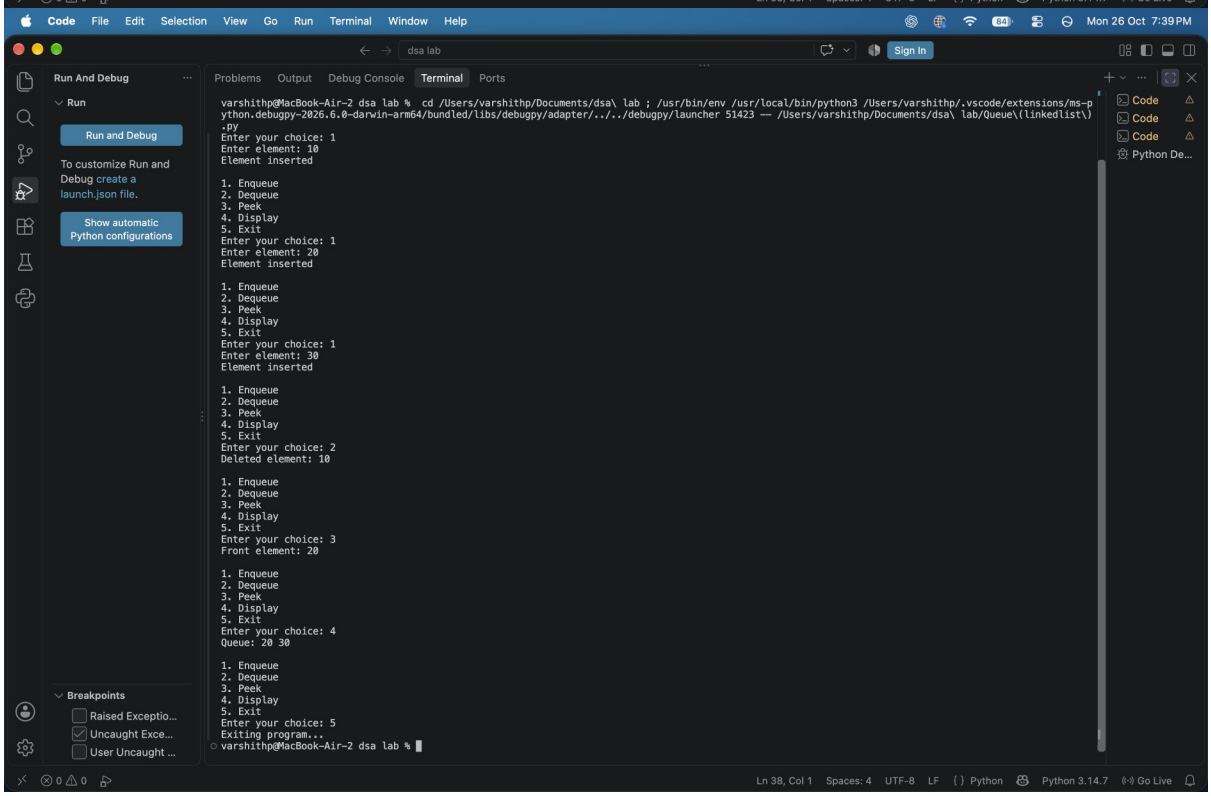
    elif choice == 4:
        q.display()

    elif choice == 5:
        print("Exiting program...")
        break

    else:
        print("Invalid choice")
```

Output:-

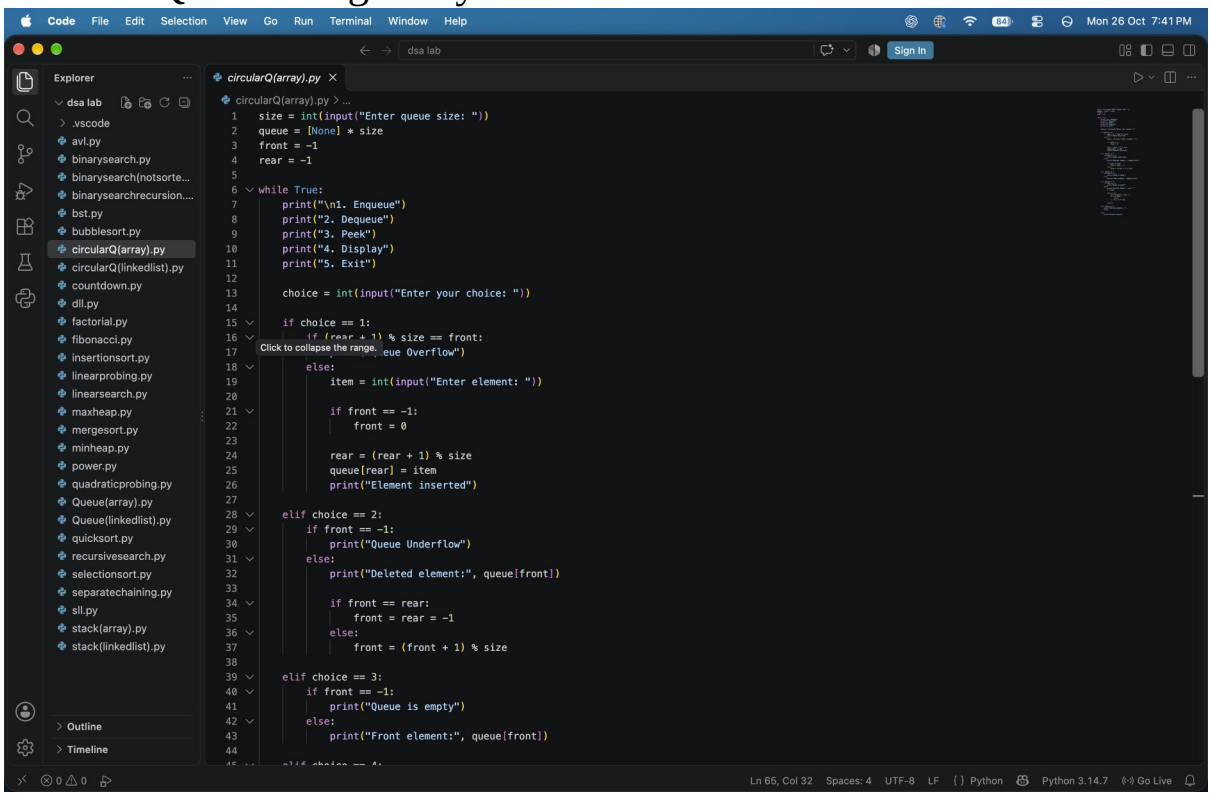

Code File Edit Selection View Go Run Terminal Window Help
Mon 26 Oct 7:38 PM
dsa lab
Sign in
Run And Debug
Run
Run and Debug
To customize Run and Debug create a launch.json file.
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Breakpoints
Raised Exceptio...
Uncaught Exce...
User Uncaught ...
Problems Output Debug Console Terminal Ports
varshithp@MacBook-Air-2 dsa lab % cd /Users/varshithp/Documents/dsa\ lab ; /usr/bin/env /usr/local/bin/python3 /Users/varshithp/.vscode/extensions/ms-python.debugpy-2026.6.0-darwin-arm64/bundled/libs/debugpy/adapter/../../debugpy/launcher 51423 -- /Users/varshithp/Documents/dsa\ lab/Queue\(linkedlist\).py
1. Enqueue
2. Dequeue
3. Peek
4. Display
5. Exit
Enter your choice: 1
Enter element: 10
Element inserted
1. Enqueue
2. Dequeue
3. Peek
4. Display
5. Exit
Enter your choice: 1
Enter element: 20
Element inserted
1. Enqueue
2. Dequeue
3. Peek
4. Display
5. Exit
Enter your choice: 1
Enter element: 30
Element inserted
1. Enqueue
2. Dequeue
3. Peek
4. Display
5. Exit
Enter your choice: 2
Deleted element: 10
1. Enqueue
2. Dequeue
3. Peek
4. Display
5. Exit
Enter your choice: 3
Front element: 20
1. Enqueue
2. Dequeue
3. Peek
4. Display
5. Exit
Enter your choice: 4
Queue: 20 30
1. Enqueue
2. Dequeue
Code
Code
Code
Python De...
Ln 38, Col 1 Spaces: 4 UTF-8 LF { } Python Python 3.14.7 Go Live


Code File Edit Selection View Go Run Terminal Window Help
Mon 26 Oct 7:39 PM
dsa lab
Sign in
Run And Debug
Run
Run and Debug
To customize Run and Debug create a launch.json file.
Show automatic Python configurations
Breakpoints
Raised Exceptio...
Uncaught Exce...
User Uncaught ...
Problems Output Debug Console Terminal Ports
varshithp@MacBook-Air-2 dsa lab % cd /Users/varshithp/Documents/dsa\ lab ; /usr/bin/env /usr/local/bin/python3 /Users/varshithp/.vscode/extensions/ms-python.debugpy-2026.6.0-darwin-arm64/bundled/libs/debugpy/adapter/../../debugpy/launcher 51423 -- /Users/varshithp/Documents/dsa\ lab/Queue\(linkedlist\).py
Enter your choice: 1
Enter element: 10
Element inserted
1. Enqueue
2. Dequeue
3. Peek
4. Display
5. Exit
Enter your choice: 1
Enter element: 20
Element inserted
1. Enqueue
2. Dequeue
3. Peek
4. Display
5. Exit
Enter your choice: 1
Enter element: 30
Element inserted
1. Enqueue
2. Dequeue
3. Peek
4. Display
5. Exit
Enter your choice: 2
Deleted element: 10
1. Enqueue
2. Dequeue
3. Peek
4. Display
5. Exit
Enter your choice: 3
Front element: 20
1. Enqueue
2. Dequeue
3. Peek
4. Display
5. Exit
Enter your choice: 4
Queue: 20 30
1. Enqueue
2. Dequeue
3. Peek
4. Display
5. Exit
Enter your choice: 5
Exiting program...
varshithp@MacBook-Air-2 dsa lab %
Code
Code
Code
Python De...
Ln 38, Col 1 Spaces: 4 UTF-8 LF { } Python Python 3.14.7 Go Live

8(B). Write a python program to implement Circular Queue and its operations such as enqueue, dequeue, peek, display with a menu driven approach. Implement using array, and linked list data structures.

Code:-

Circular Queue using Array



```
size = int(input("Enter queue size: "))
queue = [None] * size
front = -1
rear = -1

while True:
    print("\n1. Enqueue")
    print("2. Dequeue")
    print("3. Peek")
    print("4. Display")
    print("5. Exit")

    choice = int(input("Enter your choice: "))

    if choice == 1:
        if (rear + 1) % size == front:
            print("Queue Overflow")
        else:
            item = int(input("Enter element: "))

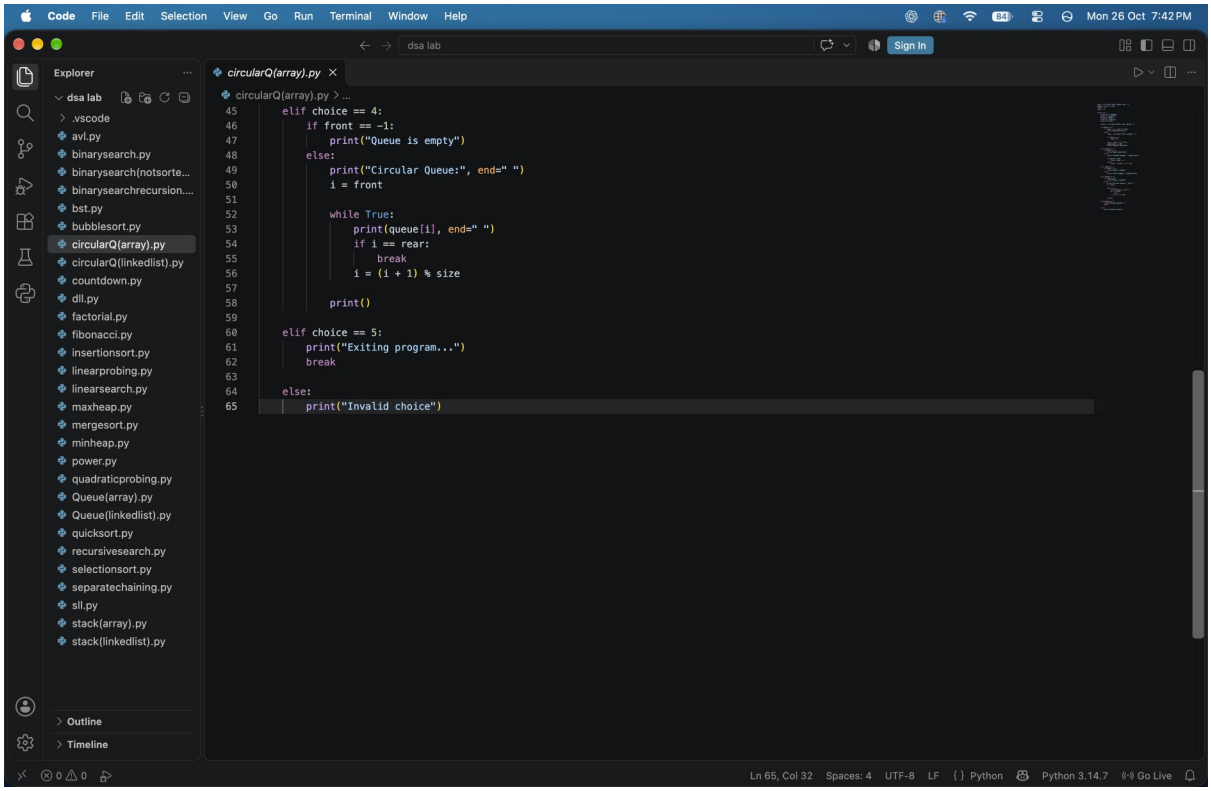
            if front == -1:
                front = 0

            rear = (rear + 1) % size
            queue[rear] = item
            print("Element inserted")

    elif choice == 2:
        if front == -1:
            print("Queue Underflow")
        else:
            print("Deleted element:", queue[front])

            if front == rear:
                front = rear = -1
            else:
                front = (front + 1) % size

    elif choice == 3:
        if front == -1:
            print("Queue is empty")
        else:
            print("Front element:", queue[front])
```



```
elif choice == 4:
    if front == -1:
        print("Queue is empty")
    else:
        print("Circular Queue:", end=" ")
        i = front

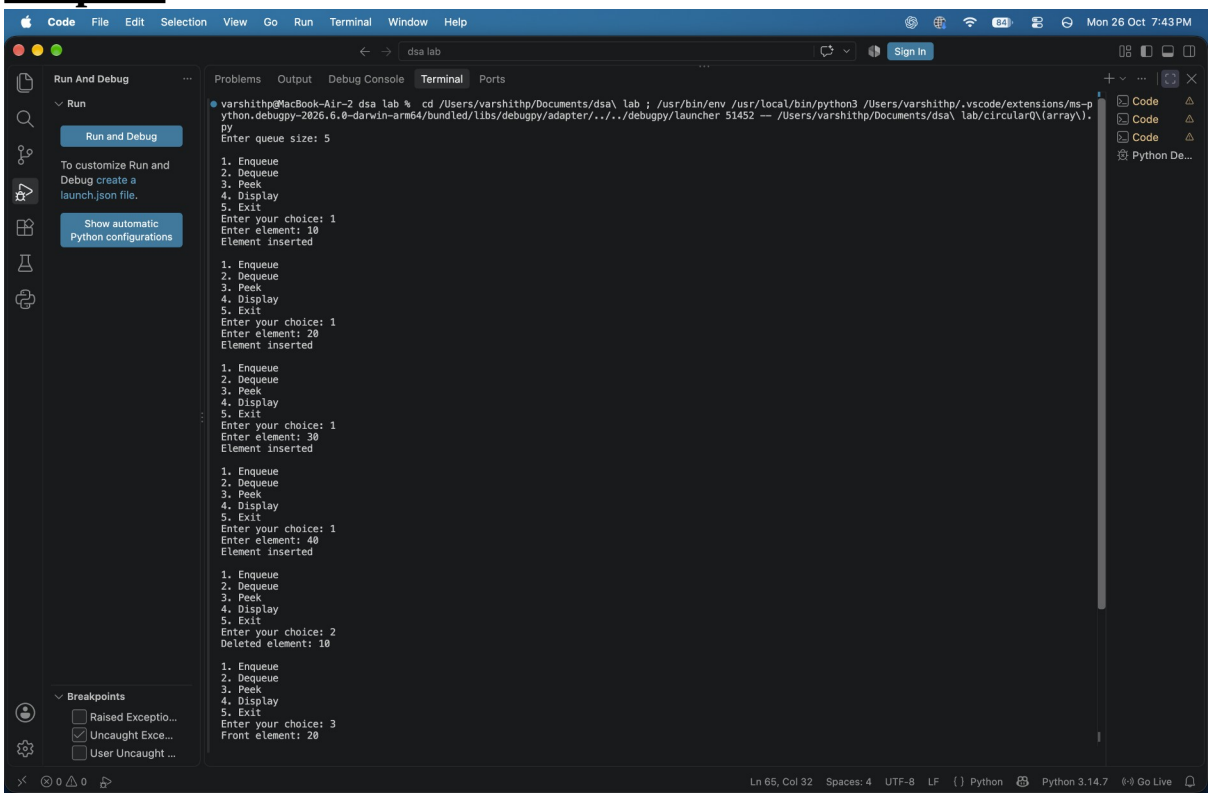
        while True:
            print(queue[i], end=" ")
            if i == rear:
                break
            i = (i + 1) % size

        print()

elif choice == 5:
    print("Exiting program...")
    break

else:
    print("Invalid choice")
```

Output:-



```
varshithp@MacBook-Air-2 dsa lab % cd /Users/varshithp/Documents/dsa\ lab ; /usr/bin/env /usr/local/bin/python3 /Users/varshithp/.vscode/extensions/ms-python.debugpy-2026.6.0-darwin-arm64/bundled/libs/debugpy/adapter/../../debugpy/launcher 51452 -- /Users/varshithp/Documents/dsa\ lab/circularQ\(array\).py
Enter queue size: 5

1. Enqueue
2. Dequeue
3. Peek
4. Display
5. Exit
Enter your choice: 1
Enter element: 10
Element inserted

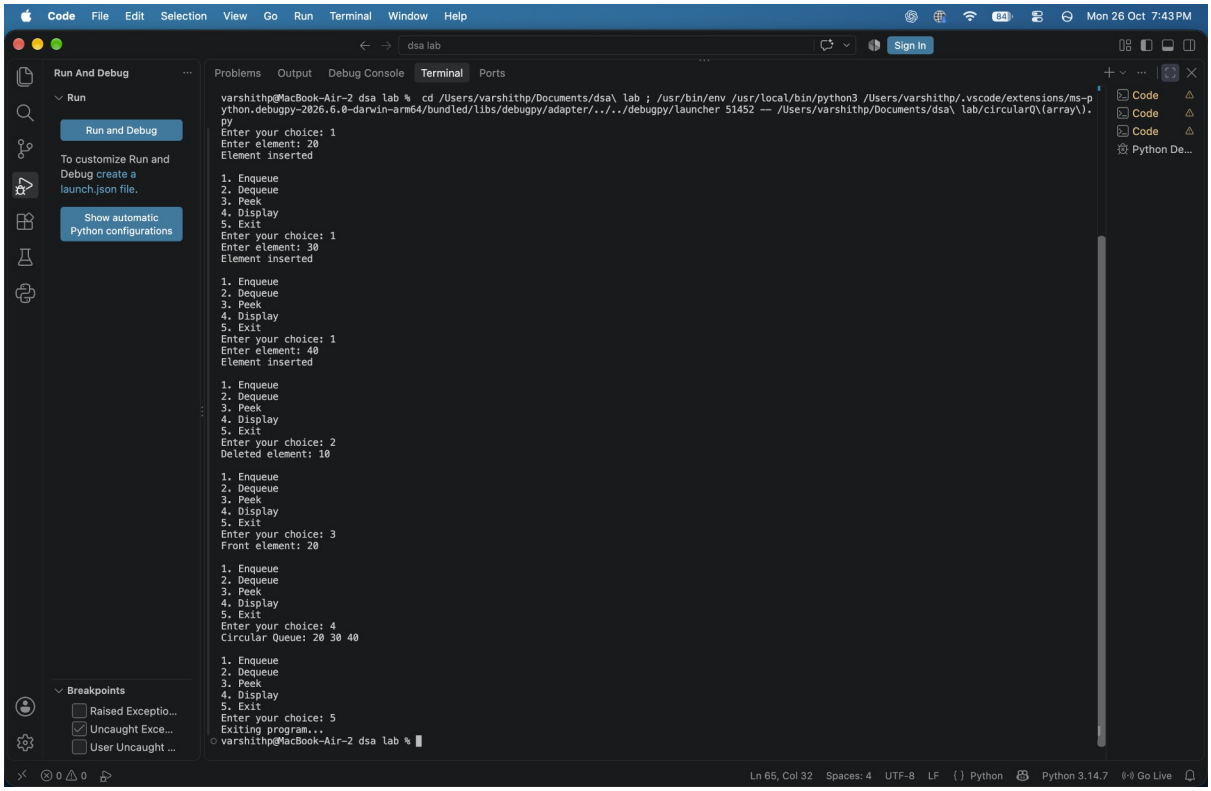
1. Enqueue
2. Dequeue
3. Peek
4. Display
5. Exit
Enter your choice: 1
Enter element: 20
Element inserted

1. Enqueue
2. Dequeue
3. Peek
4. Display
5. Exit
Enter your choice: 1
Enter element: 30
Element inserted

1. Enqueue
2. Dequeue
3. Peek
4. Display
5. Exit
Enter your choice: 1
Enter element: 40
Element inserted

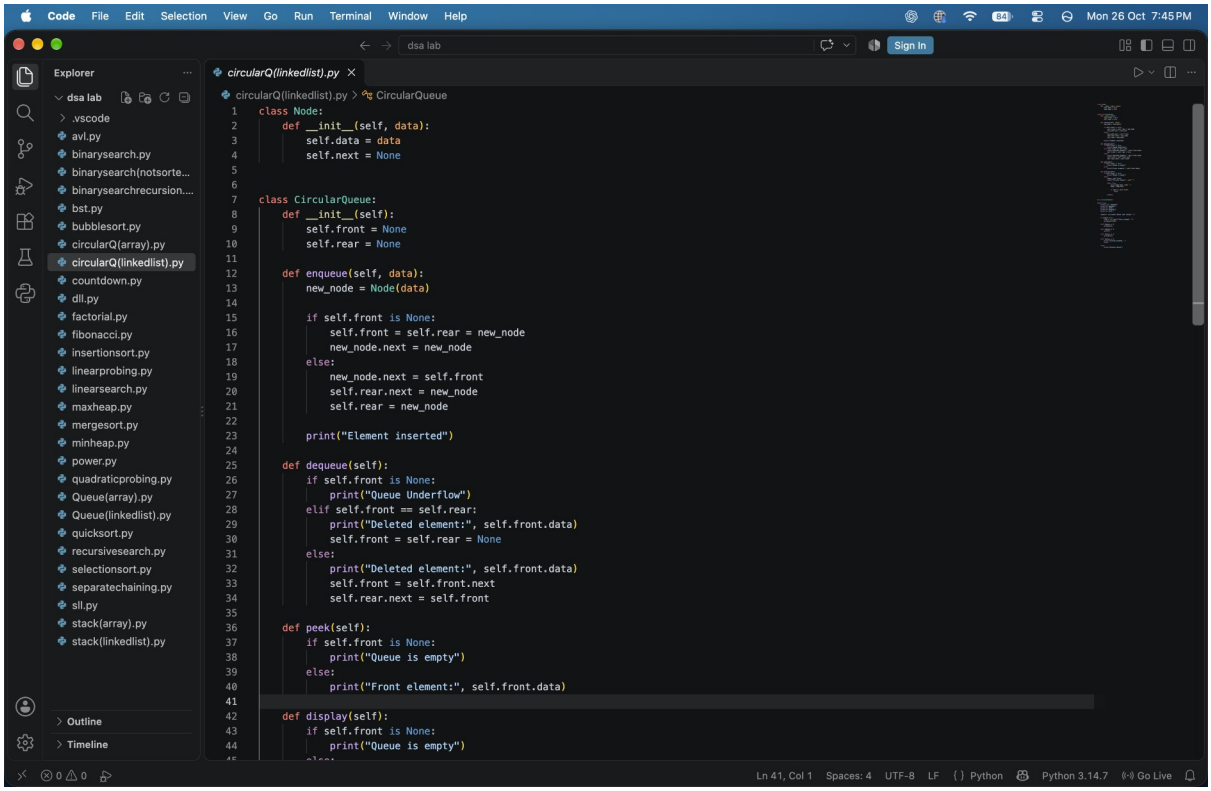
1. Enqueue
2. Dequeue
3. Peek
4. Display
5. Exit
Enter your choice: 2
Deleted element: 10

1. Enqueue
2. Dequeue
3. Peek
4. Display
5. Exit
Enter your choice: 3
Front element: 20
```



Circular Queue using Linked list

Code:-



```python
class Node:
    def __init__(self, data):
        self.data = data
        self.next = None


class CircularQueue:
    def __init__(self):
        self.front = None
        self.rear = None

    def enqueue(self, data):
        new_node = Node(data)

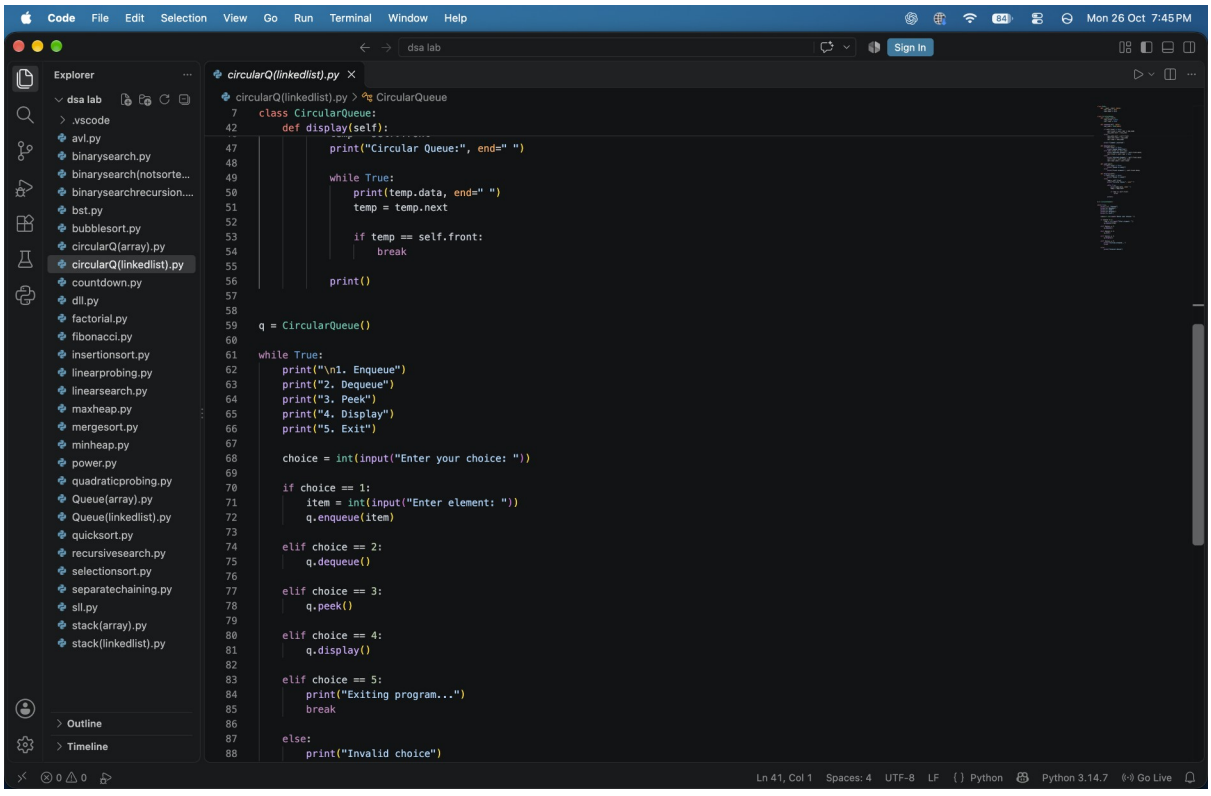
        if self.front is None:
            self.front = self.rear = new_node
            new_node.next = new_node
        else:
            new_node.next = self.front
            self.rear.next = new_node
            self.rear = new_node

        print("Element inserted")

    def dequeue(self):
        if self.front is None:
            print("Queue Underflow")
        elif self.front == self.rear:
            print("Deleted element:", self.front.data)
            self.front = self.rear = None
        else:
            print("Deleted element:", self.front.data)
            self.front = self.front.next
            self.rear.next = self.front

    def peek(self):
        if self.front is None:
            print("Queue is empty")
        else:
            print("Front element:", self.front.data)

    def display(self):
        if self.front is None:
            print("Queue is empty")
```



```python
class CircularQueue:
    def display(self):
            print("Circular Queue:", end=" ")

            while True:
                print(temp.data, end=" ")
                temp = temp.next

                if temp == self.front:
                    break

            print()


q = CircularQueue()

while True:
    print("\n1. Enqueue")
    print("2. Dequeue")
    print("3. Peek")
    print("4. Display")
    print("5. Exit")

    choice = int(input("Enter your choice: "))

    if choice == 1:
        item = int(input("Enter element: "))
        q.enqueue(item)

    elif choice == 2:
        q.dequeue()

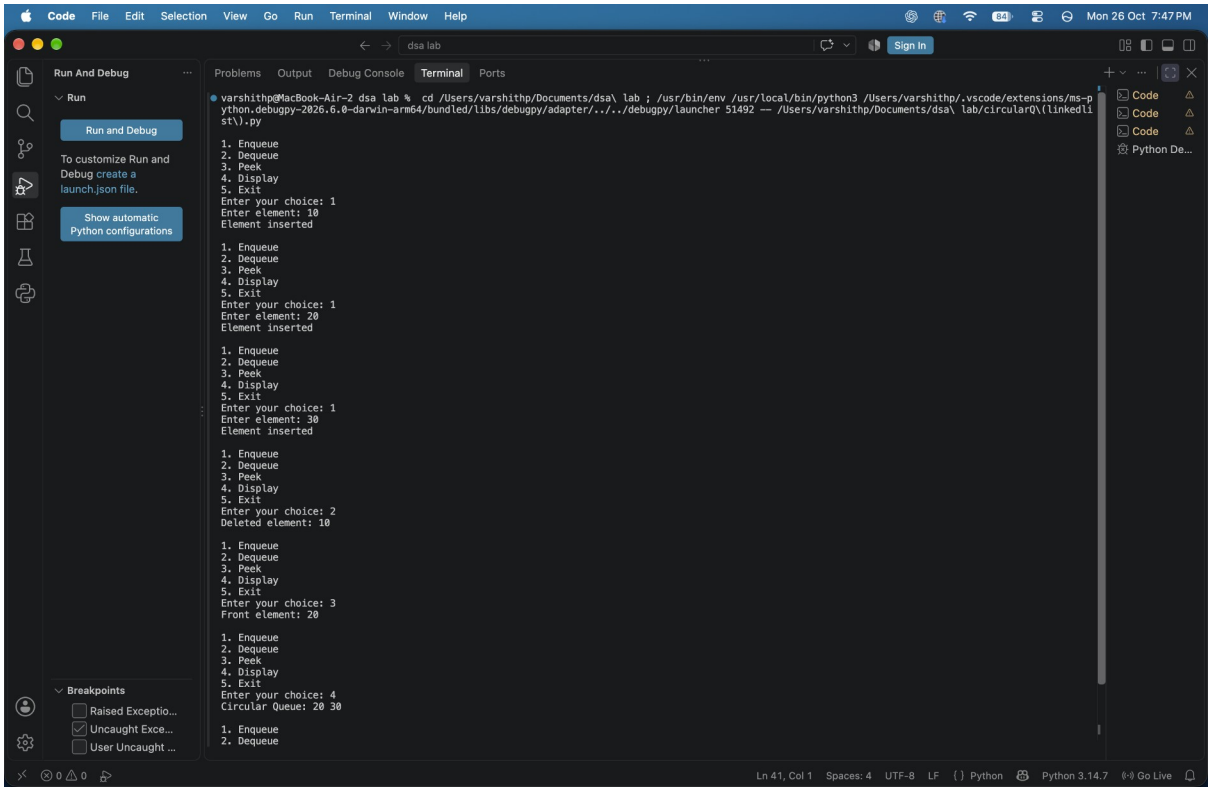
    elif choice == 3:
        q.peek()

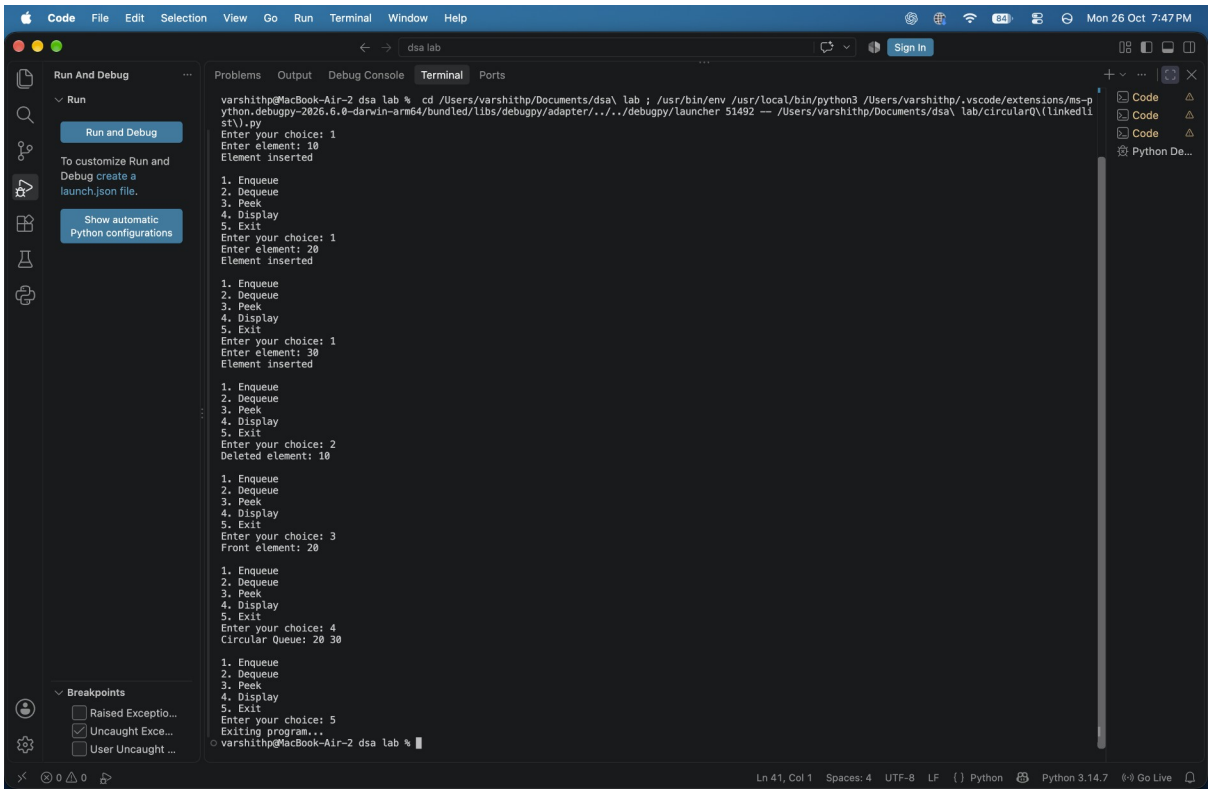
    elif choice == 4:
        q.display()

    elif choice == 5:
        print("Exiting program...")
        break

    else:
        print("Invalid choice")
```

Output:-


Code File Edit Selection View Go Run Terminal Window Help
Mon 26 Oct 7:47 PM
dsa lab
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Raised Exceptio...
Uncaught Exce...
User Uncaught ...
Problems Output Debug Console Terminal Ports
varshithp@MacBook-Air-2 dsa lab % cd /Users/varshithp/Documents/dsa\ lab ; /usr/bin/env /usr/local/bin/python3 /Users/varshithp/.vscode/extensions/ms-python.debugpy-2026.6.0-darwin-arm64/bundled/libs/debugpy/adapter/../../debugpy/launcher 51492 -- /Users/varshithp/Documents/dsa\ lab/circularQ\(linkedlist\).py
1. Enqueue
2. Dequeue
3. Peek
4. Display
5. Exit
Enter your choice: 1
Enter element: 10
Element inserted
1. Enqueue
2. Dequeue
3. Peek
4. Display
5. Exit
Enter your choice: 1
Enter element: 20
Element inserted
1. Enqueue
2. Dequeue
3. Peek
4. Display
5. Exit
Enter your choice: 1
Enter element: 30
Element inserted
1. Enqueue
2. Dequeue
3. Peek
4. Display
5. Exit
Enter your choice: 2
Deleted element: 10
1. Enqueue
2. Dequeue
3. Peek
4. Display
5. Exit
Enter your choice: 3
Front element: 20
1. Enqueue
2. Dequeue
3. Peek
4. Display
5. Exit
Enter your choice: 4
Circular Queue: 20 30
1. Enqueue
2. Dequeue
Code
Code
Code
Python De...
Ln 41, Col 1 Spaces: 4 UTF-8 LF { } Python Python 3.14.7 Go Live

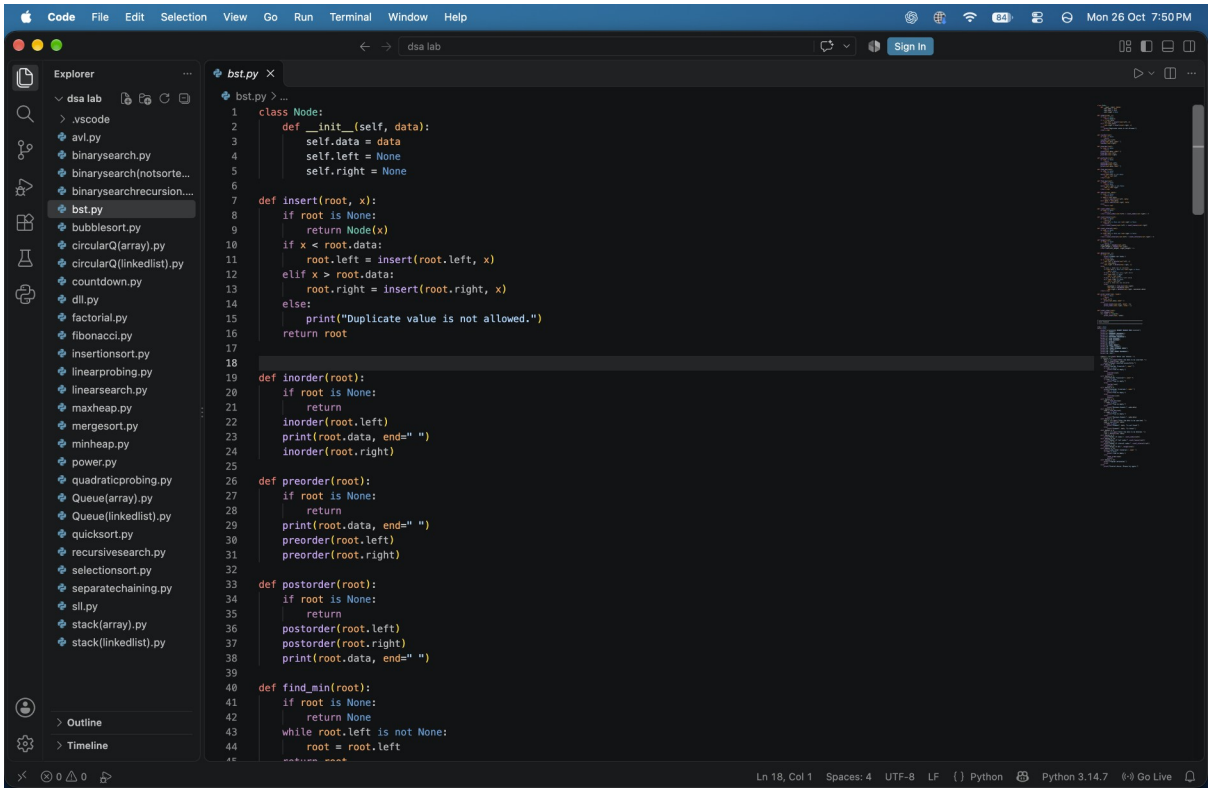

Code File Edit Selection View Go Run Terminal Window Help
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Problems Output Debug Console Terminal Ports
varshithp@MacBook-Air-2 dsa lab % cd /Users/varshithp/Documents/dsa\ lab ; /usr/bin/env /usr/local/bin/python3 /Users/varshithp/.vscode/extensions/ms-python.debugpy-2026.6.0-darwin-arm64/bundled/libs/debugpy/adapter/../../debugpy/launcher 51492 -- /Users/varshithp/Documents/dsa\ lab/circularQ\(linkedlist\).py
Enter your choice: 1
Enter element: 10
Element inserted
1. Enqueue
2. Dequeue
3. Peek
4. Display
5. Exit
Enter your choice: 1
Enter element: 20
Element inserted
1. Enqueue
2. Dequeue
3. Peek
4. Display
5. Exit
Enter your choice: 1
Enter element: 30
Element inserted
1. Enqueue
2. Dequeue
3. Peek
4. Display
5. Exit
Enter your choice: 2
Deleted element: 10
1. Enqueue
2. Dequeue
3. Peek
4. Display
5. Exit
Enter your choice: 3
Front element: 20
1. Enqueue
2. Dequeue
3. Peek
4. Display
5. Exit
Enter your choice: 4
Circular Queue: 20 30
1. Enqueue
2. Dequeue
3. Peek
4. Display
5. Exit
Enter your choice: 5
Exiting program...
varshithp@MacBook-Air-2 dsa lab %
Code
Code
Code
Python De...
Ln 41, Col 1 Spaces: 4 UTF-8 LF { } Python Python 3.14.7 Go Live

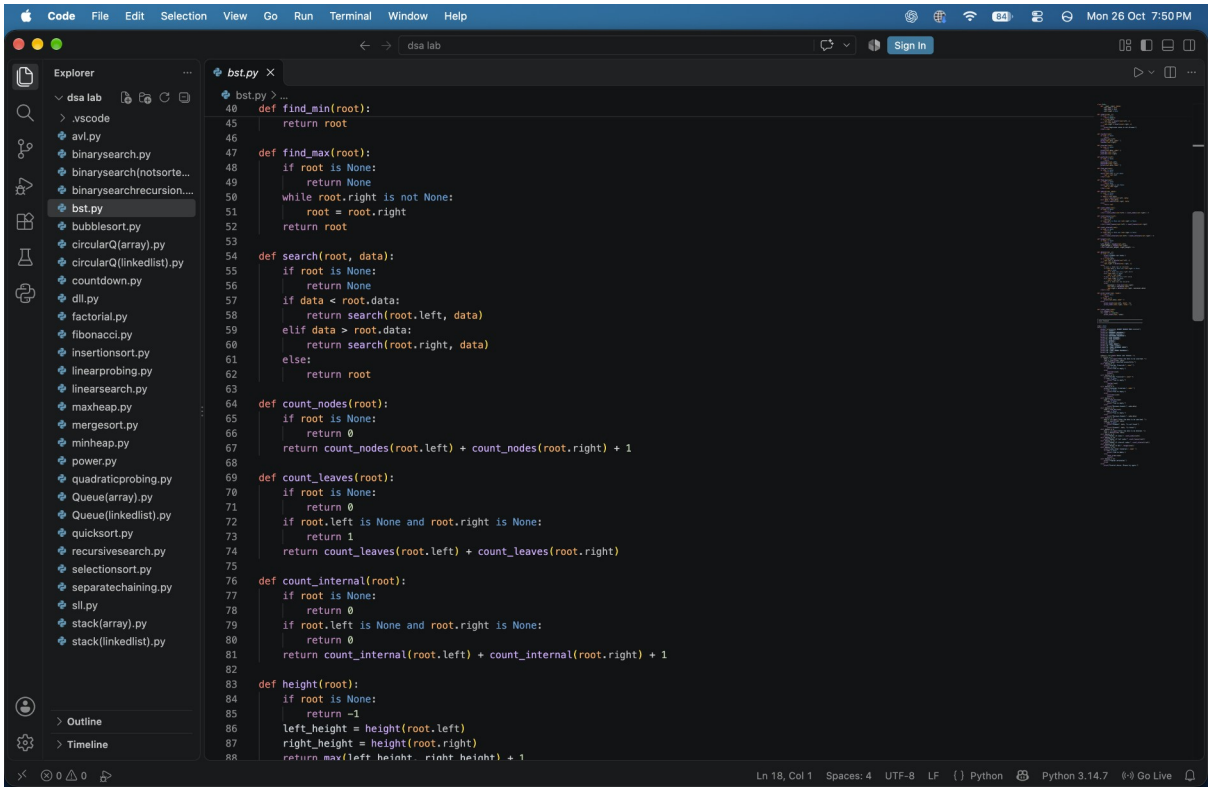
WEEK-9

9. Write a python program to implement a Binary Search Tree (BST) and perform the following operations.

1. Insertion
2. Deletion
3. Tree Traversals – in-order, pre-order and post-order
4. Count Leaf Nodes
5. Count Internal Nodes
6. Height of the Tree

Code:-





```
def find_min(root):
    return root

def find_max(root):
    if root is None:
        return None
    while root.right is not None:
        root = root.right
    return root

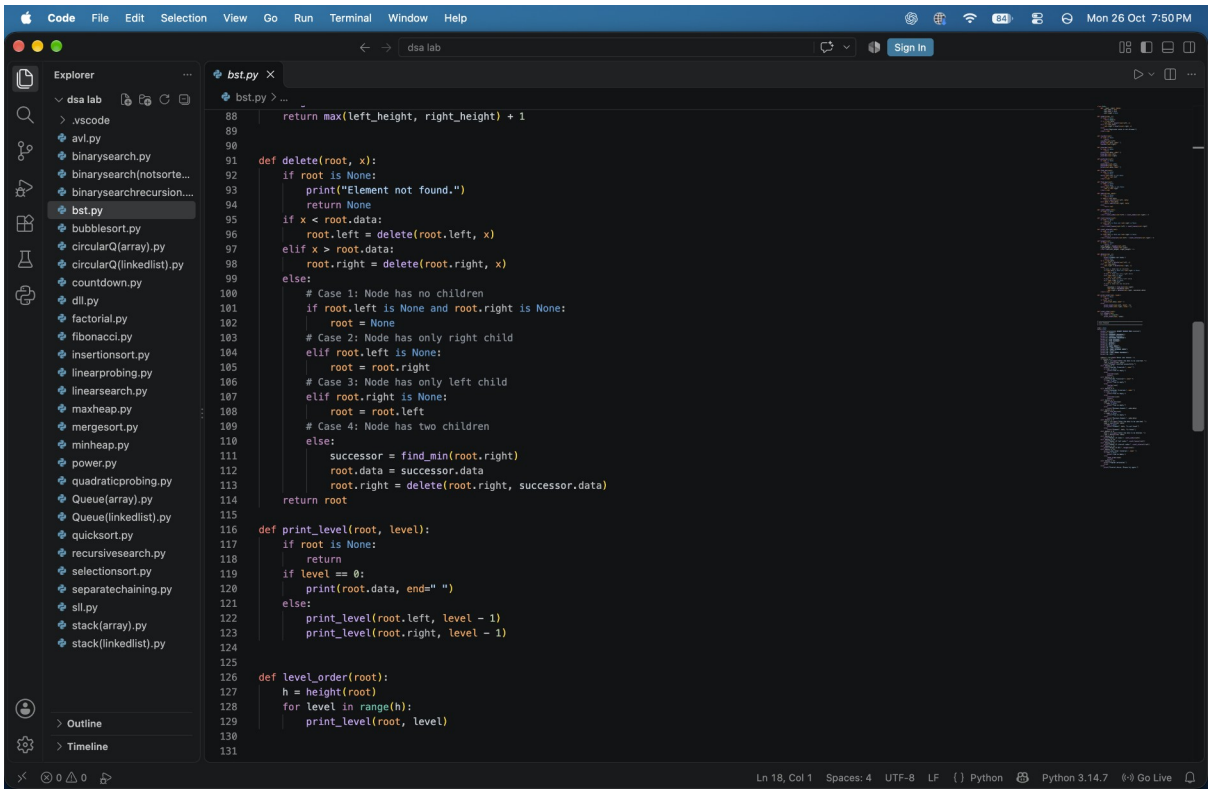
def search(root, data):
    if root is None:
        return None
    if data < root.data:
        return search(root.left, data)
    elif data > root.data:
        return search(root.right, data)
    else:
        return root

def count_nodes(root):
    if root is None:
        return 0
    return count_nodes(root.left) + count_nodes(root.right) + 1

def count_leaves(root):
    if root is None:
        return 0
    if root.left is None and root.right is None:
        return 1
    return count_leaves(root.left) + count_leaves(root.right)

def count_internal(root):
    if root is None:
        return 0
    if root.left is None and root.right is None:
        return 0
    return count_internal(root.left) + count_internal(root.right) + 1

def height(root):
    if root is None:
        return -1
    left_height = height(root.left)
    right_height = height(root.right)
```

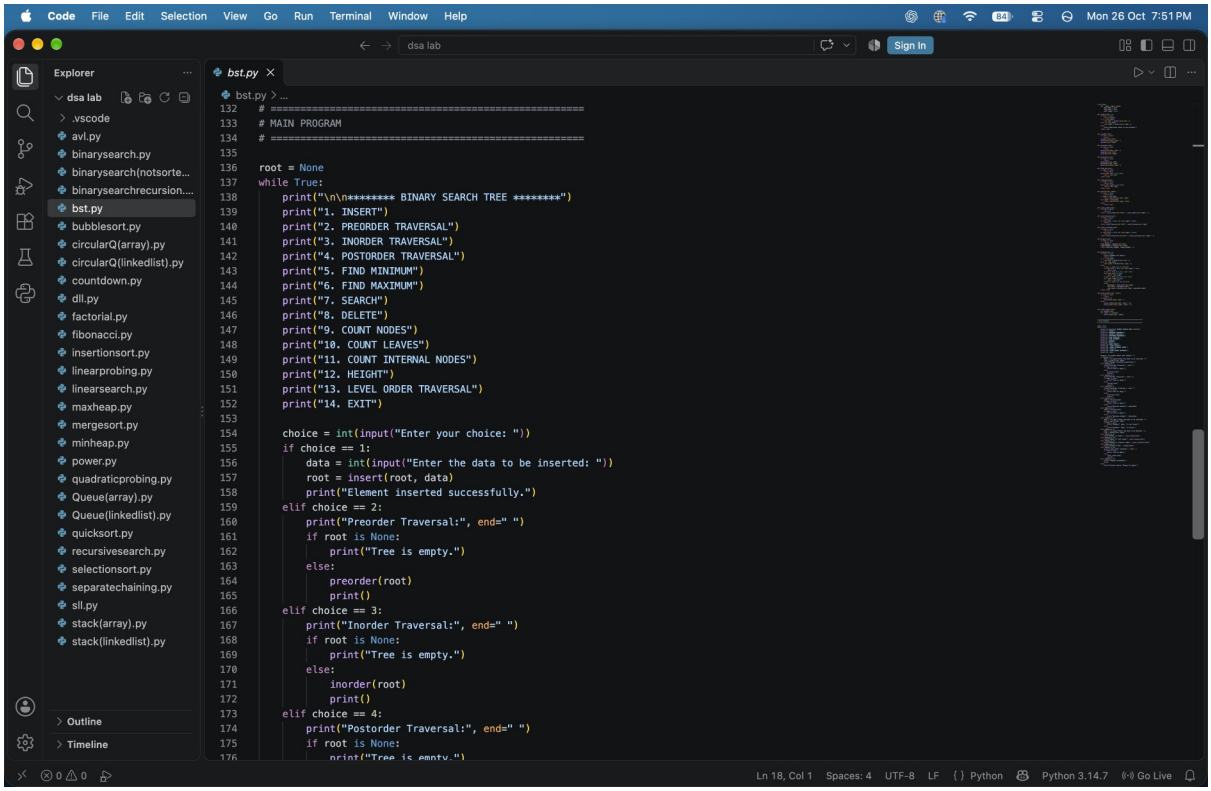


```
    return max(left_height, right_height) + 1


def delete(root, x):
    if root is None:
        print("Element not found.")
        return None
    if x < root.data:
        root.left = delete(root.left, x)
    elif x > root.data:
        root.right = delete(root.right, x)
    else:
        # Case 1: Node has no children
        if root.left is None and root.right is None:
            root = None
        # Case 2: Node has only right child
        elif root.left is None:
            root = root.right
        # Case 3: Node has only left child
        elif root.right is None:
            root = root.left
        # Case 4: Node has two children
        else:
            successor = find_min(root.right)
            root.data = successor.data
            root.right = delete(root.right, successor.data)
    return root

def print_level(root, level):
    if root is None:
        return
    if level == 0:
        print(root.data, end=" ")
    else:
        print_level(root.left, level - 1)
        print_level(root.right, level - 1)

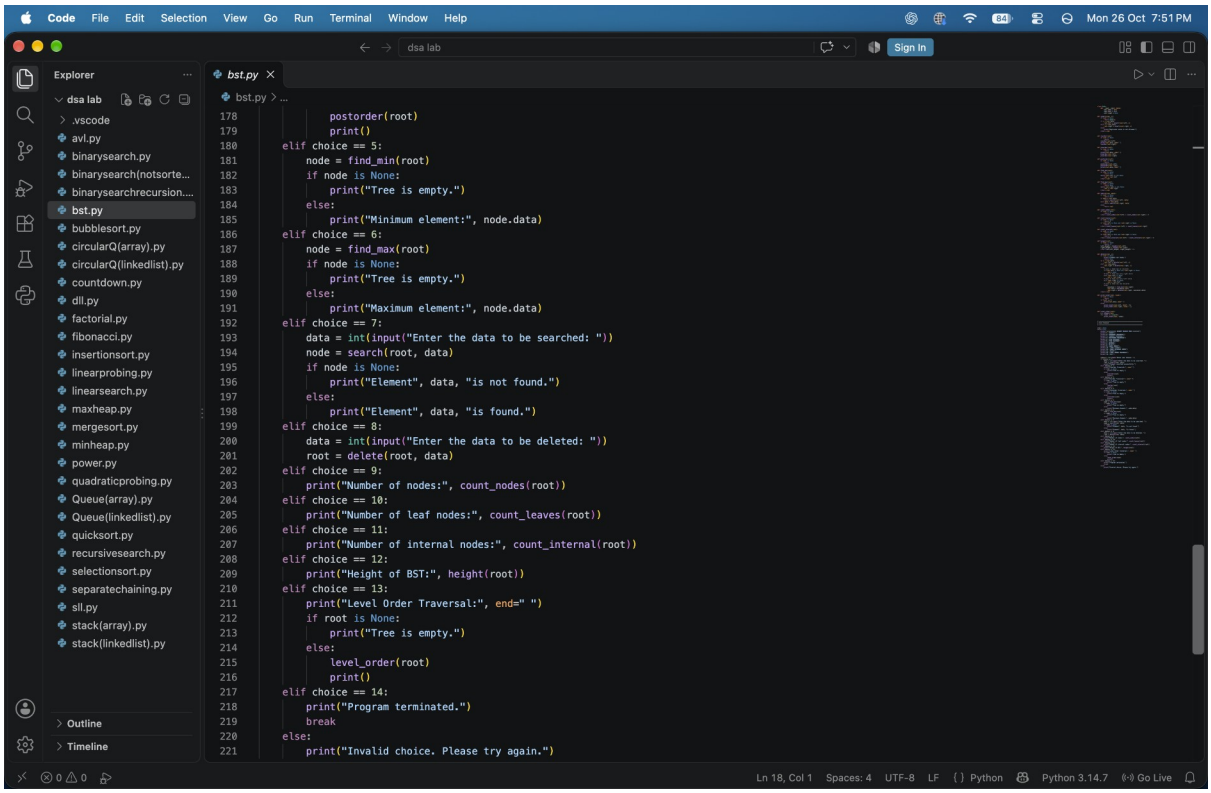

def level_order(root):
    h = height(root)
    for level in range(h):
        print_level(root, level)
```



```
# ================================================
# MAIN PROGRAM
# ================================================

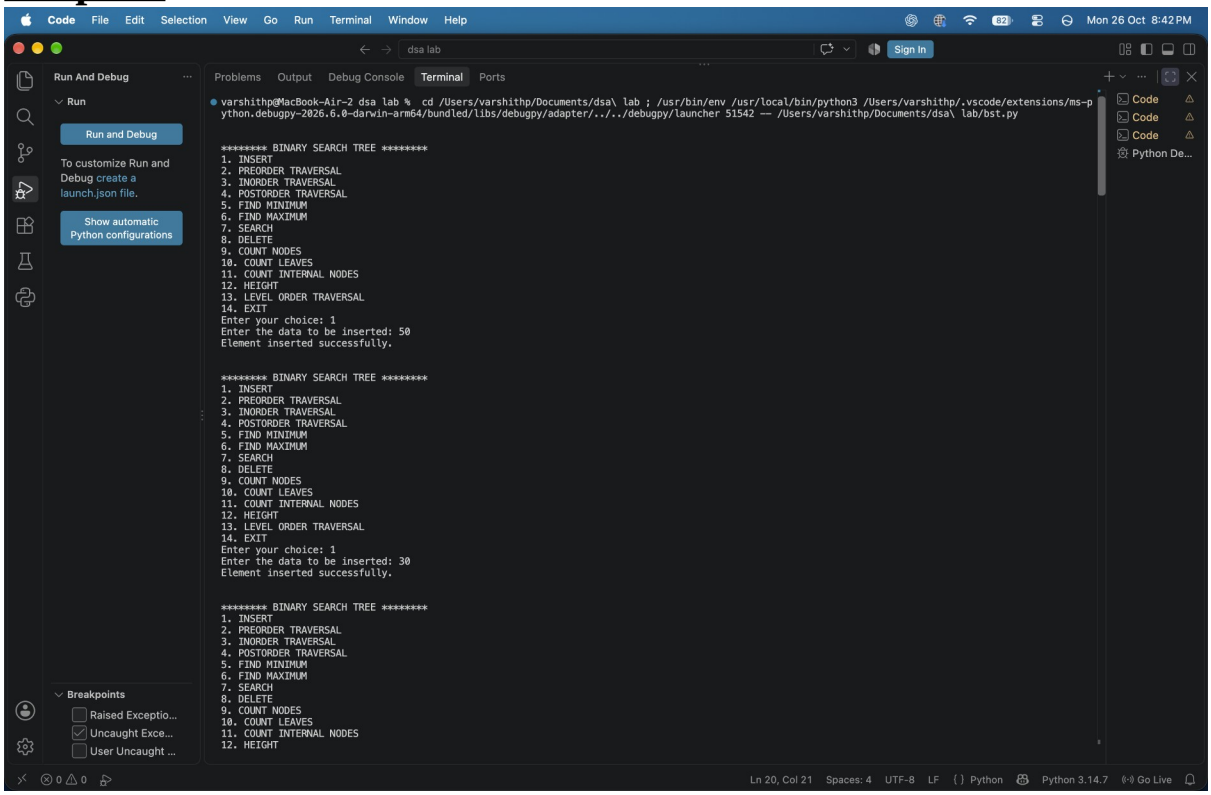
root = None
while True:
    print("\n\n******** BINARY SEARCH TREE ********")
    print("1. INSERT")
    print("2. PREORDER TRAVERSAL")
    print("3. INORDER TRAVERSAL")
    print("4. POSTORDER TRAVERSAL")
    print("5. FIND MINIMUM")
    print("6. FIND MAXIMUM")
    print("7. SEARCH")
    print("8. DELETE")
    print("9. COUNT NODES")
    print("10. COUNT LEAVES")
    print("11. COUNT INTERNAL NODES")
    print("12. HEIGHT")
    print("13. LEVEL ORDER TRAVERSAL")
    print("14. EXIT")

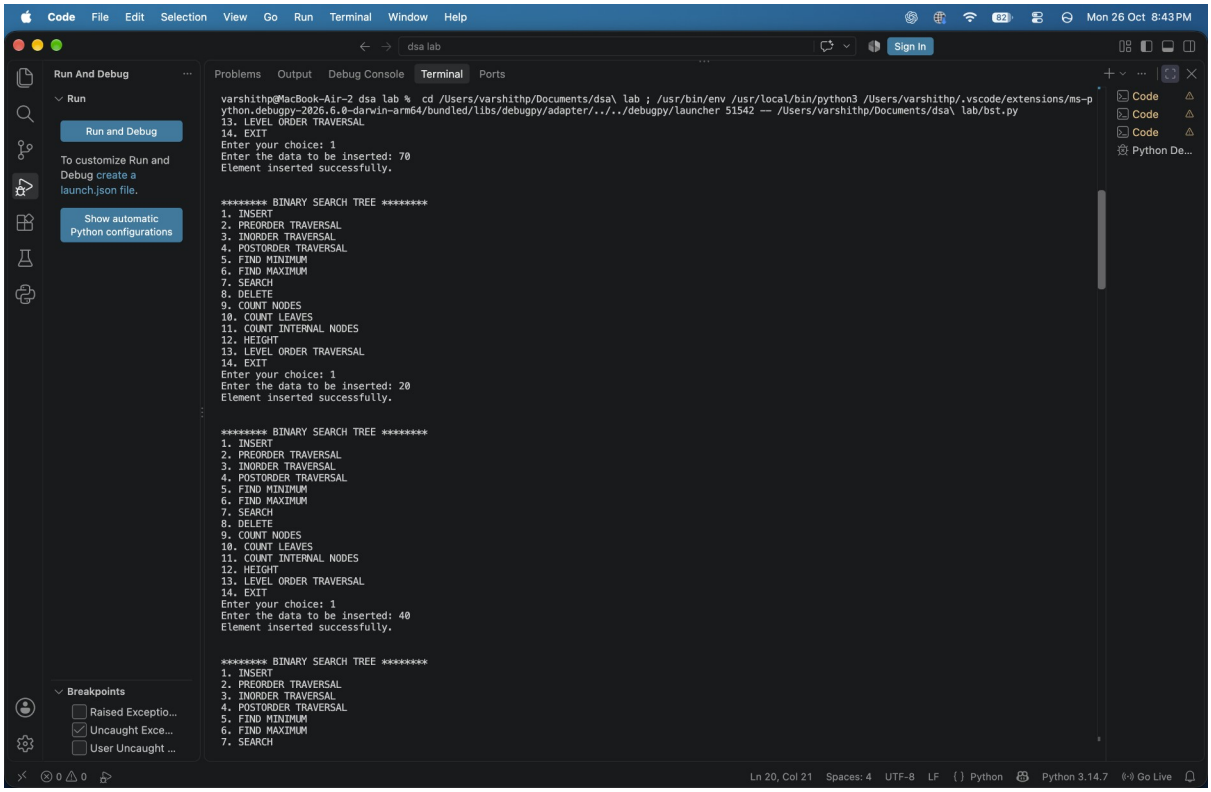
    choice = int(input("Enter your choice: "))
    if choice == 1:
        data = int(input("Enter the data to be inserted: "))
        root = insert(root, data)
        print("Element inserted successfully.")
    elif choice == 2:
        print("Preorder Traversal:", end=" ")
        if root is None:
            print("Tree is empty.")
        else:
            preorder(root)
            print()
    elif choice == 3:
        print("Inorder Traversal:", end=" ")
        if root is None:
            print("Tree is empty.")
        else:
            inorder(root)
            print()
    elif choice == 4:
        print("Postorder Traversal:", end=" ")
        if root is None:
```

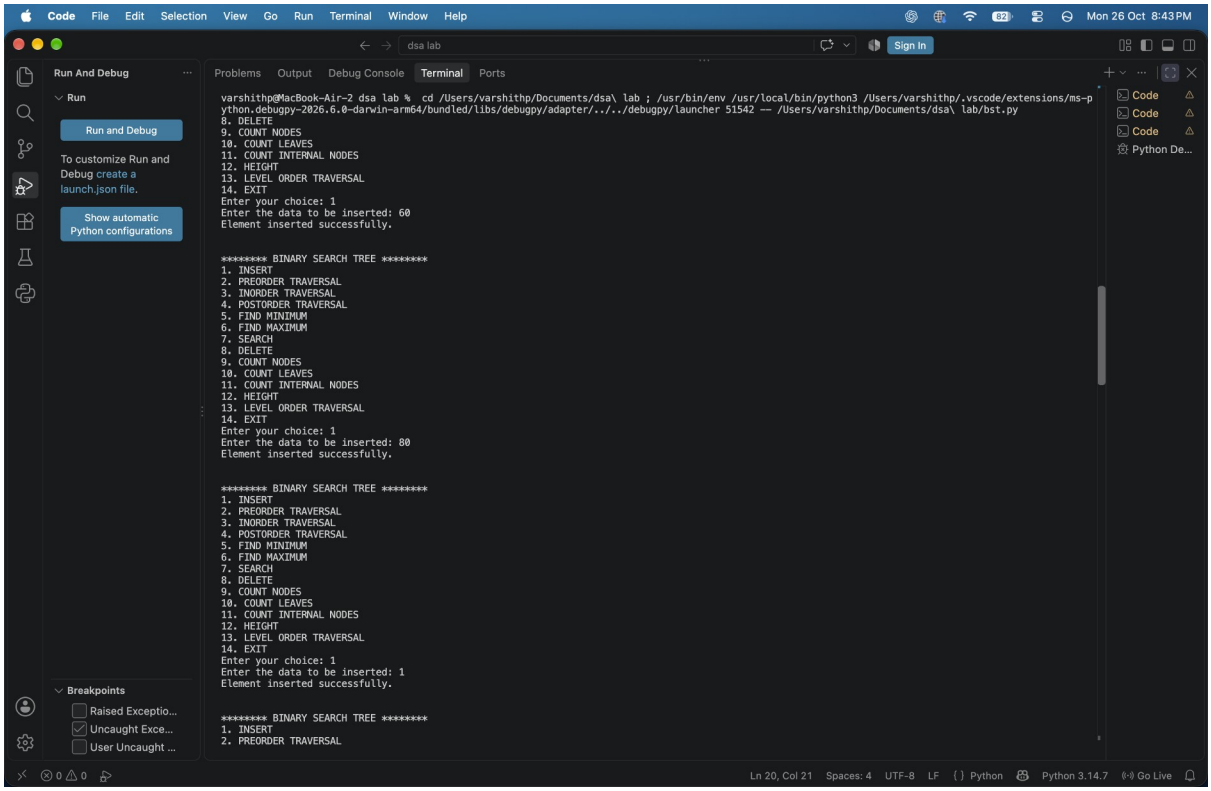


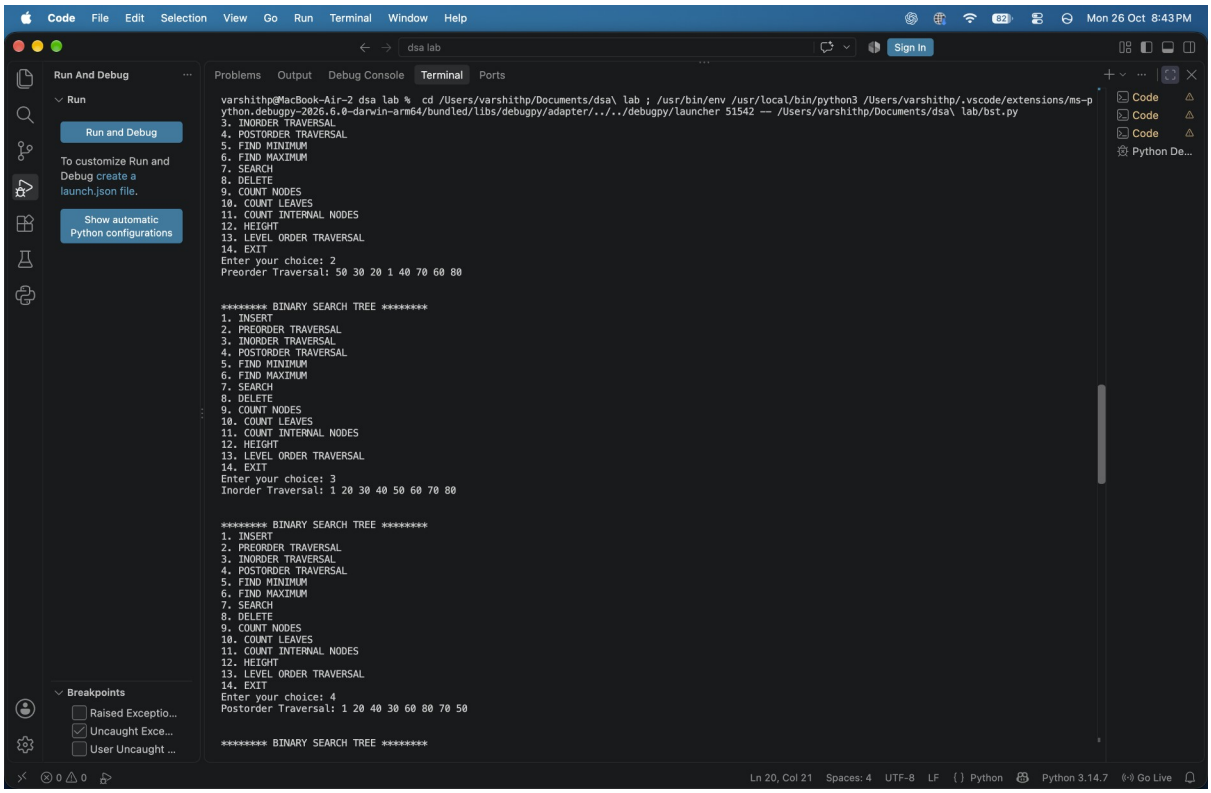
```
            postorder(root)
            print()
    elif choice == 5:
        node = find_min(root)
        if node is None:
            print("Tree is empty.")
        else:
            print("Minimum element:", node.data)
    elif choice == 6:
        node = find_max(root)
        if node is None:
            print("Tree is empty.")
        else:
            print("Maximum element:", node.data)
    elif choice == 7:
        data = int(input("Enter the data to be searched: "))
        node = search(root, data)
        if node is None:
            print("Element", data, "is not found.")
        else:
            print("Element", data, "is found.")
    elif choice == 8:
        data = int(input("Enter the data to be deleted: "))
        root = delete(root, data)
    elif choice == 9:
        print("Number of nodes:", count_nodes(root))
    elif choice == 10:
        print("Number of leaf nodes:", count_leaves(root))
    elif choice == 11:
        print("Number of internal nodes:", count_internal(root))
    elif choice == 12:
        print("Height of BST:", height(root))
    elif choice == 13:
        print("Level Order Traversal:", end=" ")
        if root is None:
            print("Tree is empty.")
        else:
            level_order(root)
            print()
    elif choice == 14:
        print("Program terminated.")
        break
    else:
        print("Invalid choice. Please try again.")
```

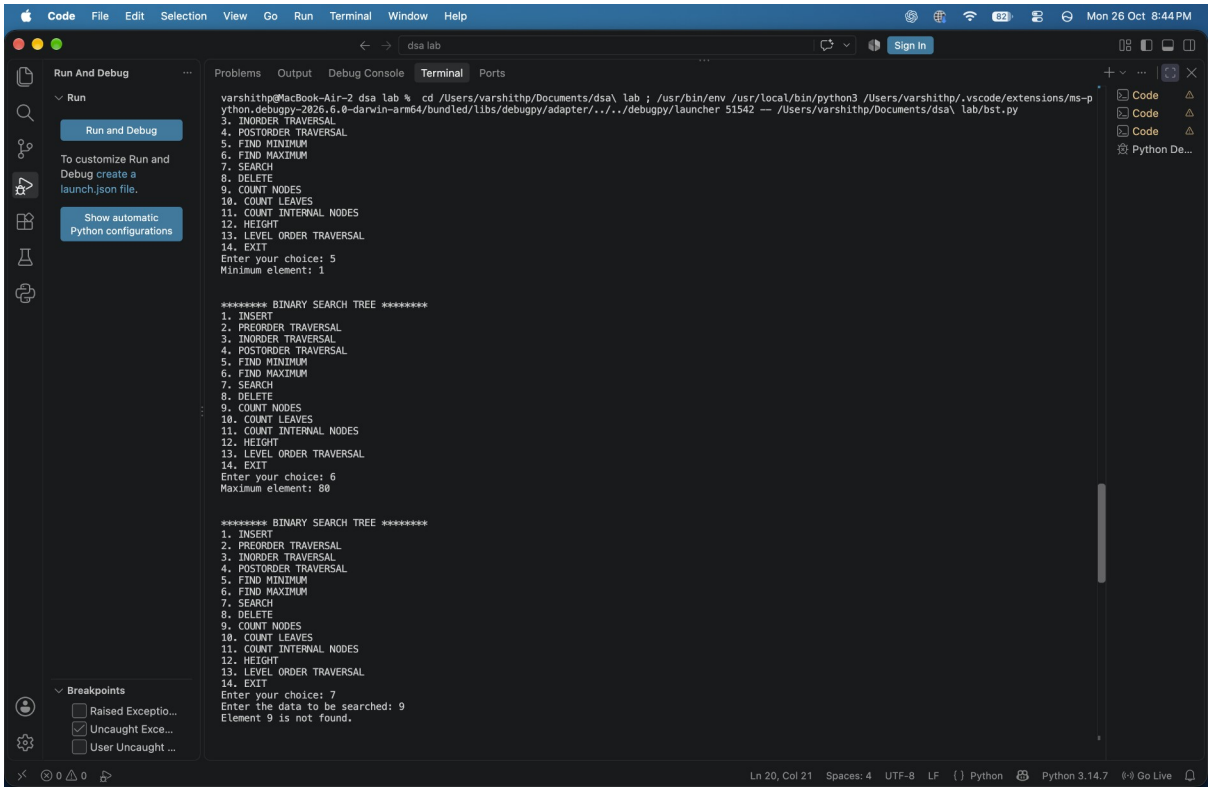
Output:-

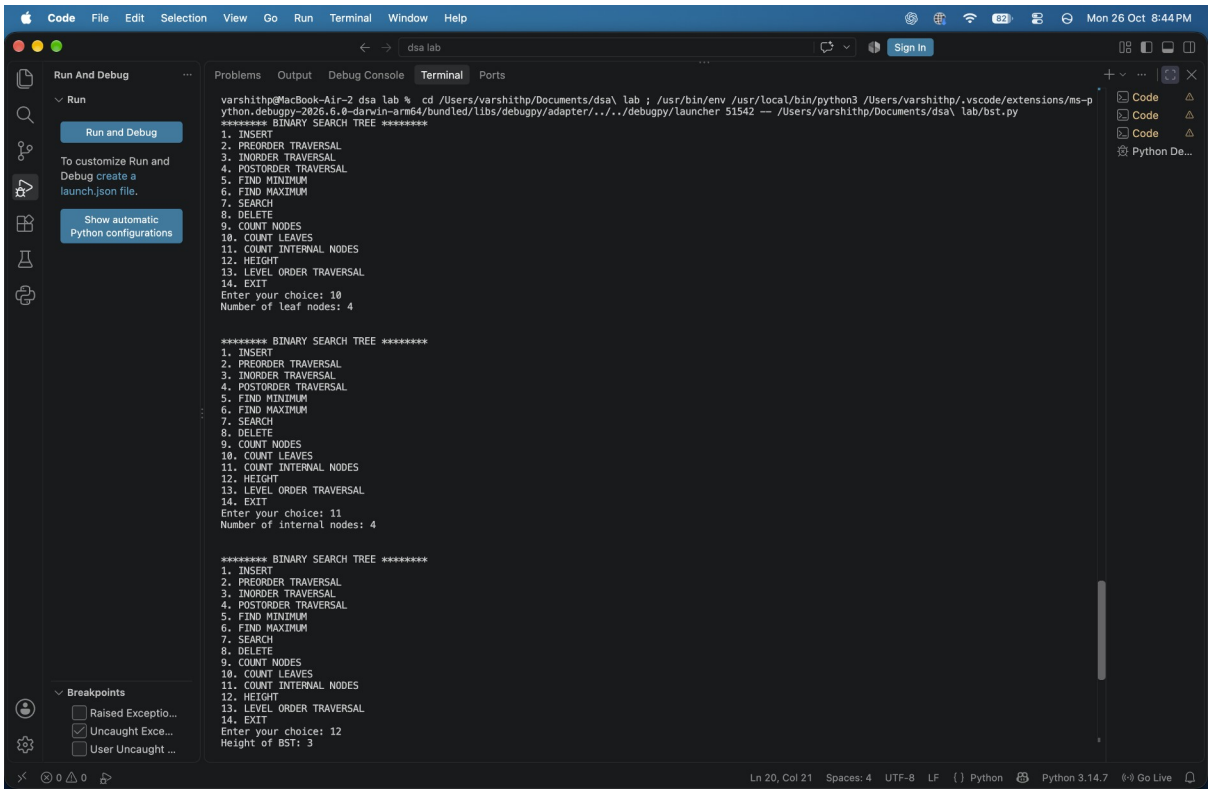

varshithp@MacBook-Air-2 dsa lab % cd /Users/varshithp/Documents/dsa\ lab ; /usr/bin/env /usr/local/bin/python3 /Users/varshithp/.vscode/extensions/ms-python.debugpy-2026.6.0-darwin-arm64/bundled/libs/debugpy/adapter/../../debugpy/launcher 51542 -- /Users/varshithp/Documents/dsa\ lab/bst.py
******** BINARY SEARCH TREE ********
1. INSERT
2. PREORDER TRAVERSAL
3. INORDER TRAVERSAL
4. POSTORDER TRAVERSAL
5. FIND MINIMUM
6. FIND MAXIMUM
7. SEARCH
8. DELETE
9. COUNT NODES
10. COUNT LEAVES
11. COUNT INTERNAL NODES
12. HEIGHT
13. LEVEL ORDER TRAVERSAL
14. EXIT
Enter your choice: 1
Enter the data to be inserted: 50
Element inserted successfully.
******** BINARY SEARCH TREE ********
1. INSERT
2. PREORDER TRAVERSAL
3. INORDER TRAVERSAL
4. POSTORDER TRAVERSAL
5. FIND MINIMUM
6. FIND MAXIMUM
7. SEARCH
8. DELETE
9. COUNT NODES
10. COUNT LEAVES
11. COUNT INTERNAL NODES
12. HEIGHT
13. LEVEL ORDER TRAVERSAL
14. EXIT
Enter your choice: 1
Enter the data to be inserted: 30
Element inserted successfully.
******** BINARY SEARCH TREE ********
1. INSERT
2. PREORDER TRAVERSAL
3. INORDER TRAVERSAL
4. POSTORDER TRAVERSAL
5. FIND MINIMUM
6. FIND MAXIMUM
7. SEARCH
8. DELETE
9. COUNT NODES
10. COUNT LEAVES
11. COUNT INTERNAL NODES
12. HEIGHT

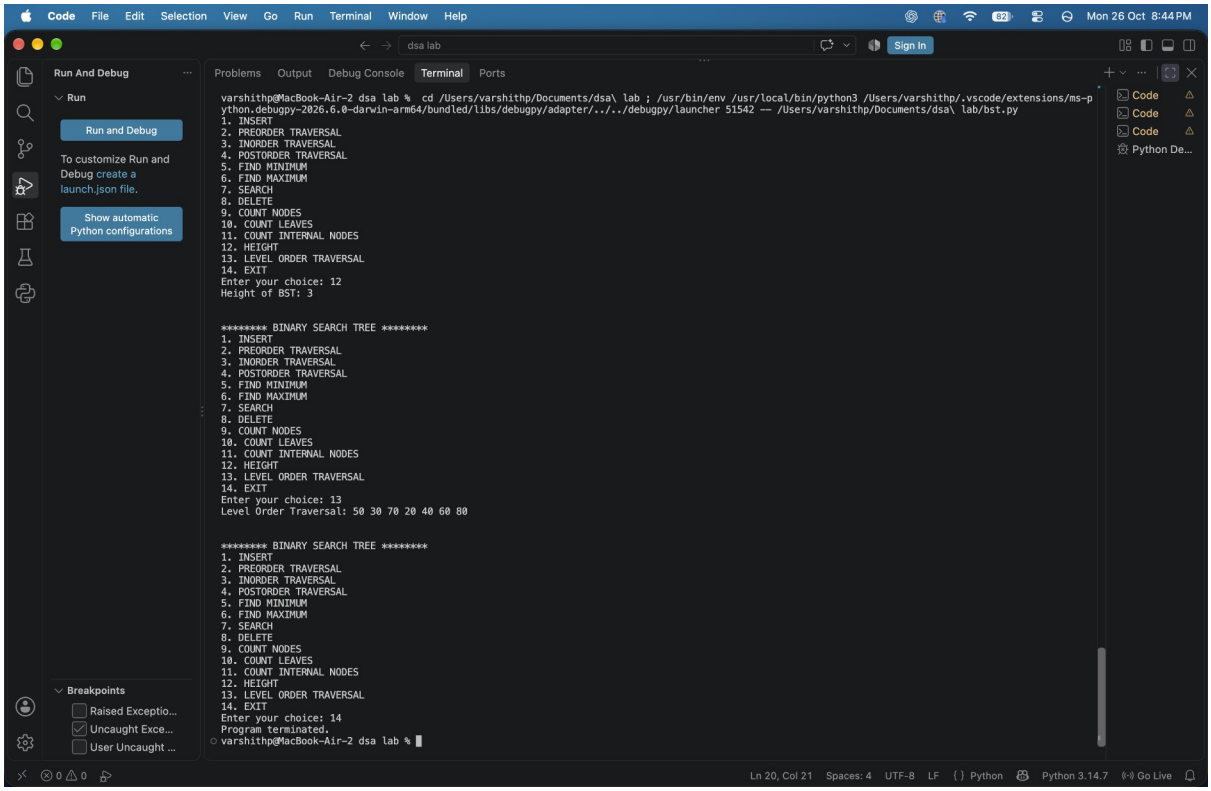

varshithp@MacBook-Air-2 dsa lab % cd /Users/varshithp/Documents/dsa\ lab ; /usr/bin/env /usr/local/bin/python3 /Users/varshithp/.vscode/extensions/ms-python.debugpy-2026.6.0-darwin-arm64/bundled/libs/debugpy/adapter/../../debugpy/launcher 51542 -- /Users/varshithp/Documents/dsa\ lab/bst.py
13. LEVEL ORDER TRAVERSAL
14. EXIT
Enter your choice: 1
Enter the data to be inserted: 70
Element inserted successfully.
******** BINARY SEARCH TREE ********
1. INSERT
2. PREORDER TRAVERSAL
3. INORDER TRAVERSAL
4. POSTORDER TRAVERSAL
5. FIND MINIMUM
6. FIND MAXIMUM
7. SEARCH
8. DELETE
9. COUNT NODES
10. COUNT LEAVES
11. COUNT INTERNAL NODES
12. HEIGHT
13. LEVEL ORDER TRAVERSAL
14. EXIT
Enter your choice: 1
Enter the data to be inserted: 20
Element inserted successfully.
******** BINARY SEARCH TREE ********
1. INSERT
2. PREORDER TRAVERSAL
3. INORDER TRAVERSAL
4. POSTORDER TRAVERSAL
5. FIND MINIMUM
6. FIND MAXIMUM
7. SEARCH
8. DELETE
9. COUNT NODES
10. COUNT LEAVES
11. COUNT INTERNAL NODES
12. HEIGHT
13. LEVEL ORDER TRAVERSAL
14. EXIT
Enter your choice: 1
Enter the data to be inserted: 40
Element inserted successfully.
******** BINARY SEARCH TREE ********
1. INSERT
2. PREORDER TRAVERSAL
3. INORDER TRAVERSAL
4. POSTORDER TRAVERSAL
5. FIND MINIMUM
6. FIND MAXIMUM
7. SEARCH


varshithp@MacBook-Air-2 dsa lab % cd /Users/varshithp/Documents/dsa\ lab ; /usr/bin/env /usr/local/bin/python3 /Users/varshithp/.vscode/extensions/ms-python.debugpy-2026.6.0-darwin-arm64/bundled/libs/debugpy/adapter/../../debugpy/launcher 51542 -- /Users/varshithp/Documents/dsa\ lab/bst.py
8. DELETE
9. COUNT NODES
10. COUNT LEAVES
11. COUNT INTERNAL NODES
12. HEIGHT
13. LEVEL ORDER TRAVERSAL
14. EXIT
Enter your choice: 1
Enter the data to be inserted: 60
Element inserted successfully.
******** BINARY SEARCH TREE ********
1. INSERT
2. PREORDER TRAVERSAL
3. INORDER TRAVERSAL
4. POSTORDER TRAVERSAL
5. FIND MINIMUM
6. FIND MAXIMUM
7. SEARCH
8. DELETE
9. COUNT NODES
10. COUNT LEAVES
11. COUNT INTERNAL NODES
12. HEIGHT
13. LEVEL ORDER TRAVERSAL
14. EXIT
Enter your choice: 1
Enter the data to be inserted: 80
Element inserted successfully.
******** BINARY SEARCH TREE ********
1. INSERT
2. PREORDER TRAVERSAL
3. INORDER TRAVERSAL
4. POSTORDER TRAVERSAL
5. FIND MINIMUM
6. FIND MAXIMUM
7. SEARCH
8. DELETE
9. COUNT NODES
10. COUNT LEAVES
11. COUNT INTERNAL NODES
12. HEIGHT
13. LEVEL ORDER TRAVERSAL
14. EXIT
Enter your choice: 1
Enter the data to be inserted: 1
Element inserted successfully.
******** BINARY SEARCH TREE ********
1. INSERT
2. PREORDER TRAVERSAL


varshithp@MacBook-Air-2 dsa lab % cd /Users/varshithp/Documents/dsa\ lab ; /usr/bin/env /usr/local/bin/python3 /Users/varshithp/.vscode/extensions/ms-python.debugpy-2026.6.0-darwin-arm64/bundled/libs/debugpy/adapter/../../debugpy/launcher 51542 -- /Users/varshithp/Documents/dsa\ lab/bst.py
3. INORDER TRAVERSAL
4. POSTORDER TRAVERSAL
5. FIND MINIMUM
6. FIND MAXIMUM
7. SEARCH
8. DELETE
9. COUNT NODES
10. COUNT LEAVES
11. COUNT INTERNAL NODES
12. HEIGHT
13. LEVEL ORDER TRAVERSAL
14. EXIT
Enter your choice: 2
Preorder Traversal: 50 30 20 1 40 70 60 80
******** BINARY SEARCH TREE ********
1. INSERT
2. PREORDER TRAVERSAL
3. INORDER TRAVERSAL
4. POSTORDER TRAVERSAL
5. FIND MINIMUM
6. FIND MAXIMUM
7. SEARCH
8. DELETE
9. COUNT NODES
10. COUNT LEAVES
11. COUNT INTERNAL NODES
12. HEIGHT
13. LEVEL ORDER TRAVERSAL
14. EXIT
Enter your choice: 3
Inorder Traversal: 1 20 30 40 50 60 70 80
******** BINARY SEARCH TREE ********
1. INSERT
2. PREORDER TRAVERSAL
3. INORDER TRAVERSAL
4. POSTORDER TRAVERSAL
5. FIND MINIMUM
6. FIND MAXIMUM
7. SEARCH
8. DELETE
9. COUNT NODES
10. COUNT LEAVES
11. COUNT INTERNAL NODES
12. HEIGHT
13. LEVEL ORDER TRAVERSAL
14. EXIT
Enter your choice: 4
Postorder Traversal: 1 20 40 30 60 80 70 50
******** BINARY SEARCH TREE ********


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User Uncaught ...
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varshithp@MacBook-Air-2 dsa lab % cd /Users/varshithp/Documents/dsa\ lab ; /usr/bin/env /usr/local/bin/python3 /Users/varshithp/.vscode/extensions/ms-python.debugpy-2026.6.0-darwin-arm64/bundled/libs/debugpy/adapter/../../debugpy/launcher 51542 -- /Users/varshithp/Documents/dsa\ lab/bst.py
3. INORDER TRAVERSAL
4. POSTORDER TRAVERSAL
5. FIND MINIMUM
6. FIND MAXIMUM
7. SEARCH
8. DELETE
9. COUNT NODES
10. COUNT LEAVES
11. COUNT INTERNAL NODES
12. HEIGHT
13. LEVEL ORDER TRAVERSAL
14. EXIT
Enter your choice: 5
Minimum element: 1
******** BINARY SEARCH TREE ********
1. INSERT
2. PREORDER TRAVERSAL
3. INORDER TRAVERSAL
4. POSTORDER TRAVERSAL
5. FIND MINIMUM
6. FIND MAXIMUM
7. SEARCH
8. DELETE
9. COUNT NODES
10. COUNT LEAVES
11. COUNT INTERNAL NODES
12. HEIGHT
13. LEVEL ORDER TRAVERSAL
14. EXIT
Enter your choice: 6
Maximum element: 80
******** BINARY SEARCH TREE ********
1. INSERT
2. PREORDER TRAVERSAL
3. INORDER TRAVERSAL
4. POSTORDER TRAVERSAL
5. FIND MINIMUM
6. FIND MAXIMUM
7. SEARCH
8. DELETE
9. COUNT NODES
10. COUNT LEAVES
11. COUNT INTERNAL NODES
12. HEIGHT
13. LEVEL ORDER TRAVERSAL
14. EXIT
Enter your choice: 7
Enter the data to be searched: 9
Element 9 is not found.
Code
Code
Code
Python De...
Ln 20, Col 21 Spaces: 4 UTF-8 LF { } Python Python 3.14.7 Go Live


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Problems Output Debug Console Terminal Ports
varshithp@MacBook-Air-2 dsa lab % cd /Users/varshithp/Documents/dsa\ lab ; /usr/bin/env /usr/local/bin/python3 /Users/varshithp/.vscode/extensions/ms-python.debugpy-2026.6.0-darwin-arm64/bundled/libs/debugpy/adapter/../../debugpy/launcher 51542 -- /Users/varshithp/Documents/dsa\ lab/bst.py
******** BINARY SEARCH TREE ********
1. INSERT
2. PREORDER TRAVERSAL
3. INORDER TRAVERSAL
4. POSTORDER TRAVERSAL
5. FIND MINIMUM
6. FIND MAXIMUM
7. SEARCH
8. DELETE
9. COUNT NODES
10. COUNT LEAVES
11. COUNT INTERNAL NODES
12. HEIGHT
13. LEVEL ORDER TRAVERSAL
14. EXIT
Enter your choice: 10
Number of leaf nodes: 4
******** BINARY SEARCH TREE ********
1. INSERT
2. PREORDER TRAVERSAL
3. INORDER TRAVERSAL
4. POSTORDER TRAVERSAL
5. FIND MINIMUM
6. FIND MAXIMUM
7. SEARCH
8. DELETE
9. COUNT NODES
10. COUNT LEAVES
11. COUNT INTERNAL NODES
12. HEIGHT
13. LEVEL ORDER TRAVERSAL
14. EXIT
Enter your choice: 11
Number of internal nodes: 4
******** BINARY SEARCH TREE ********
1. INSERT
2. PREORDER TRAVERSAL
3. INORDER TRAVERSAL
4. POSTORDER TRAVERSAL
5. FIND MINIMUM
6. FIND MAXIMUM
7. SEARCH
8. DELETE
9. COUNT NODES
10. COUNT LEAVES
11. COUNT INTERNAL NODES
12. HEIGHT
13. LEVEL ORDER TRAVERSAL
14. EXIT
Enter your choice: 12
Height of BST: 3
Code
Code
Code
Python De...
Ln 20, Col 21 Spaces: 4 UTF-8 LF { } Python Python 3.14.7 Go Live

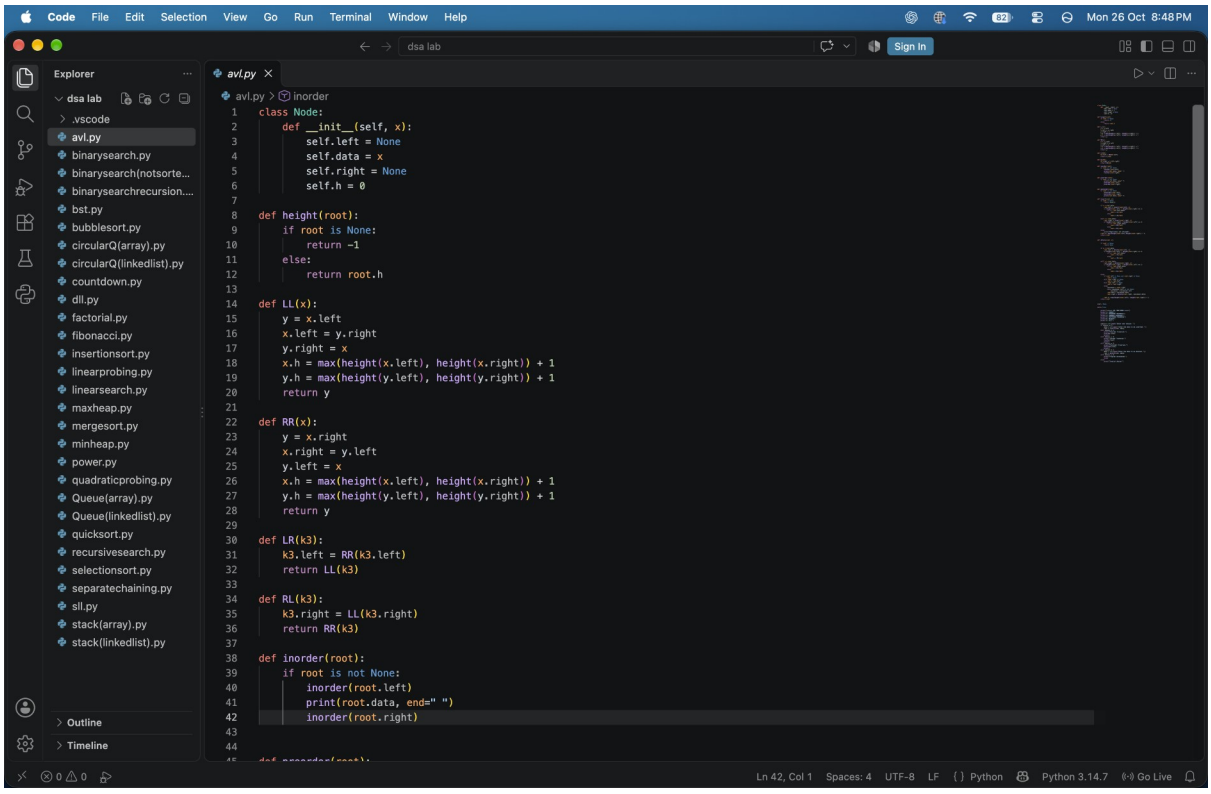

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Sign In
Run And Debug
Run
Run and Debug
To customize Run and Debug create a launch.json file.
Show automatic Python configurations
Breakpoints
Raised Exceptio...
Uncaught Exce...
User Uncaught ...
Problems Output Debug Console Terminal Ports
varshithp@MacBook-Air-2 dsa lab % cd /Users/varshithp/Documents/dsa\ lab ; /usr/bin/env /usr/local/bin/python3 /Users/varshithp/.vscode/extensions/ms-python.debugpy-2026.6.0-darwin-arm64/bundled/libs/debugpy/adapter/../../debugpy/launcher 51542 -- /Users/varshithp/Documents/dsa\ lab/bst.py
1. INSERT
2. PREORDER TRAVERSAL
3. INORDER TRAVERSAL
4. POSTORDER TRAVERSAL
5. FIND MINIMUM
6. FIND MAXIMUM
7. SEARCH
8. DELETE
9. COUNT NODES
10. COUNT LEAVES
11. COUNT INTERNAL NODES
12. HEIGHT
13. LEVEL ORDER TRAVERSAL
14. EXIT
Enter your choice: 12
Height of BST: 3
******** BINARY SEARCH TREE ********
1. INSERT
2. PREORDER TRAVERSAL
3. INORDER TRAVERSAL
4. POSTORDER TRAVERSAL
5. FIND MINIMUM
6. FIND MAXIMUM
7. SEARCH
8. DELETE
9. COUNT NODES
10. COUNT LEAVES
11. COUNT INTERNAL NODES
12. HEIGHT
13. LEVEL ORDER TRAVERSAL
14. EXIT
Enter your choice: 13
Level Order Traversal: 50 30 70 20 40 60 80
******** BINARY SEARCH TREE ********
1. INSERT
2. PREORDER TRAVERSAL
3. INORDER TRAVERSAL
4. POSTORDER TRAVERSAL
5. FIND MINIMUM
6. FIND MAXIMUM
7. SEARCH
8. DELETE
9. COUNT NODES
10. COUNT LEAVES
11. COUNT INTERNAL NODES
12. HEIGHT
13. LEVEL ORDER TRAVERSAL
14. EXIT
Enter your choice: 14
Program terminated.
varshithp@MacBook-Air-2 dsa lab %
Code
Code
Code
Python De...
Ln 20, Col 21 Spaces: 4 UTF-8 LF Python Python 3.14.7 Go Live

WEEK-10

10. Write a menu driven python program to implement an AVL Tree and perform the following operations.

1. Insertion
2. Deletion
3. Tree Traversals
4. Count Leaf Nodes
5. Count Internal Nodes
6. Height of the Tree

Code:-



```python
class Node:
    def __init__(self, x):
        self.left = None
        self.data = x
        self.right = None
        self.h = 0

def height(root):
    if root is None:
        return -1
    else:
        return root.h

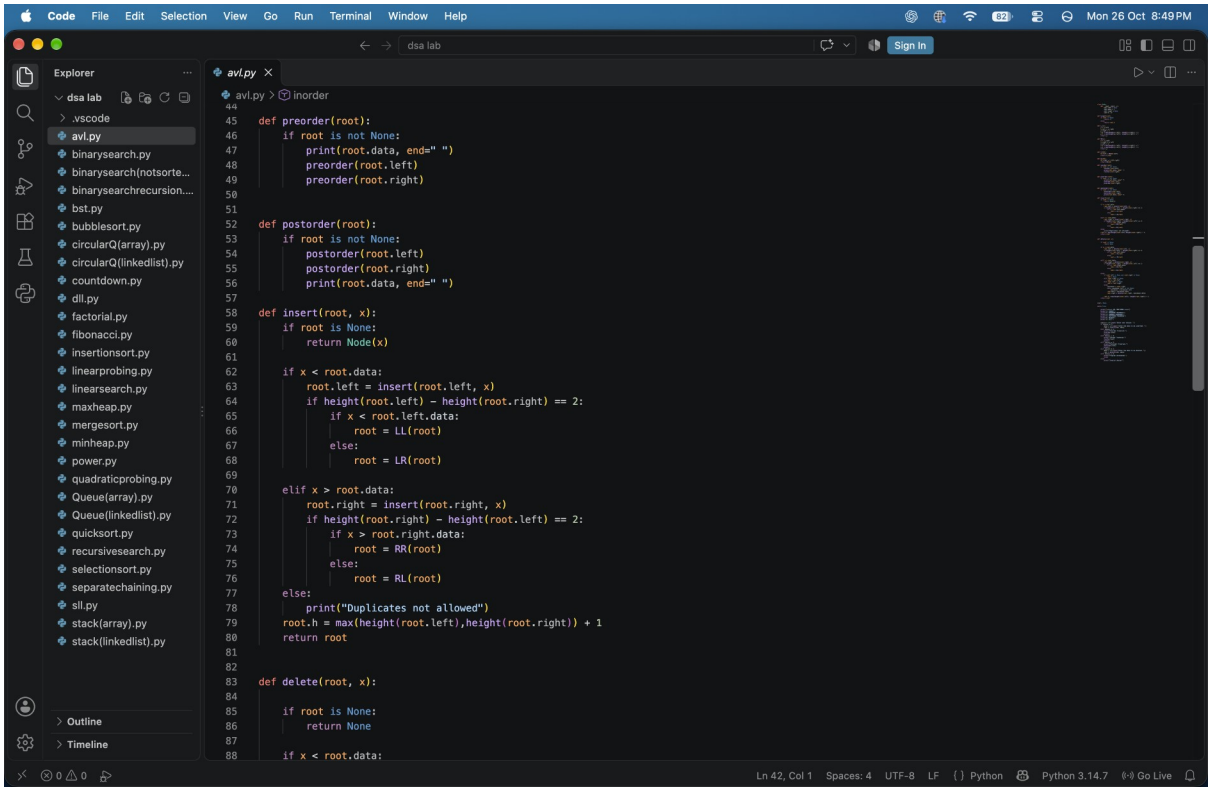
def LL(x):
    y = x.left
    x.left = y.right
    y.right = x
    x.h = max(height(x.left), height(x.right)) + 1
    y.h = max(height(y.left), height(y.right)) + 1
    return y

def RR(x):
    y = x.right
    x.right = y.left
    y.left = x
    x.h = max(height(x.left), height(x.right)) + 1
    y.h = max(height(y.left), height(y.right)) + 1
    return y

def LR(k3):
    k3.left = RR(k3.left)
    return LL(k3)

def RL(k3):
    k3.right = LL(k3.right)
    return RR(k3)

def inorder(root):
    if root is not None:
        inorder(root.left)
        print(root.data, end=" ")
        inorder(root.right)
```



```
def preorder(root):
    if root is not None:
        print(root.data, end=" ")
        preorder(root.left)
        preorder(root.right)


def postorder(root):
    if root is not None:
        postorder(root.left)
        postorder(root.right)
        print(root.data, end=" ")

def insert(root, x):
    if root is None:
        return Node(x)

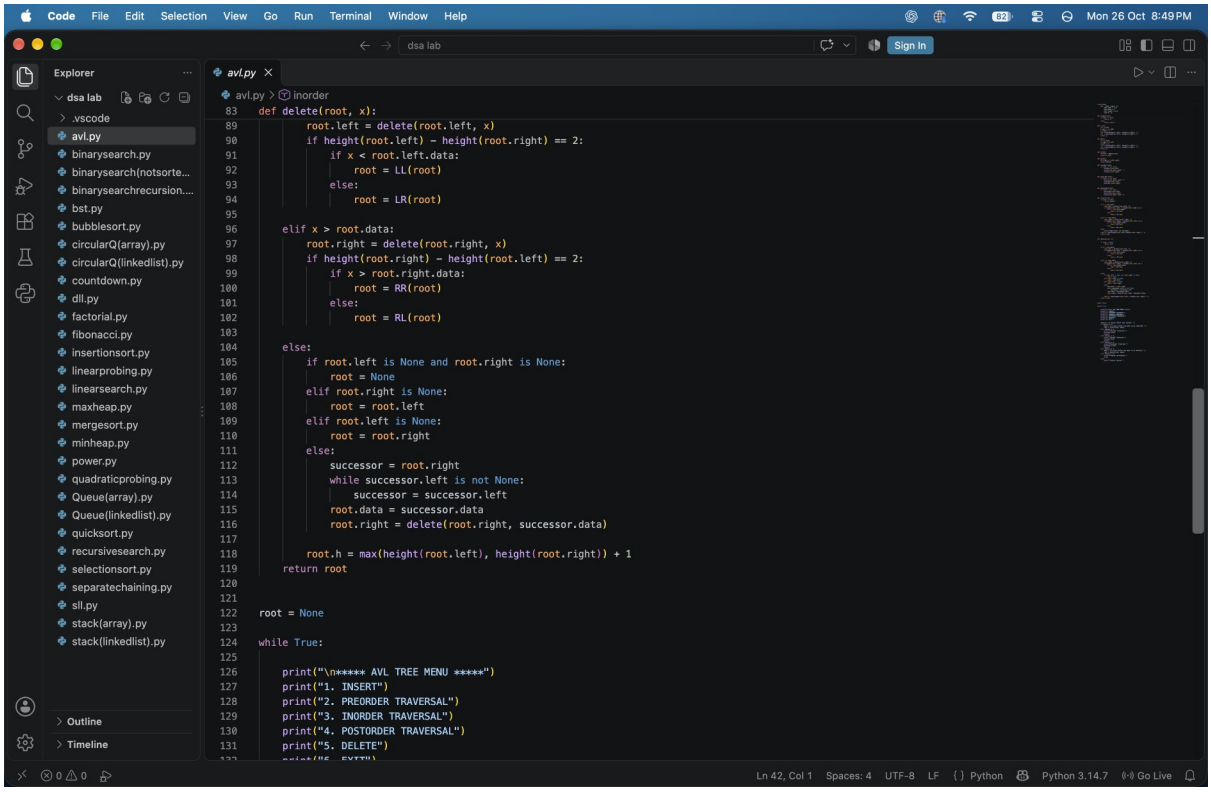
    if x < root.data:
        root.left = insert(root.left, x)
        if height(root.left) - height(root.right) == 2:
            if x < root.left.data:
                root = LL(root)
            else:
                root = LR(root)

    elif x > root.data:
        root.right = insert(root.right, x)
        if height(root.right) - height(root.left) == 2:
            if x > root.right.data:
                root = RR(root)
            else:
                root = RL(root)
    else:
        print("Duplicates not allowed")
    root.h = max(height(root.left),height(root.right)) + 1
    return root


def delete(root, x):

    if root is None:
        return None

    if x < root.data:
```



```
def delete(root, x):
        root.left = delete(root.left, x)
        if height(root.left) - height(root.right) == 2:
            if x < root.left.data:
                root = LL(root)
            else:
                root = LR(root)

    elif x > root.data:
        root.right = delete(root.right, x)
        if height(root.right) - height(root.left) == 2:
            if x > root.right.data:
                root = RR(root)
            else:
                root = RL(root)

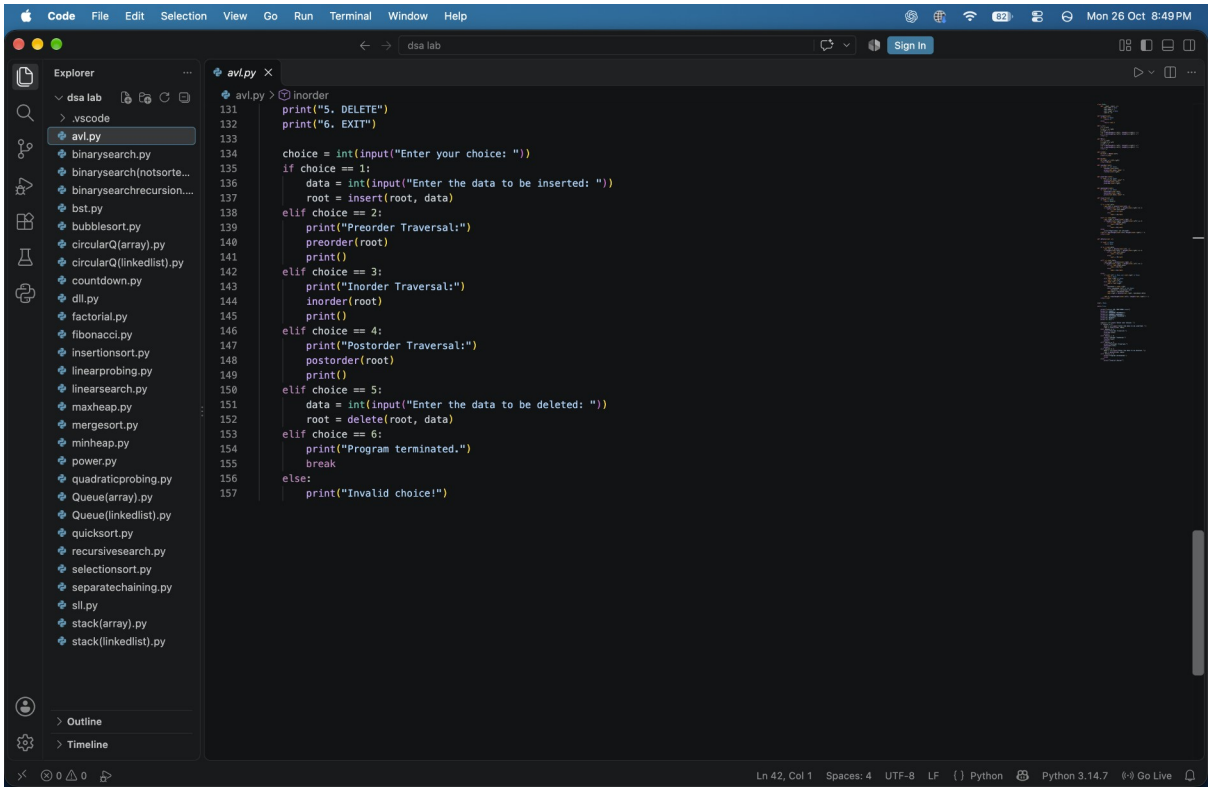
    else:
        if root.left is None and root.right is None:
            root = None
        elif root.right is None:
            root = root.left
        elif root.left is None:
            root = root.right
        else:
            successor = root.right
            while successor.left is not None:
                successor = successor.left
            root.data = successor.data
            root.right = delete(root.right, successor.data)

        root.h = max(height(root.left), height(root.right)) + 1
    return root

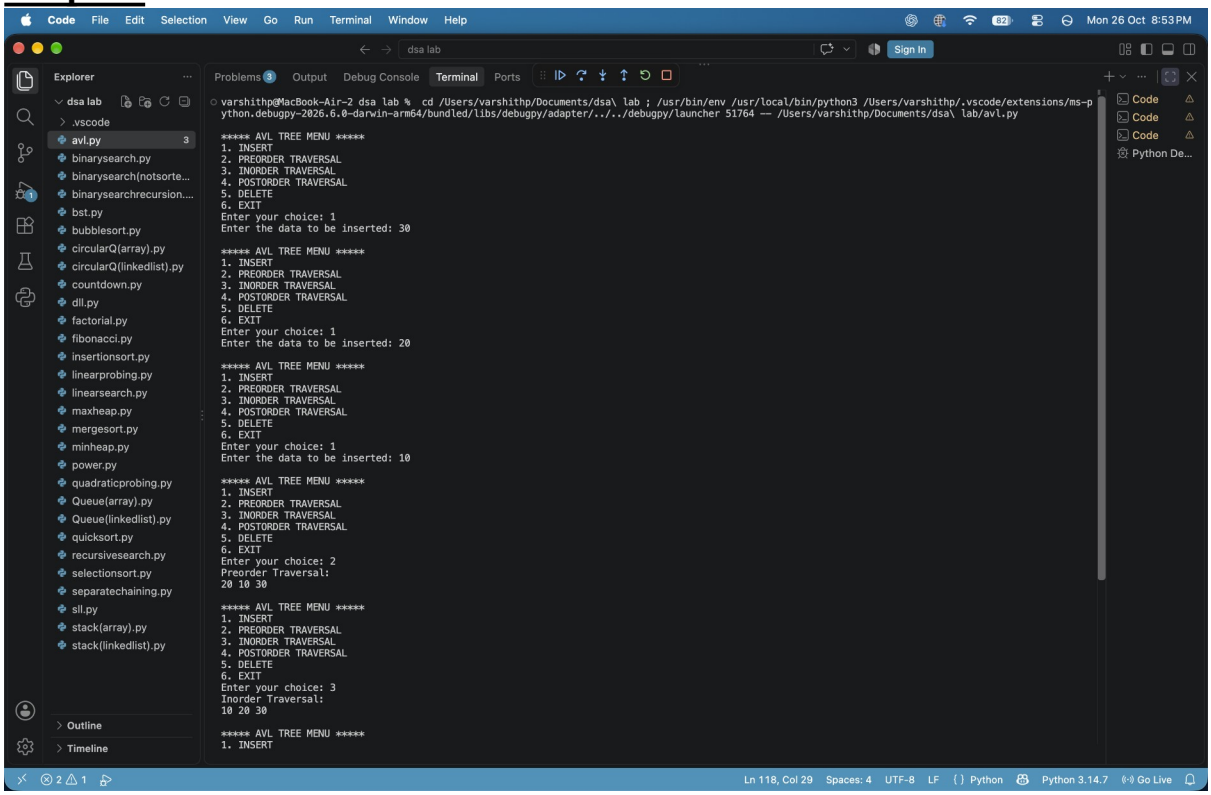

root = None

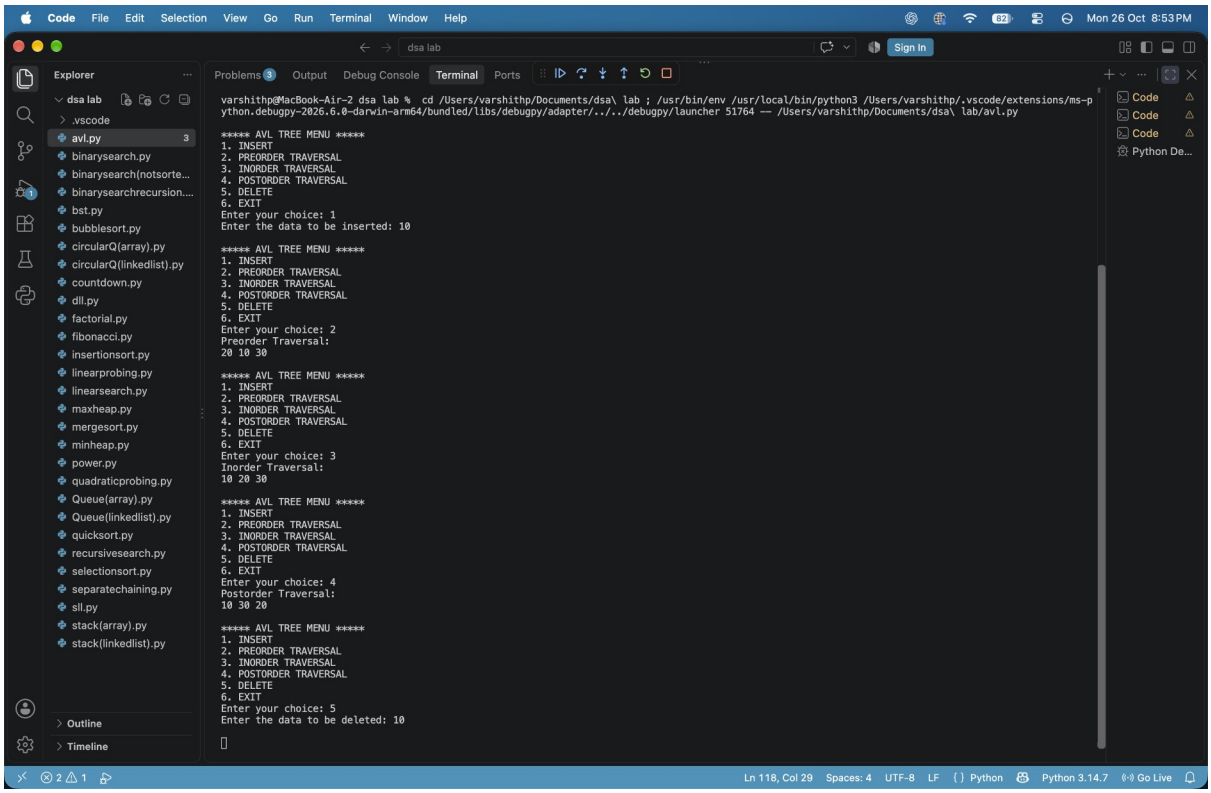
while True:

    print("\n***** AVL TREE MENU *****")
    print("1. INSERT")
    print("2. PREORDER TRAVERSAL")
    print("3. INORDER TRAVERSAL")
    print("4. POSTORDER TRAVERSAL")
    print("5. DELETE")
```


Code File Edit Selection View Go Run Terminal Window Help
Mon 26 Oct 8:49 PM
dsa lab
Sign In
Explorer
dsa lab
.vscode
avl.py
binarysearch.py
binarysearch(notsorte...
binarysearchrecursion....
bst.py
bubblesort.py
circularQ(array).py
circularQ(linkedlist).py
countdown.py
dll.py
factorial.py
fibonacci.py
insertionsort.py
linearprobing.py
linearsearch.py
maxheap.py
mergesort.py
minheap.py
power.py
quadraticprobing.py
Queue(array).py
Queue(linkedlist).py
quicksort.py
recursivesearch.py
selectionsort.py
separatechaining.py
sll.py
stack(array).py
stack(linkedlist).py
Outline
Timeline
avl.py
avl.py > inorder
print("5. DELETE")
print("6. EXIT")
choice = int(input("Enter your choice: "))
if choice == 1:
data = int(input("Enter the data to be inserted: "))
root = insert(root, data)
elif choice == 2:
print("Preorder Traversal:")
preorder(root)
print()
elif choice == 3:
print("Inorder Traversal:")
inorder(root)
print()
elif choice == 4:
print("Postorder Traversal:")
postorder(root)
print()
elif choice == 5:
data = int(input("Enter the data to be deleted: "))
root = delete(root, data)
elif choice == 6:
print("Program terminated.")
break
else:
print("Invalid choice!")
Ln 42, Col 1 Spaces: 4 UTF-8 LF Python Python 3.14.7 Go Live

Output:-


varshithp@MacBook-Air-2 dsa lab % cd /Users/varshithp/Documents/dsa\ lab ; /usr/bin/env /usr/local/bin/python3 /Users/varshithp/.vscode/extensions/ms-python.debugpy-2026.6.0-darwin-arm64/bundled/libs/debugpy/adapter/../../debugpy/launcher 51764 -- /Users/varshithp/Documents/dsa\ lab/avl.py
***** AVL TREE MENU *****
1. INSERT
2. PREORDER TRAVERSAL
3. INORDER TRAVERSAL
4. POSTORDER TRAVERSAL
5. DELETE
6. EXIT
Enter your choice: 1
Enter the data to be inserted: 30
***** AVL TREE MENU *****
1. INSERT
2. PREORDER TRAVERSAL
3. INORDER TRAVERSAL
4. POSTORDER TRAVERSAL
5. DELETE
6. EXIT
Enter your choice: 1
Enter the data to be inserted: 20
***** AVL TREE MENU *****
1. INSERT
2. PREORDER TRAVERSAL
3. INORDER TRAVERSAL
4. POSTORDER TRAVERSAL
5. DELETE
6. EXIT
Enter your choice: 1
Enter the data to be inserted: 10
***** AVL TREE MENU *****
1. INSERT
2. PREORDER TRAVERSAL
3. INORDER TRAVERSAL
4. POSTORDER TRAVERSAL
5. DELETE
6. EXIT
Enter your choice: 2
Preorder Traversal:
20 10 30
***** AVL TREE MENU *****
1. INSERT
2. PREORDER TRAVERSAL
3. INORDER TRAVERSAL
4. POSTORDER TRAVERSAL
5. DELETE
6. EXIT
Enter your choice: 3
Inorder Traversal:
10 20 30
***** AVL TREE MENU *****
1. INSERT

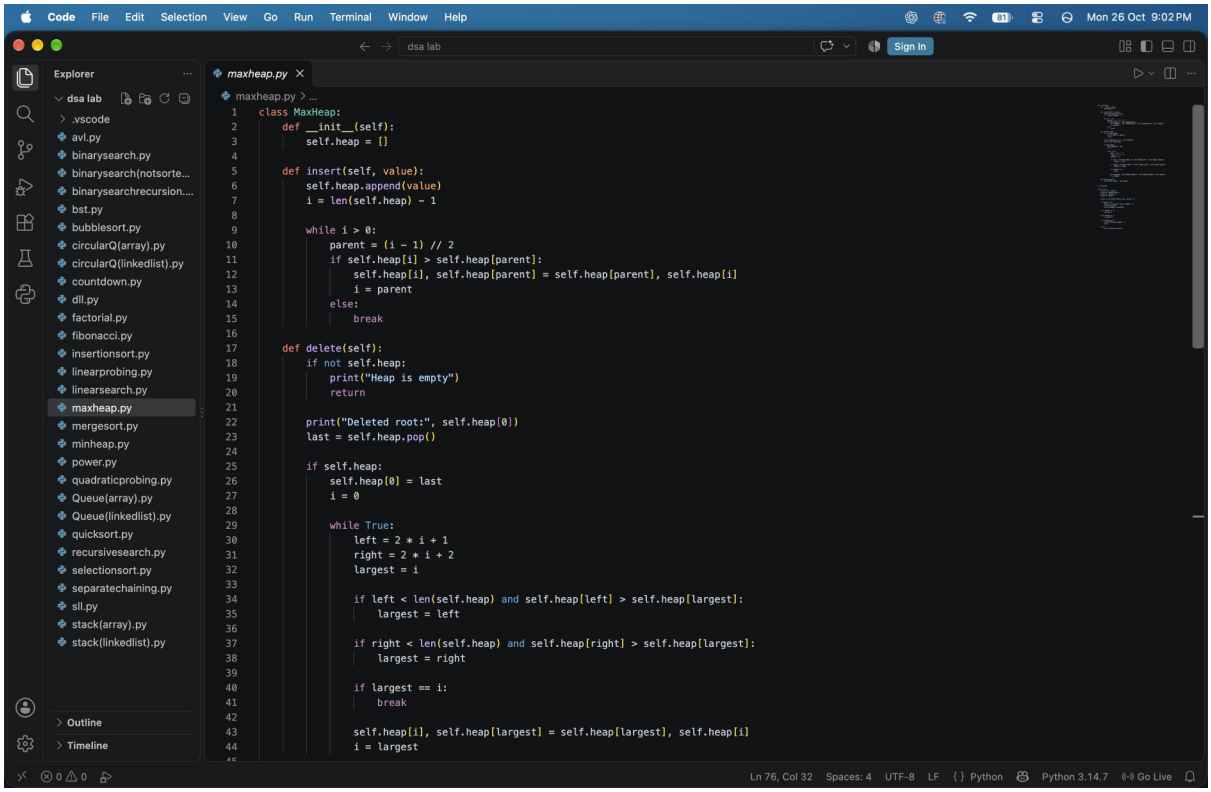

varshithp@MacBook-Air-2 dsa lab % cd /Users/varshithp/Documents/dsa\ lab ; /usr/bin/env /usr/local/bin/python3 /Users/varshithp/.vscode/extensions/ms-python.debugpy-2026.6.0-darwin-arm64/bundled/libs/debugpy/adapter/../../debugpy/launcher 51764 -- /Users/varshithp/Documents/dsa\ lab/avl.py
***** AVL TREE MENU *****
1. INSERT
2. PREORDER TRAVERSAL
3. INORDER TRAVERSAL
4. POSTORDER TRAVERSAL
5. DELETE
6. EXIT
Enter your choice: 1
Enter the data to be inserted: 10
***** AVL TREE MENU *****
1. INSERT
2. PREORDER TRAVERSAL
3. INORDER TRAVERSAL
4. POSTORDER TRAVERSAL
5. DELETE
6. EXIT
Enter your choice: 2
Preorder Traversal:
20 10 30
***** AVL TREE MENU *****
1. INSERT
2. PREORDER TRAVERSAL
3. INORDER TRAVERSAL
4. POSTORDER TRAVERSAL
5. DELETE
6. EXIT
Enter your choice: 3
Inorder Traversal:
10 20 30
***** AVL TREE MENU *****
1. INSERT
2. PREORDER TRAVERSAL
3. INORDER TRAVERSAL
4. POSTORDER TRAVERSAL
5. DELETE
6. EXIT
Enter your choice: 4
Postorder Traversal:
10 30 20
***** AVL TREE MENU *****
1. INSERT
2. PREORDER TRAVERSAL
3. INORDER TRAVERSAL
4. POSTORDER TRAVERSAL
5. DELETE
6. EXIT
Enter your choice: 5
Enter the data to be deleted: 10

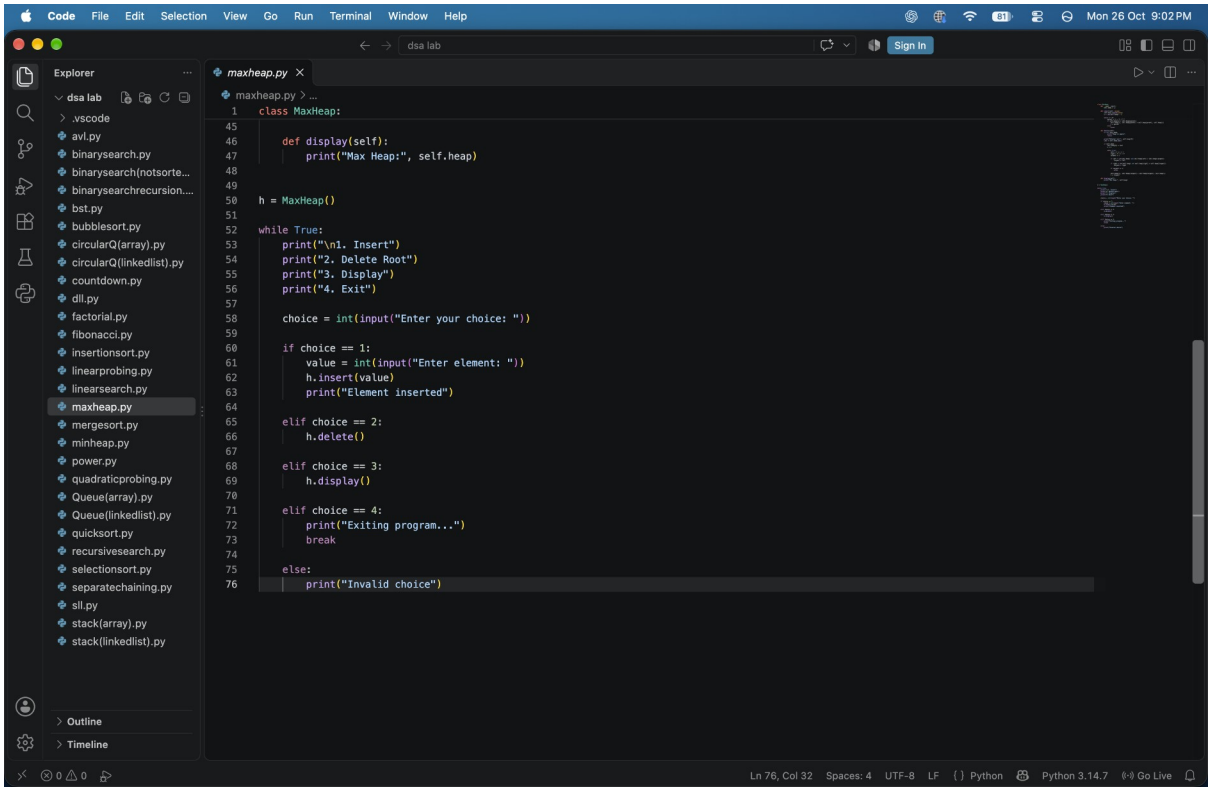
WEEK-11

11. Write a menu-driven Python program to implement a Binary Heap (Max Heap and Min Heap) and perform the following operations:
1. Insertion of an element into the Heap.
2. Deletion of the root element from the Heap.
3. Display the array representation of the Heap.
4. Exit the program.

Code:-

Max Heap





```python
class MaxHeap:

    def display(self):
        print("Max Heap:", self.heap)


h = MaxHeap()

while True:
    print("\n1. Insert")
    print("2. Delete Root")
    print("3. Display")
    print("4. Exit")

    choice = int(input("Enter your choice: "))

    if choice == 1:
        value = int(input("Enter element: "))
        h.insert(value)
        print("Element inserted")

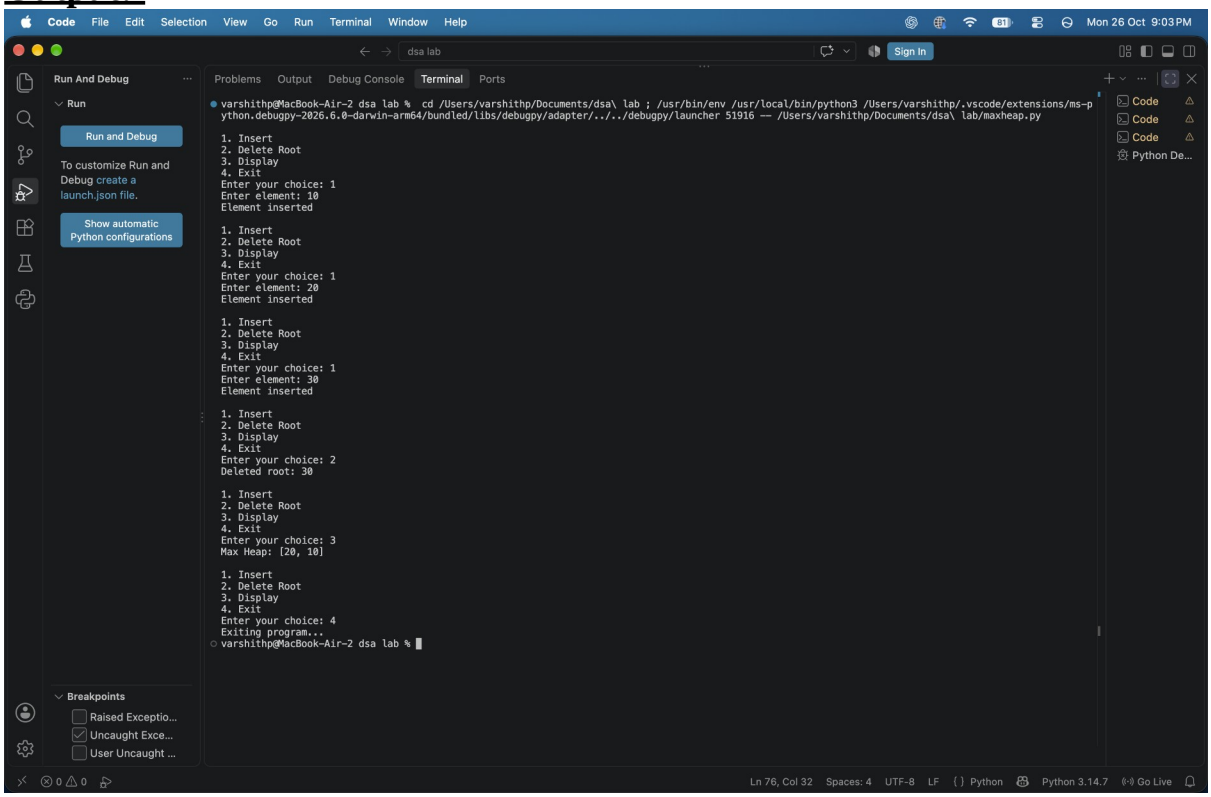
    elif choice == 2:
        h.delete()

    elif choice == 3:
        h.display()

    elif choice == 4:
        print("Exiting program...")
        break

    else:
        print("Invalid choice")
```

Output:-



```
varshithp@MacBook-Air-2 dsa lab % cd /Users/varshithp/Documents/dsa\ lab ; /usr/bin/env /usr/local/bin/python3 /Users/varshithp/.vscode/extensions/ms-python.debugpy-2026.6.0-darwin-arm64/bundled/libs/debugpy/adapter/../../debugpy/launcher 51916 -- /Users/varshithp/Documents/dsa\ lab/maxheap.py

1. Insert
2. Delete Root
3. Display
4. Exit
Enter your choice: 1
Enter element: 10
Element inserted

1. Insert
2. Delete Root
3. Display
4. Exit
Enter your choice: 1
Enter element: 20
Element inserted

1. Insert
2. Delete Root
3. Display
4. Exit
Enter your choice: 1
Enter element: 30
Element inserted

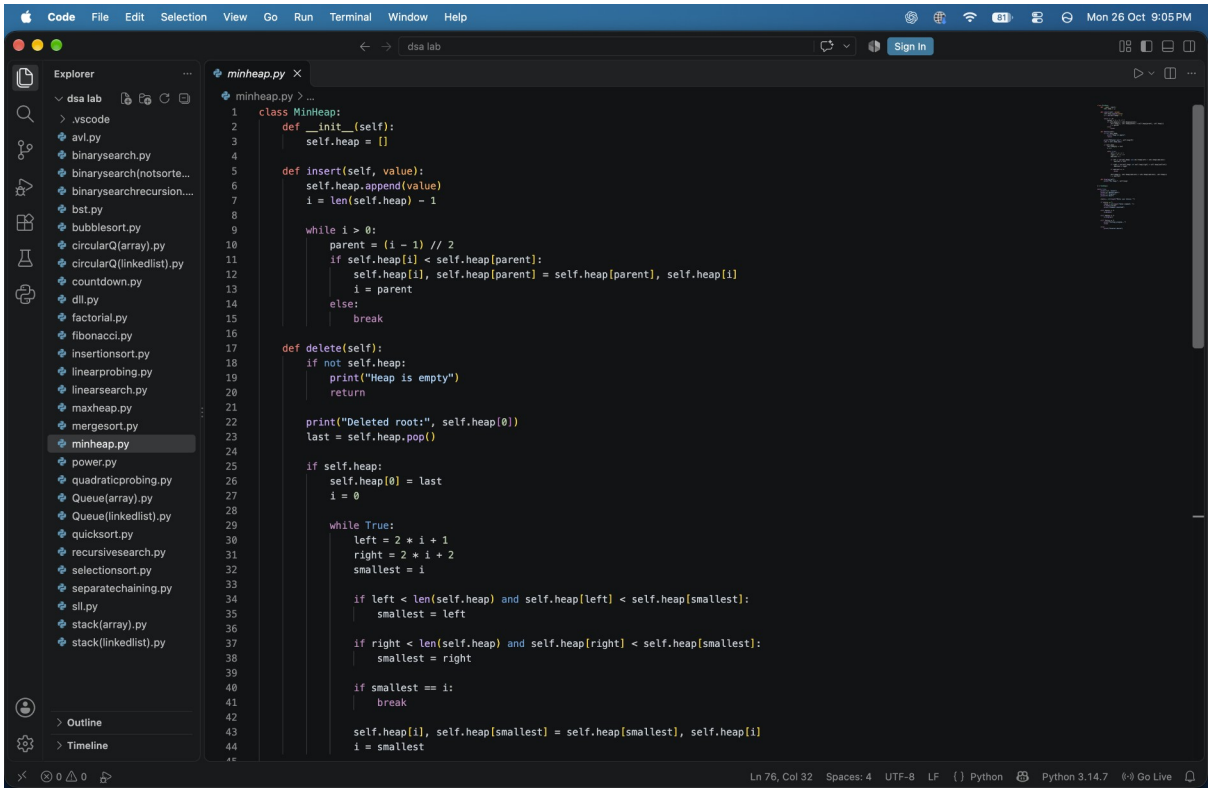
1. Insert
2. Delete Root
3. Display
4. Exit
Enter your choice: 2
Deleted root: 30

1. Insert
2. Delete Root
3. Display
4. Exit
Enter your choice: 3
Max Heap: [20, 10]

1. Insert
2. Delete Root
3. Display
4. Exit
Enter your choice: 4
Exiting program...
varshithp@MacBook-Air-2 dsa lab %
```

Min Heap

Code:-



```python
class MinHeap:
    def __init__(self):
        self.heap = []

    def insert(self, value):
        self.heap.append(value)
        i = len(self.heap) - 1

        while i > 0:
            parent = (i - 1) // 2
            if self.heap[i] < self.heap[parent]:
                self.heap[i], self.heap[parent] = self.heap[parent], self.heap[i]
                i = parent
            else:
                break

    def delete(self):
        if not self.heap:
            print("Heap is empty")
            return

        print("Deleted root:", self.heap[0])
        last = self.heap.pop()

        if self.heap:
            self.heap[0] = last
            i = 0

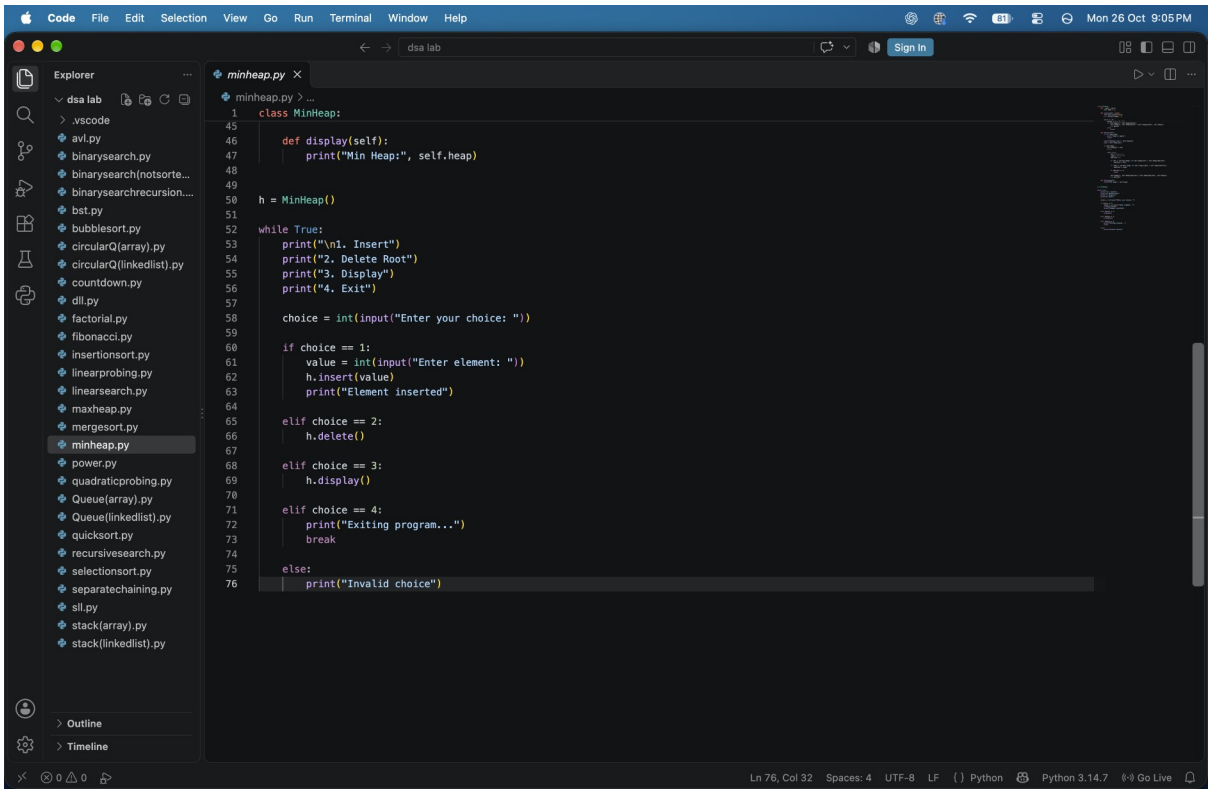
            while True:
                left = 2 * i + 1
                right = 2 * i + 2
                smallest = i

                if left < len(self.heap) and self.heap[left] < self.heap[smallest]:
                    smallest = left

                if right < len(self.heap) and self.heap[right] < self.heap[smallest]:
                    smallest = right

                if smallest == i:
                    break

                self.heap[i], self.heap[smallest] = self.heap[smallest], self.heap[i]
                i = smallest
```



```python
class MinHeap:

    def display(self):
        print("Min Heap:", self.heap)


h = MinHeap()

while True:
    print("\n1. Insert")
    print("2. Delete Root")
    print("3. Display")
    print("4. Exit")

    choice = int(input("Enter your choice: "))

    if choice == 1:
        value = int(input("Enter element: "))
        h.insert(value)
        print("Element inserted")

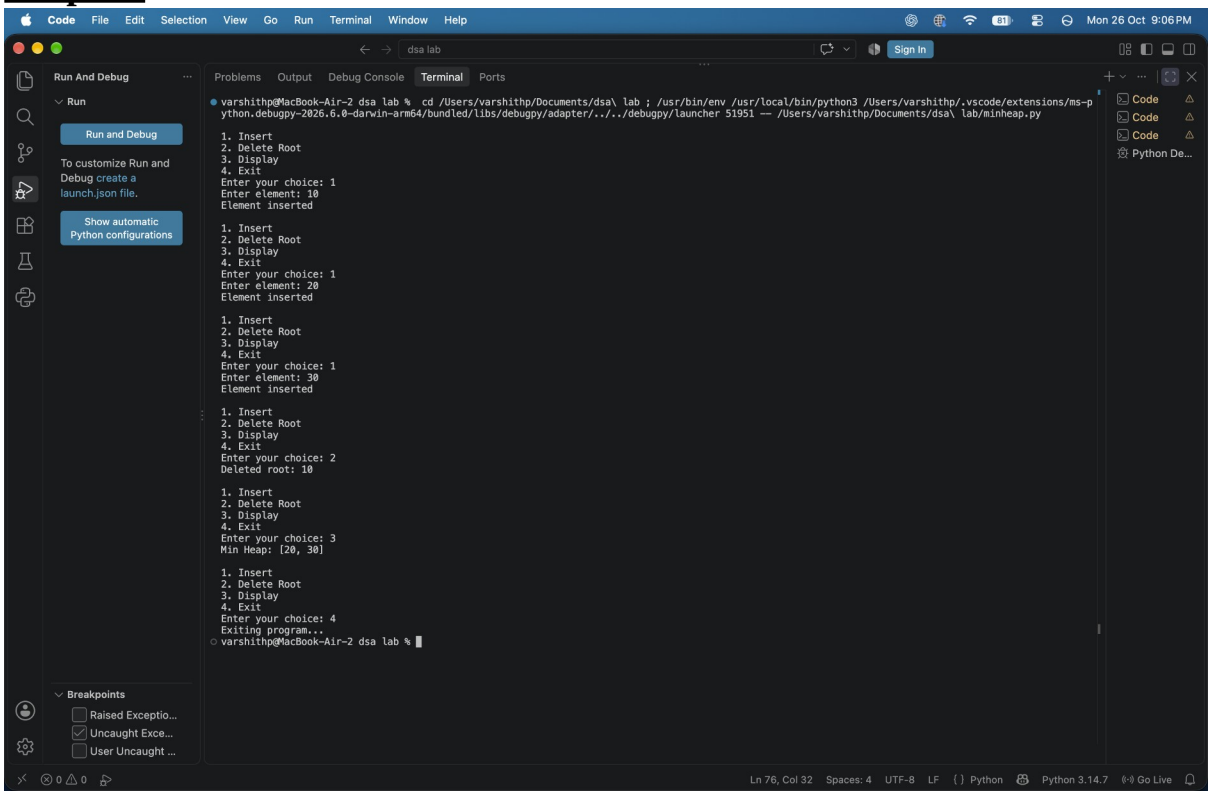
    elif choice == 2:
        h.delete()

    elif choice == 3:
        h.display()

    elif choice == 4:
        print("Exiting program...")
        break

    else:
        print("Invalid choice")
```

Output:-

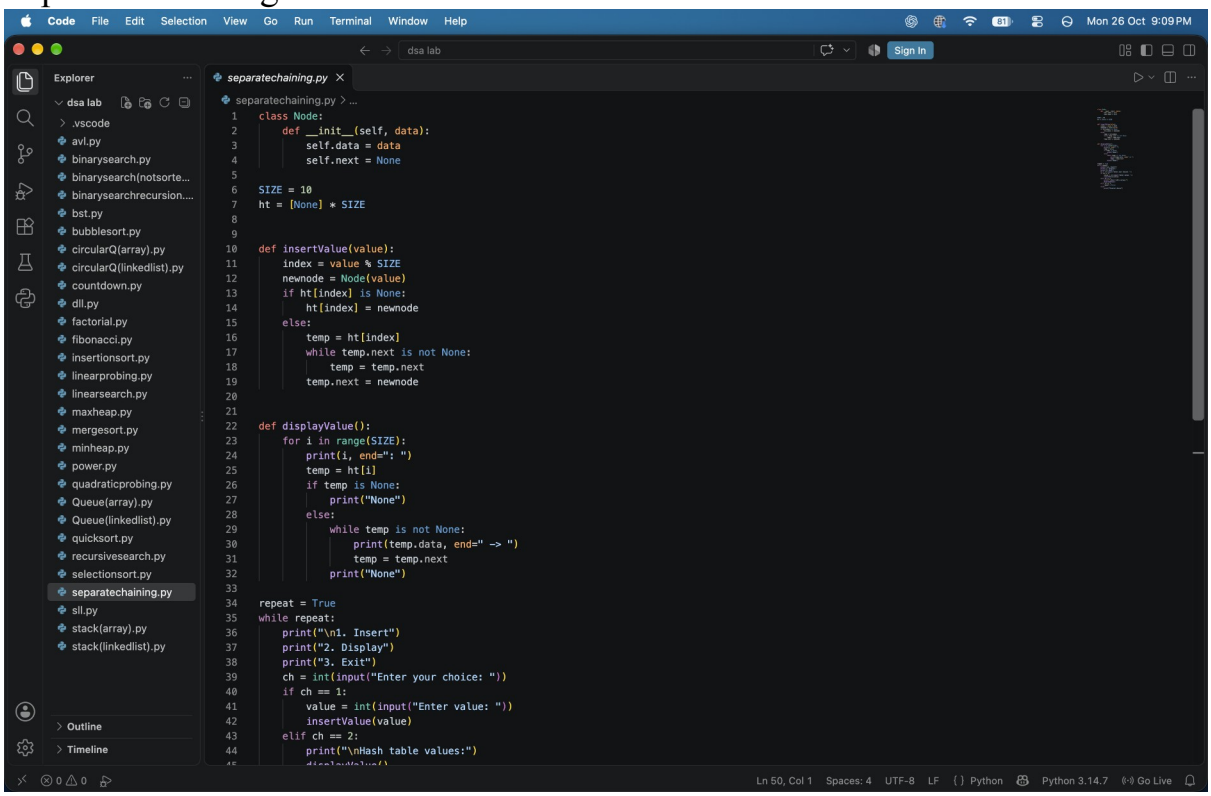

Code File Edit Selection View Go Run Terminal Window Help
Mon 26 Oct 9:06 PM
dsa lab
Sign in
Run And Debug
Run
Run and Debug
To customize Run and Debug create a launch.json file.
Show automatic Python configurations
Breakpoints
Raised Exceptio...
Uncaught Exce...
User Uncaught ...
Problems Output Debug Console Terminal Ports
varshithp@MacBook-Air-2 dsa lab % cd /Users/varshithp/Documents/dsa\ lab ; /usr/bin/env /usr/local/bin/python3 /Users/varshithp/.vscode/extensions/ms-python.debugpy-2026.6.0-darwin-arm64/bundled/libs/debugpy/adapter/../../debugpy/launcher 51951 -- /Users/varshithp/Documents/dsa\ lab/minheap.py
1. Insert
2. Delete Root
3. Display
4. Exit
Enter your choice: 1
Enter element: 10
Element inserted
1. Insert
2. Delete Root
3. Display
4. Exit
Enter your choice: 1
Enter element: 20
Element inserted
1. Insert
2. Delete Root
3. Display
4. Exit
Enter your choice: 1
Enter element: 30
Element inserted
1. Insert
2. Delete Root
3. Display
4. Exit
Enter your choice: 2
Deleted root: 10
1. Insert
2. Delete Root
3. Display
4. Exit
Enter your choice: 3
Min Heap: [20, 30]
1. Insert
2. Delete Root
3. Display
4. Exit
Enter your choice: 4
Exiting program...
varshithp@MacBook-Air-2 dsa lab %
Code
Code
Code
Python De...
Ln 76, Col 32 Spaces: 4 UTF-8 LF { } Python Python 3.14.7 Go Live

WEEK-12

12. Write a program to implement the following three collision resolution techniques in a Hash Table:
1. Separate Chaining
2. Linear Probing
3. Quadratic Probing

Code:-

Separate Chaining



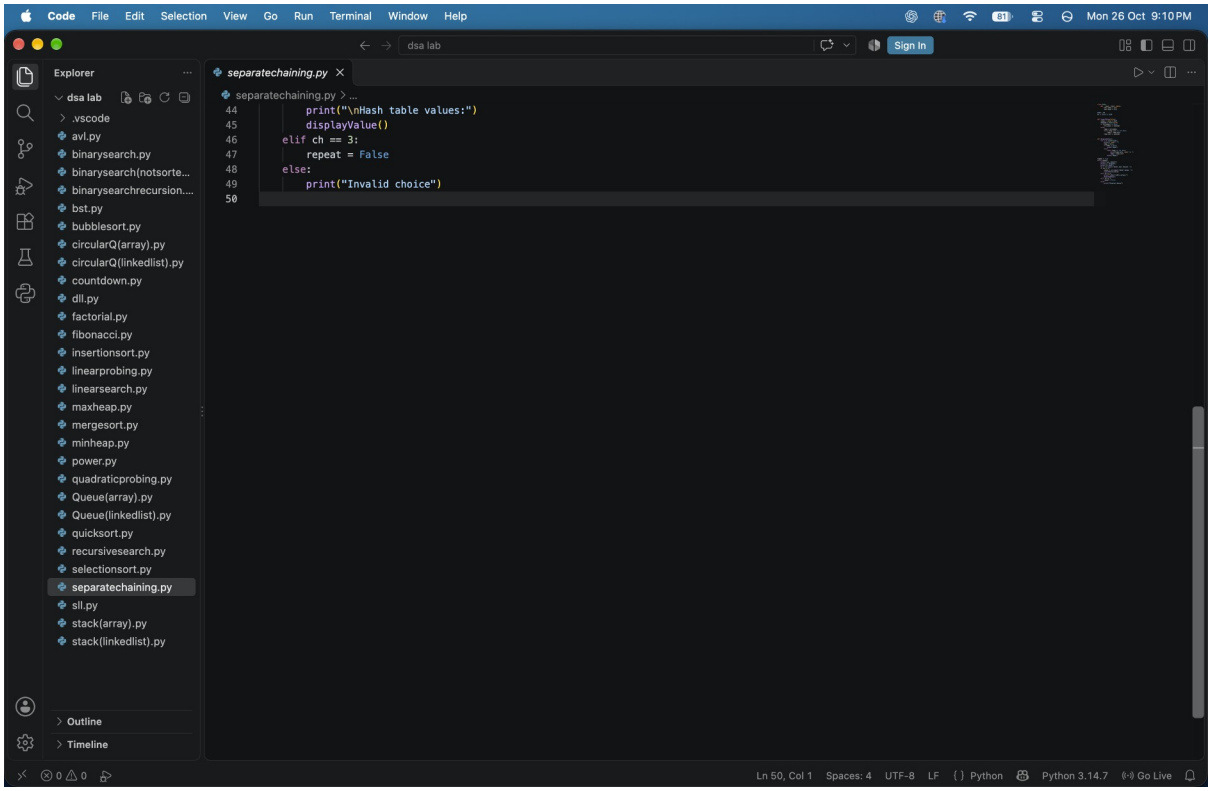
```
class Node:
    def __init__(self, data):
        self.data = data
        self.next = None

SIZE = 10
ht = [None] * SIZE

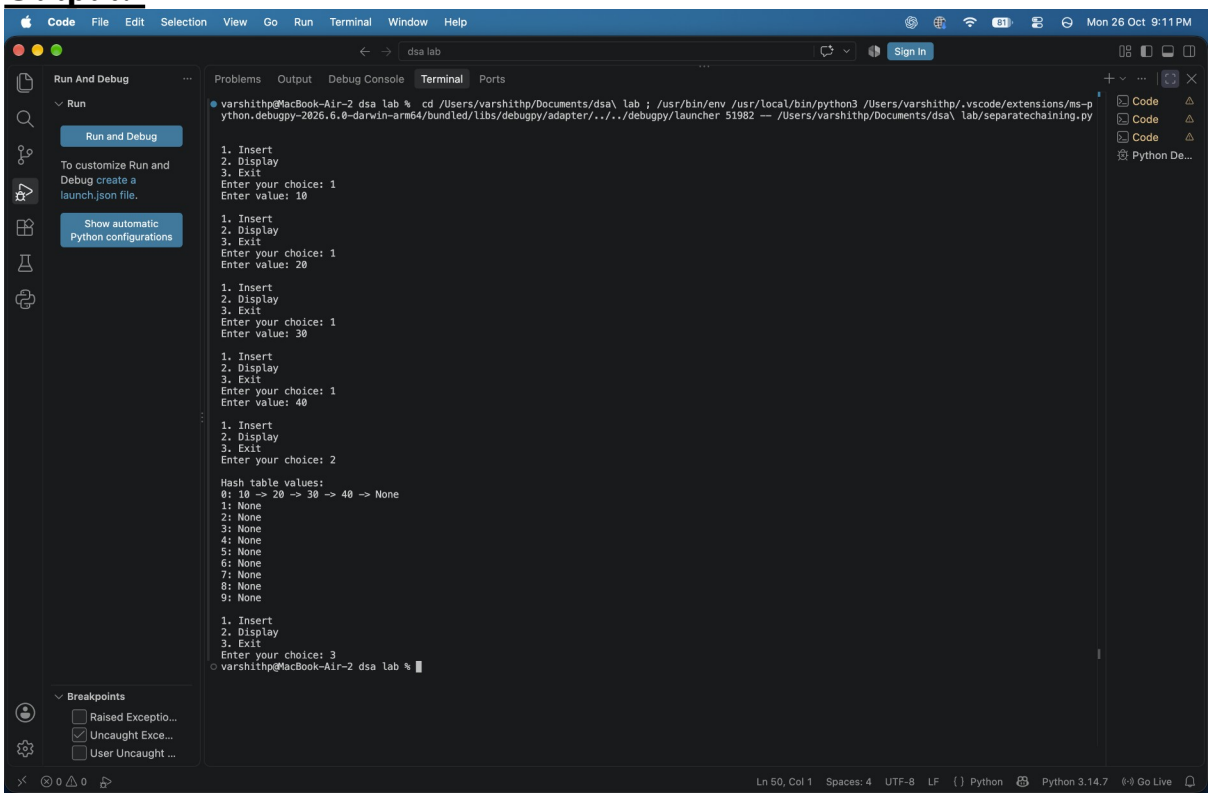

def insertValue(value):
    index = value % SIZE
    newnode = Node(value)
    if ht[index] is None:
        ht[index] = newnode
    else:
        temp = ht[index]
        while temp.next is not None:
            temp = temp.next
        temp.next = newnode


def displayValue():
    for i in range(SIZE):
        print(i, end=": ")
        temp = ht[i]
        if temp is None:
            print("None")
        else:
            while temp is not None:
                print(temp.data, end=" -> ")
                temp = temp.next
            print("None")

repeat = True
while repeat:
    print("\n1. Insert")
    print("2. Display")
    print("3. Exit")
    ch = int(input("Enter your choice: "))
    if ch == 1:
        value = int(input("Enter value: "))
        insertValue(value)
    elif ch == 2:
        print("\nHash table values:")
```

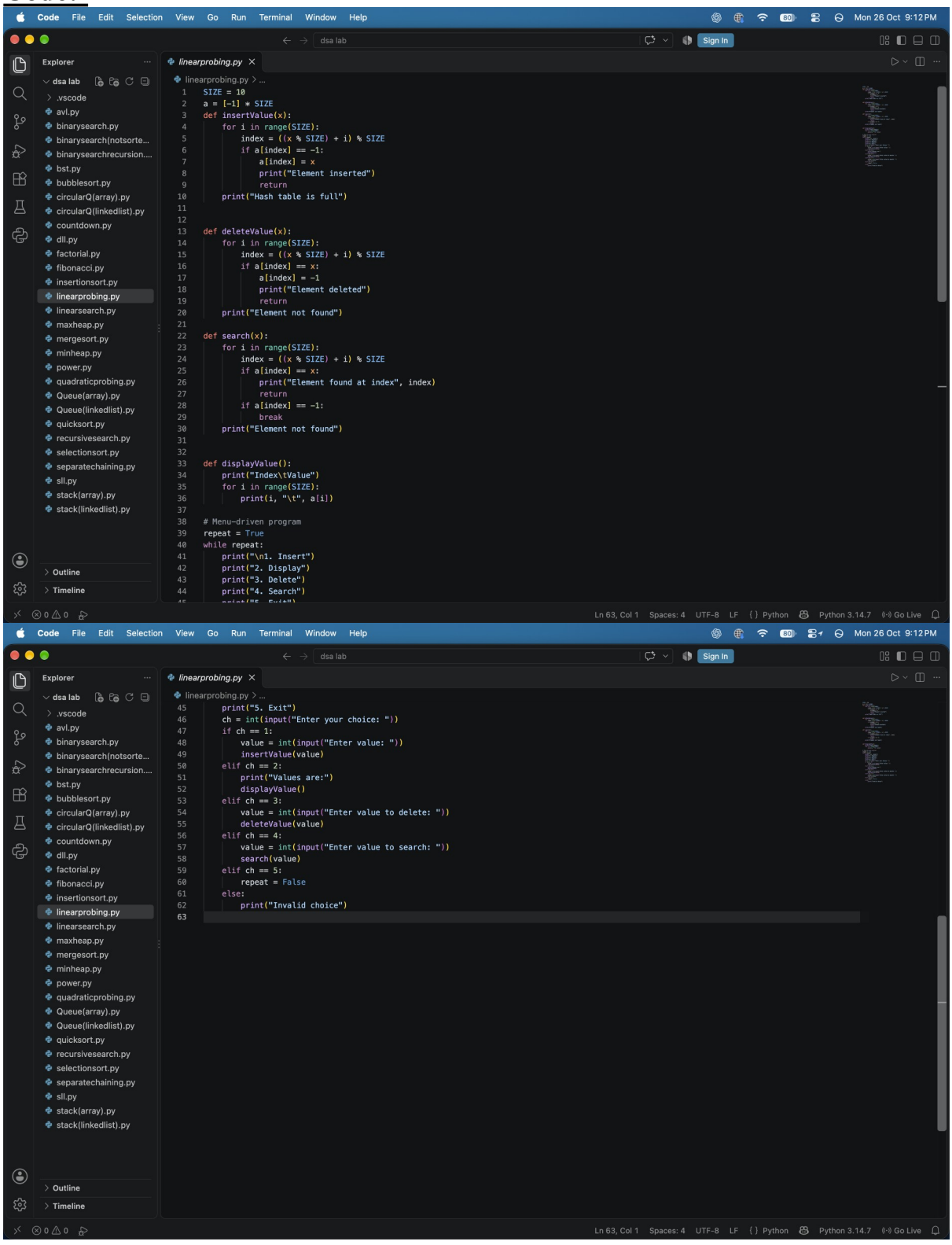

Code File Edit Selection View Go Run Terminal Window Help
Mon 26 Oct 9:10 PM
dsa lab
Sign in
Explorer
dsa lab
separatechaining.py
separatechaining.py > ...
44 print("\nHash table values:")
45 displayValue()
46 elif ch == 3:
47 repeat = False
48 else:
49 print("Invalid choice")
50
.vscode
avl.py
binarysearch.py
binarysearch(notsorte...
binarysearchrecursion....
bst.py
bubblesort.py
circularQ(array).py
circularQ(linkedlist).py
countdown.py
dll.py
factorial.py
fibonacci.py
insertionsort.py
linearprobing.py
linearsearch.py
maxheap.py
mergesort.py
minheap.py
power.py
quadraticprobing.py
Queue(array).py
Queue(linkedlist).py
quicksort.py
recursivesearch.py
selectionsort.py
separatechaining.py
sll.py
stack(array).py
stack(linkedlist).py
Outline
Timeline
Ln 50, Col 1 Spaces: 4 UTF-8 LF { } Python Python 3.14.7 Go Live

Output:-


Code File Edit Selection View Go Run Terminal Window Help
Mon 26 Oct 9:11 PM
dsa lab
Sign in
Run And Debug
Run
Run and Debug
To customize Run and Debug create a launch.json file.
Show automatic Python configurations
Problems Output Debug Console Terminal Ports
varshithp@MacBook-Air-2 dsa lab % cd /Users/varshithp/Documents/dsa\ lab ; /usr/bin/env /usr/local/bin/python3 /Users/varshithp/.vscode/extensions/ms-python.debugpy-2026.6.0-darwin-arm64/bundled/libs/debugpy/adapter/../../debugpy/launcher 51982 -- /Users/varshithp/Documents/dsa\ lab/separatechaining.py
1. Insert
2. Display
3. Exit
Enter your choice: 1
Enter value: 10
1. Insert
2. Display
3. Exit
Enter your choice: 1
Enter value: 20
1. Insert
2. Display
3. Exit
Enter your choice: 1
Enter value: 30
1. Insert
2. Display
3. Exit
Enter your choice: 1
Enter value: 40
1. Insert
2. Display
3. Exit
Enter your choice: 2
Hash table values:
0: 10 -> 20 -> 30 -> 40 -> None
1: None
2: None
3: None
4: None
5: None
6: None
7: None
8: None
9: None
1. Insert
2. Display
3. Exit
Enter your choice: 3
varshithp@MacBook-Air-2 dsa lab %
Code
Code
Code
Python De...
Breakpoints
Raised Exceptio...
Uncaught Exce...
User Uncaught ...
Ln 50, Col 1 Spaces: 4 UTF-8 LF { } Python Python 3.14.7 Go Live

Linear Probing

Code:-



```
SIZE = 10
a = [-1] * SIZE
def insertValue(x):
    for i in range(SIZE):
        index = ((x % SIZE) + i) % SIZE
        if a[index] == -1:
            a[index] = x
            print("Element inserted")
            return
    print("Hash table is full")


def deleteValue(x):
    for i in range(SIZE):
        index = ((x % SIZE) + i) % SIZE
        if a[index] == x:
            a[index] = -1
            print("Element deleted")
            return
    print("Element not found")

def search(x):
    for i in range(SIZE):
        index = ((x % SIZE) + i) % SIZE
        if a[index] == x:
            print("Element found at index", index)
            return
        if a[index] == -1:
            break
    print("Element not found")

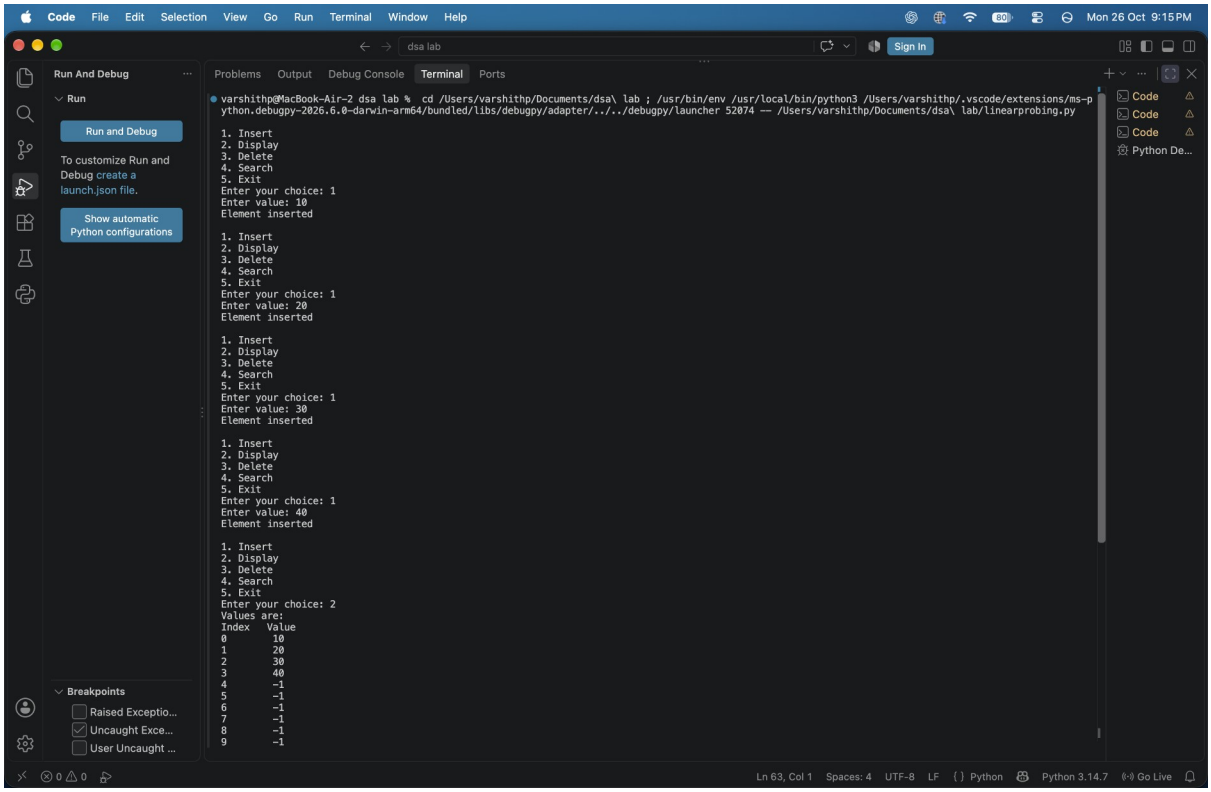

def displayValue():
    print("Index\tValue")
    for i in range(SIZE):
        print(i, "\t", a[i])

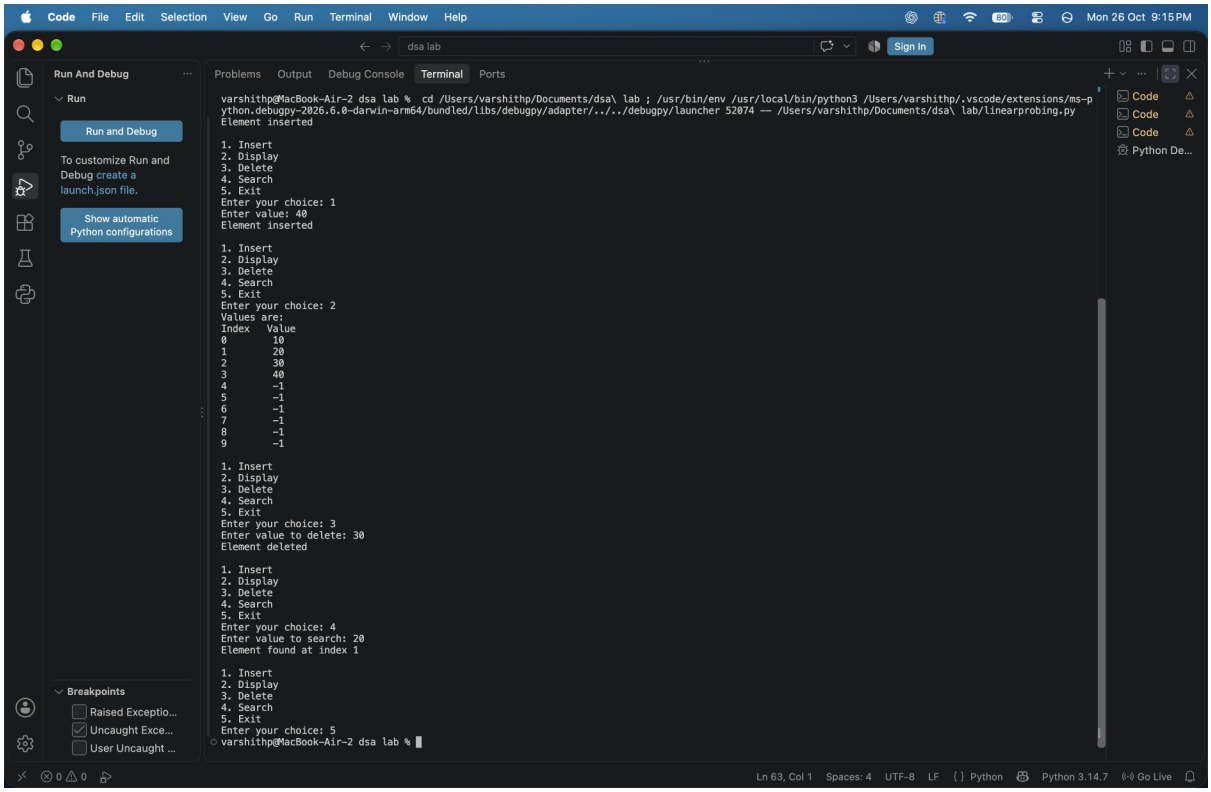
# Menu-driven program
repeat = True
while repeat:
    print("\n1. Insert")
    print("2. Display")
    print("3. Delete")
    print("4. Search")
```



```
    print("5. Exit")
    ch = int(input("Enter your choice: "))
    if ch == 1:
        value = int(input("Enter value: "))
        insertValue(value)
    elif ch == 2:
        print("Values are:")
        displayValue()
    elif ch == 3:
        value = int(input("Enter value to delete: "))
        deleteValue(value)
    elif ch == 4:
        value = int(input("Enter value to search: "))
        search(value)
    elif ch == 5:
        repeat = False
    else:
        print("Invalid choice")
```

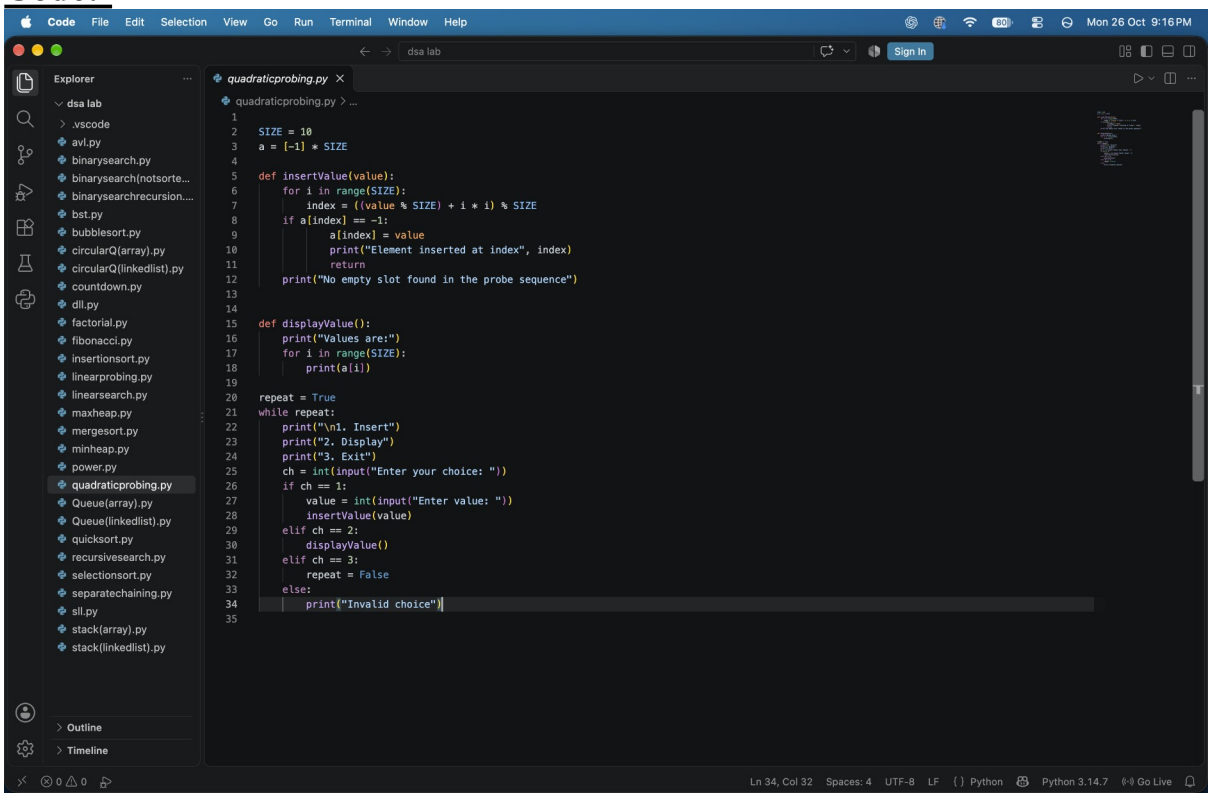
Output:-


varshithp@MacBook-Air-2 dsa lab % cd /Users/varshithp/Documents/dsa\ lab ; /usr/bin/env /usr/local/bin/python3 /Users/varshithp/.vscode/extensions/ms-python.debugpy-2026.6.0-darwin-arm64/bundled/libs/debugpy/adapter/../../debugpy/launcher 52074 -- /Users/varshithp/Documents/dsa\ lab/linearprobing.py
1. Insert
2. Display
3. Delete
4. Search
5. Exit
Enter your choice: 1
Enter value: 10
Element inserted
1. Insert
2. Display
3. Delete
4. Search
5. Exit
Enter your choice: 1
Enter value: 20
Element inserted
1. Insert
2. Display
3. Delete
4. Search
5. Exit
Enter your choice: 1
Enter value: 30
Element inserted
1. Insert
2. Display
3. Delete
4. Search
5. Exit
Enter your choice: 1
Enter value: 40
Element inserted
1. Insert
2. Display
3. Delete
4. Search
5. Exit
Enter your choice: 2
Values are:
Index Value
0 10
1 20
2 30
3 40
4 -1
5 -1
6 -1
7 -1
8 -1
9 -1


varshithp@MacBook-Air-2 dsa lab % cd /Users/varshithp/Documents/dsa\ lab ; /usr/bin/env /usr/local/bin/python3 /Users/varshithp/.vscode/extensions/ms-python.debugpy-2026.6.0-darwin-arm64/bundled/libs/debugpy/adapter/../../debugpy/launcher 52074 -- /Users/varshithp/Documents/dsa\ lab/linearprobing.py
Element inserted
1. Insert
2. Display
3. Delete
4. Search
5. Exit
Enter your choice: 1
Enter value: 40
Element inserted
1. Insert
2. Display
3. Delete
4. Search
5. Exit
Enter your choice: 2
Values are:
Index Value
0 10
1 20
2 30
3 40
4 -1
5 -1
6 -1
7 -1
8 -1
9 -1
1. Insert
2. Display
3. Delete
4. Search
5. Exit
Enter your choice: 3
Enter value to delete: 30
Element deleted
1. Insert
2. Display
3. Delete
4. Search
5. Exit
Enter your choice: 4
Enter value to search: 20
Element found at index 1
1. Insert
2. Display
3. Delete
4. Search
5. Exit
Enter your choice: 5
varshithp@MacBook-Air-2 dsa lab %

Quadratic Probing

Code:-



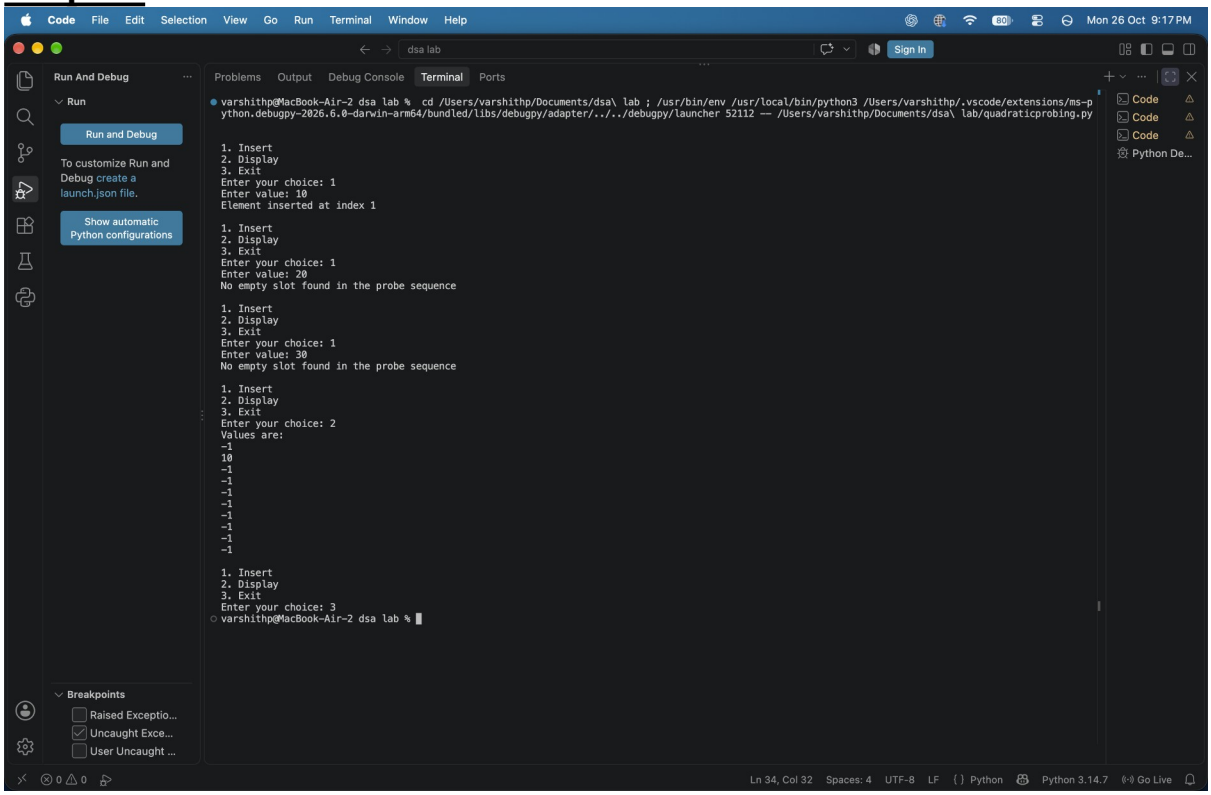
```
SIZE = 10
a = [-1] * SIZE

def insertValue(value):
    for i in range(SIZE):
        index = ((value % SIZE) + i * i) % SIZE
    if a[index] == -1:
            a[index] = value
            print("Element inserted at index", index)
            return
    print("No empty slot found in the probe sequence")


def displayValue():
    print("Values are:")
    for i in range(SIZE):
        print(a[i])

repeat = True
while repeat:
    print("\n1. Insert")
    print("2. Display")
    print("3. Exit")
    ch = int(input("Enter your choice: "))
    if ch == 1:
        value = int(input("Enter value: "))
        insertValue(value)
    elif ch == 2:
        displayValue()
    elif ch == 3:
        repeat = False
    else:
        print("Invalid choice")
```

Output:-



```
1. Insert
2. Display
3. Exit
Enter your choice: 1
Enter value: 10
Element inserted at index 1

1. Insert
2. Display
3. Exit
Enter your choice: 1
Enter value: 20
No empty slot found in the probe sequence

1. Insert
2. Display
3. Exit
Enter your choice: 1
Enter value: 30
No empty slot found in the probe sequence

1. Insert
2. Display
3. Exit
Enter your choice: 2
Values are:
-1
10
-1
-1
-1
-1
-1
-1
-1
-1

1. Insert
2. Display
3. Exit
Enter your choice: 3
varshithp@MacBook-Air-2 dsa lab %
```